Resilient Communication Networks

Designing Resilient Communication Networks

Redundancy Mechanisms



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Disclaimer: some of these slides have been derived from Günter Schäfer's lecture „Protection of Communication Infrastructures", Mathias Fischers Lecture „Resilient Networks", Matthias Hollick's Lecture „ Ringvorlesung KRITIS", and Ralf Steinmetz's Lecture "Communication Networks II"

L3-Routing (Intra Autonomous System)

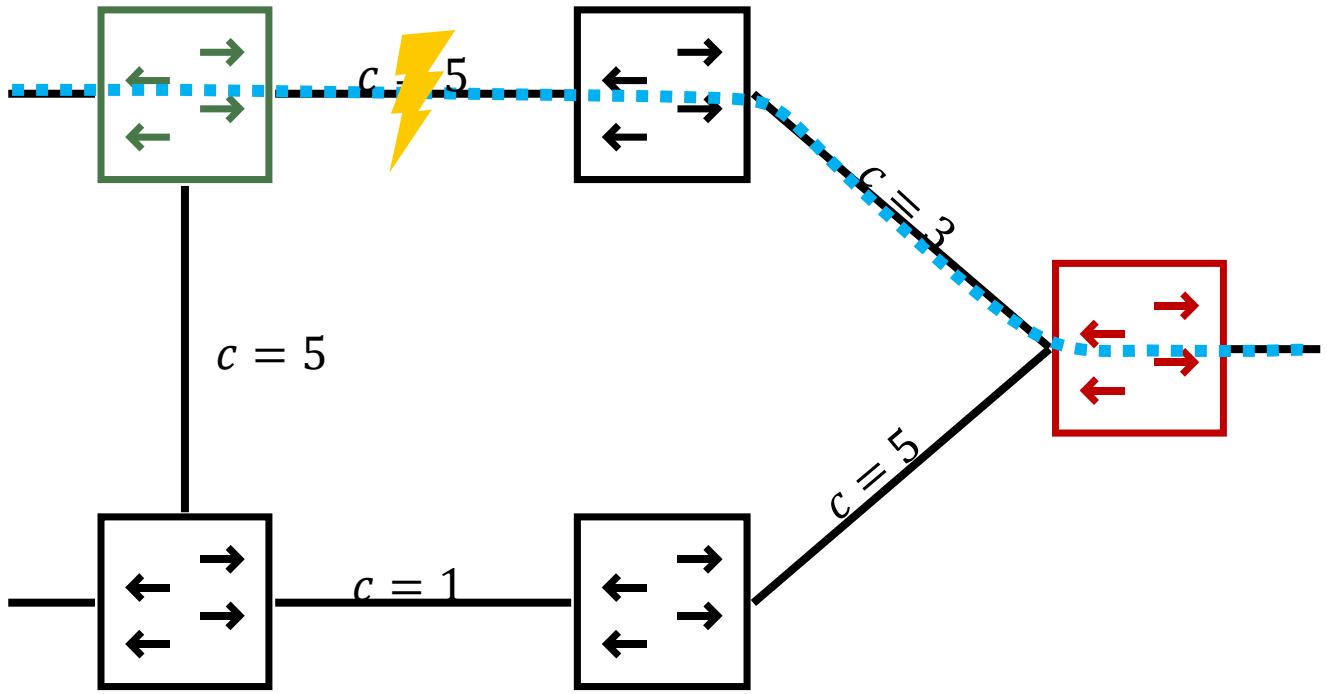
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Based on different routing protocols

- Distance-vector based, e.g., Routing Information Protocol (RIP), Interior Gateway Routing Protocol (IGRP)
- Link-State based, e.g., Open Shortest Path First (OSPF) or Intermediate System to Intermediate System (IS-IS)

All of those protocols build routing tables using a cost metric based on

- Number of hops
- Bandwidth
- Latency



After a failure, these routing tables need to converge again

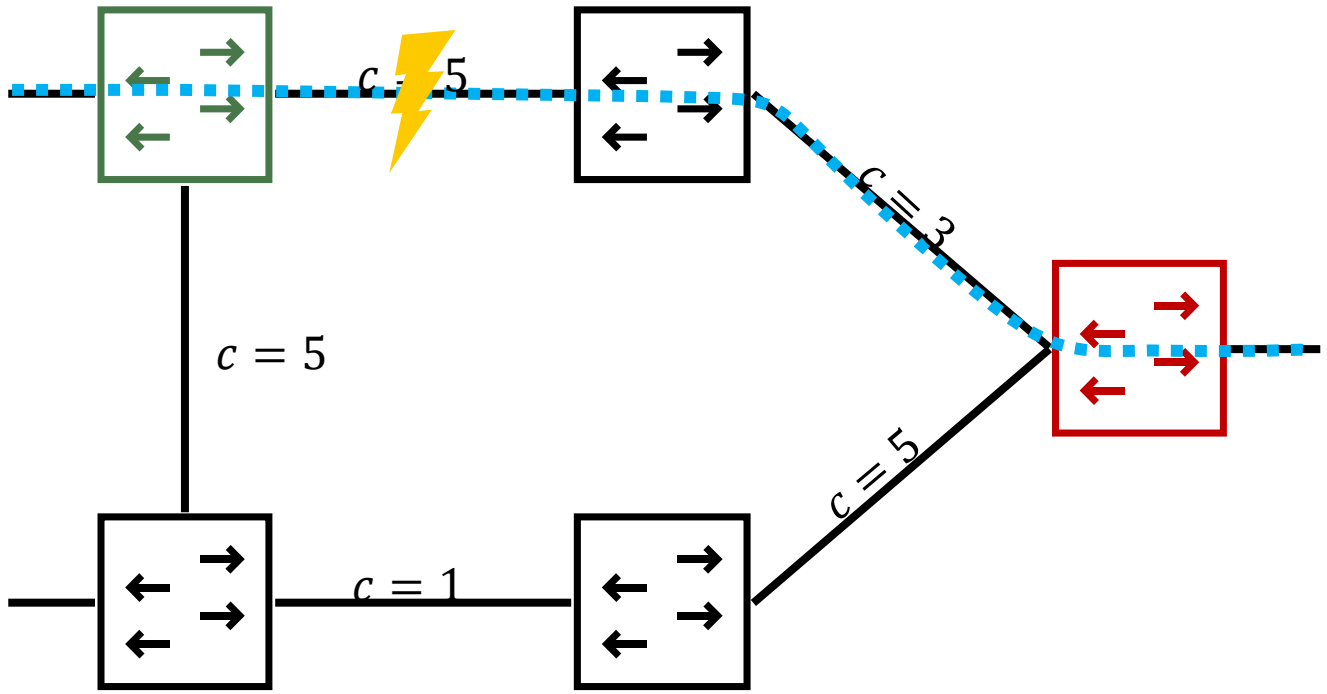
Loop-Free Alternates

After a failure, the routing tables still contain (now potentially invalid) routes

- Neighbor routers can react to the failure directly
- But all other routers need to notice and react

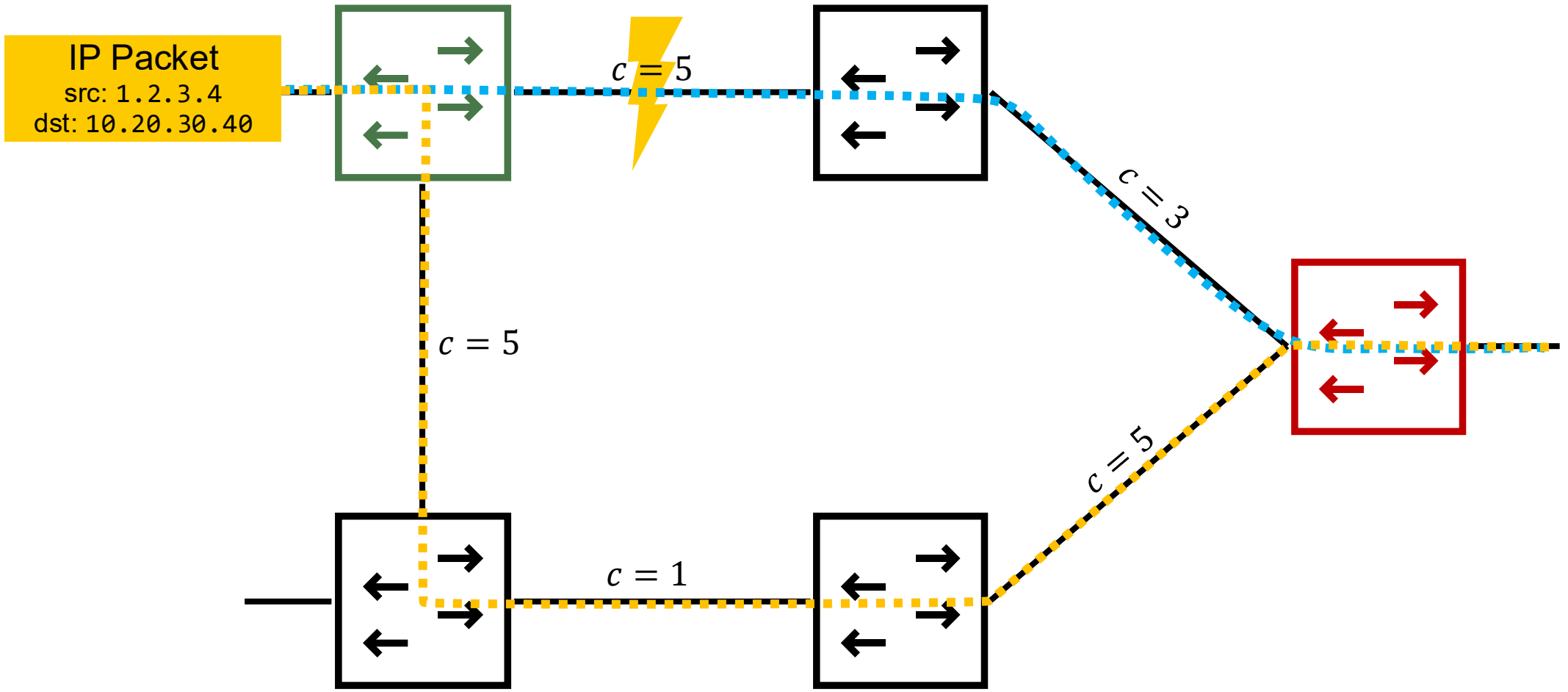
During that time, traffic utilizing the failed link would be lost

- This holds true even if alternative paths exist
- Loop-Free Alternates as defined in RFC 5286 avoid this issue by preparing alternative routes

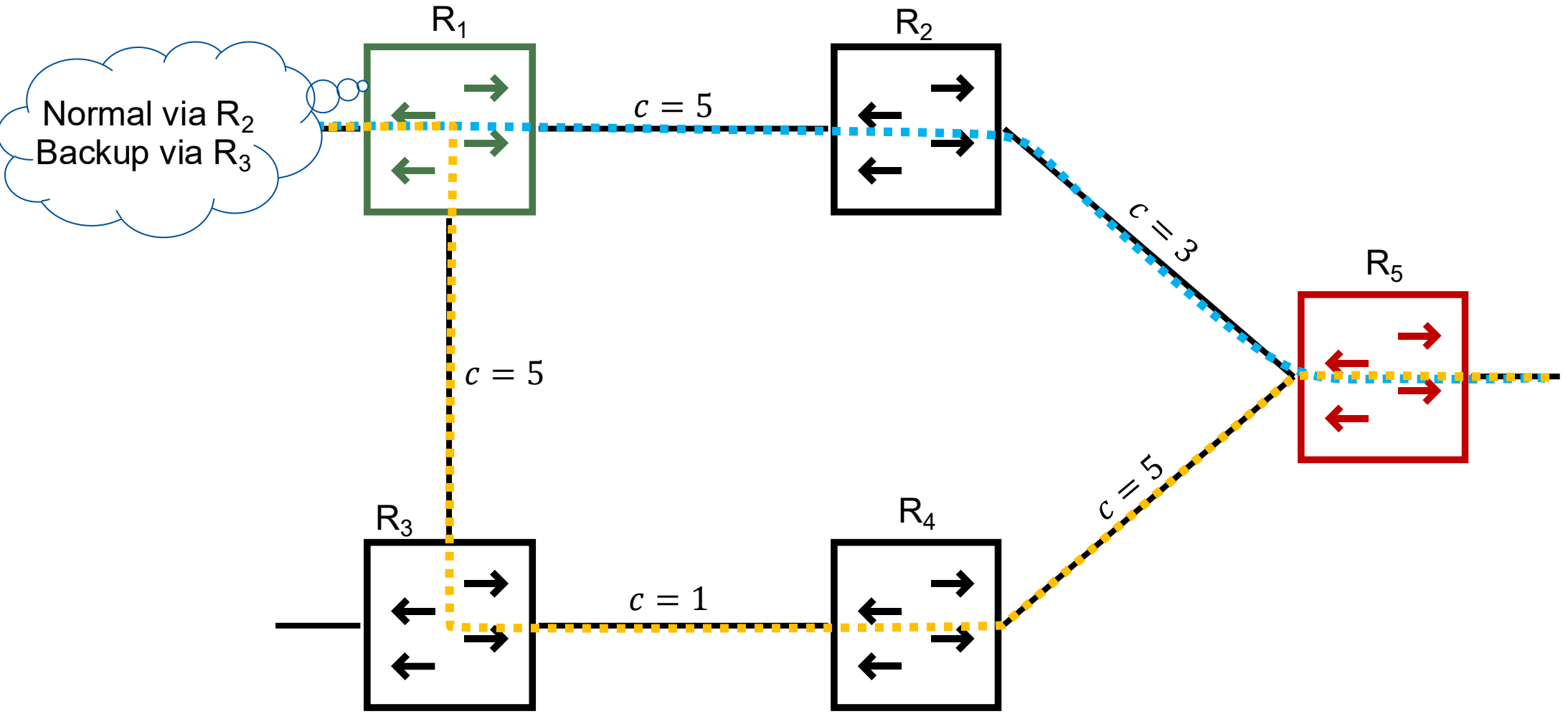


General idea: Calculate a backup path that does not create a loop

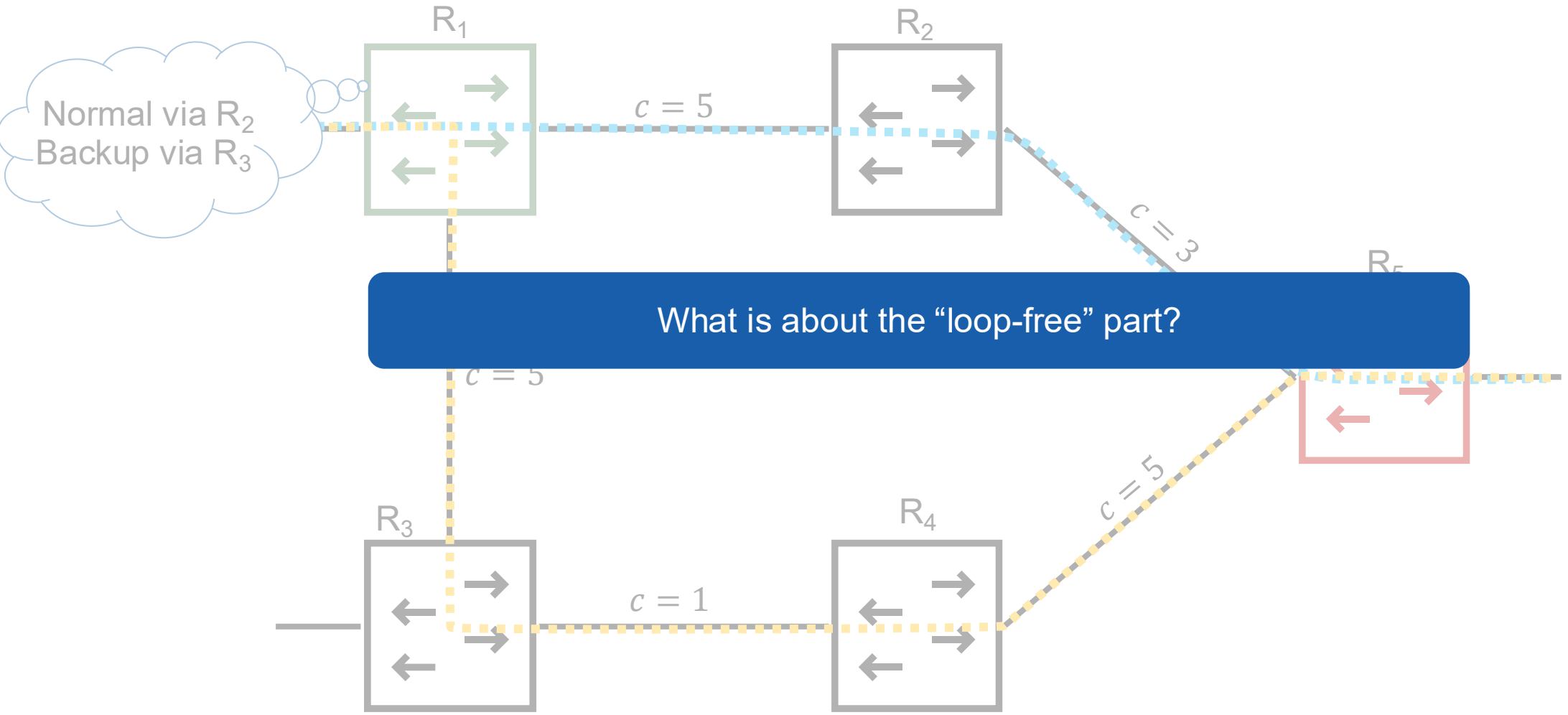
Loop-Free Alternates



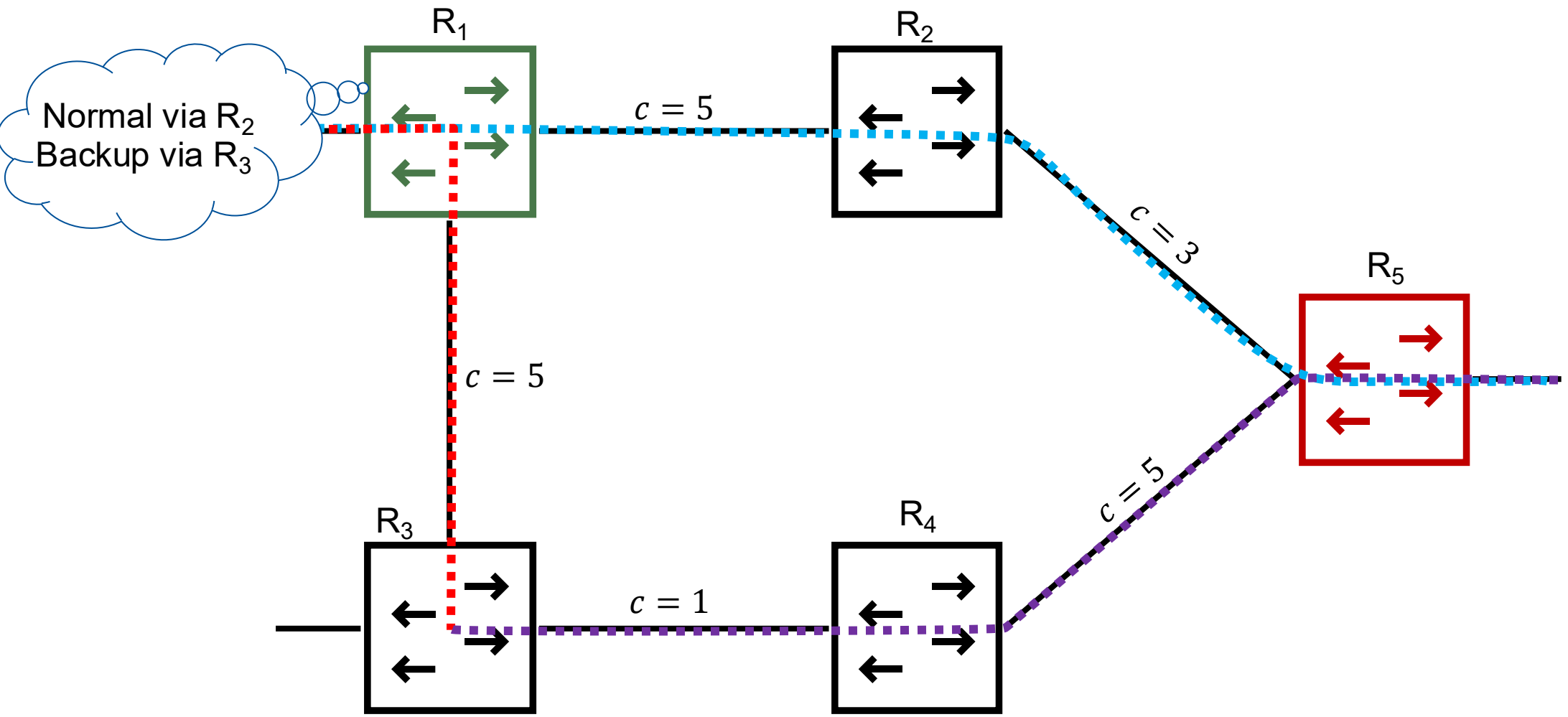
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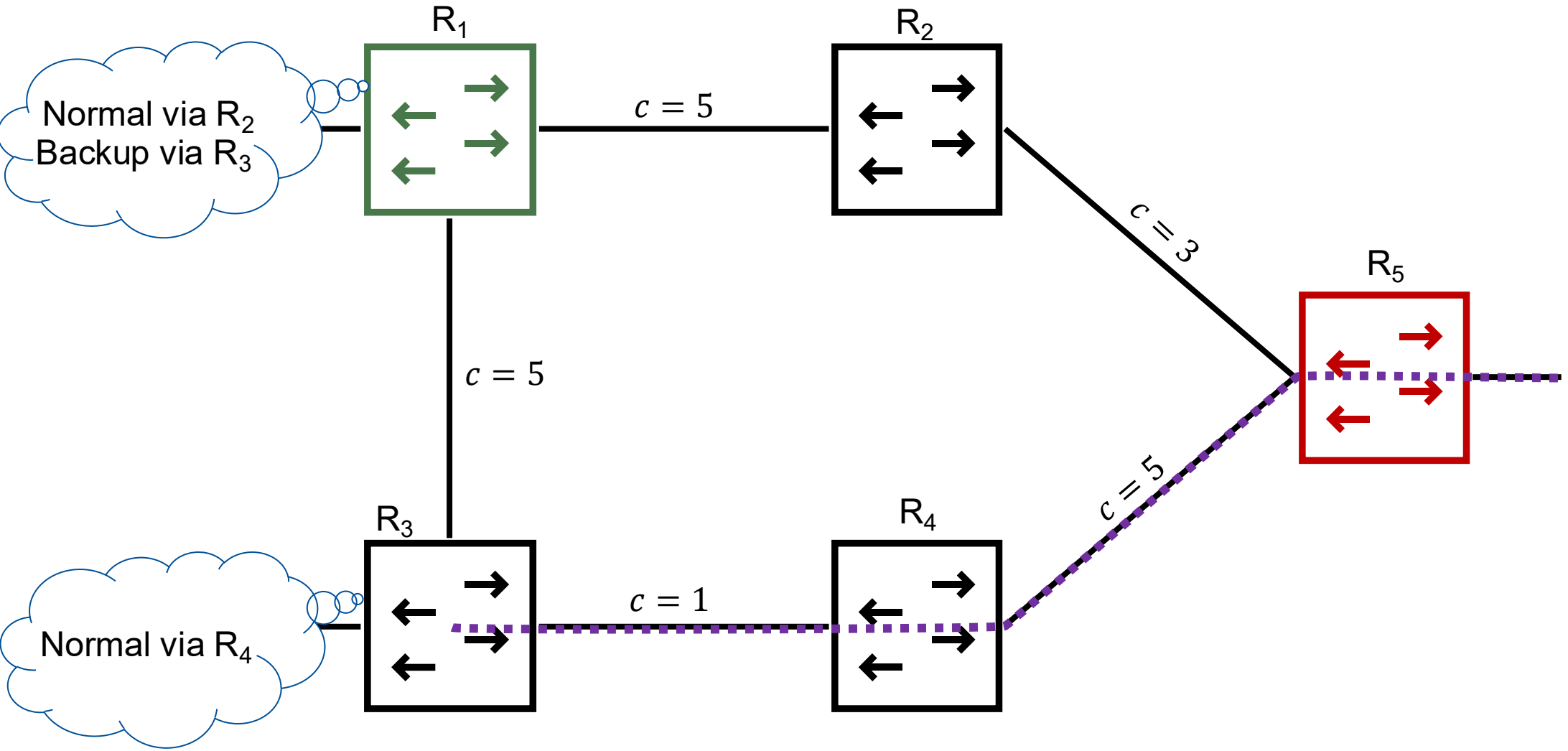
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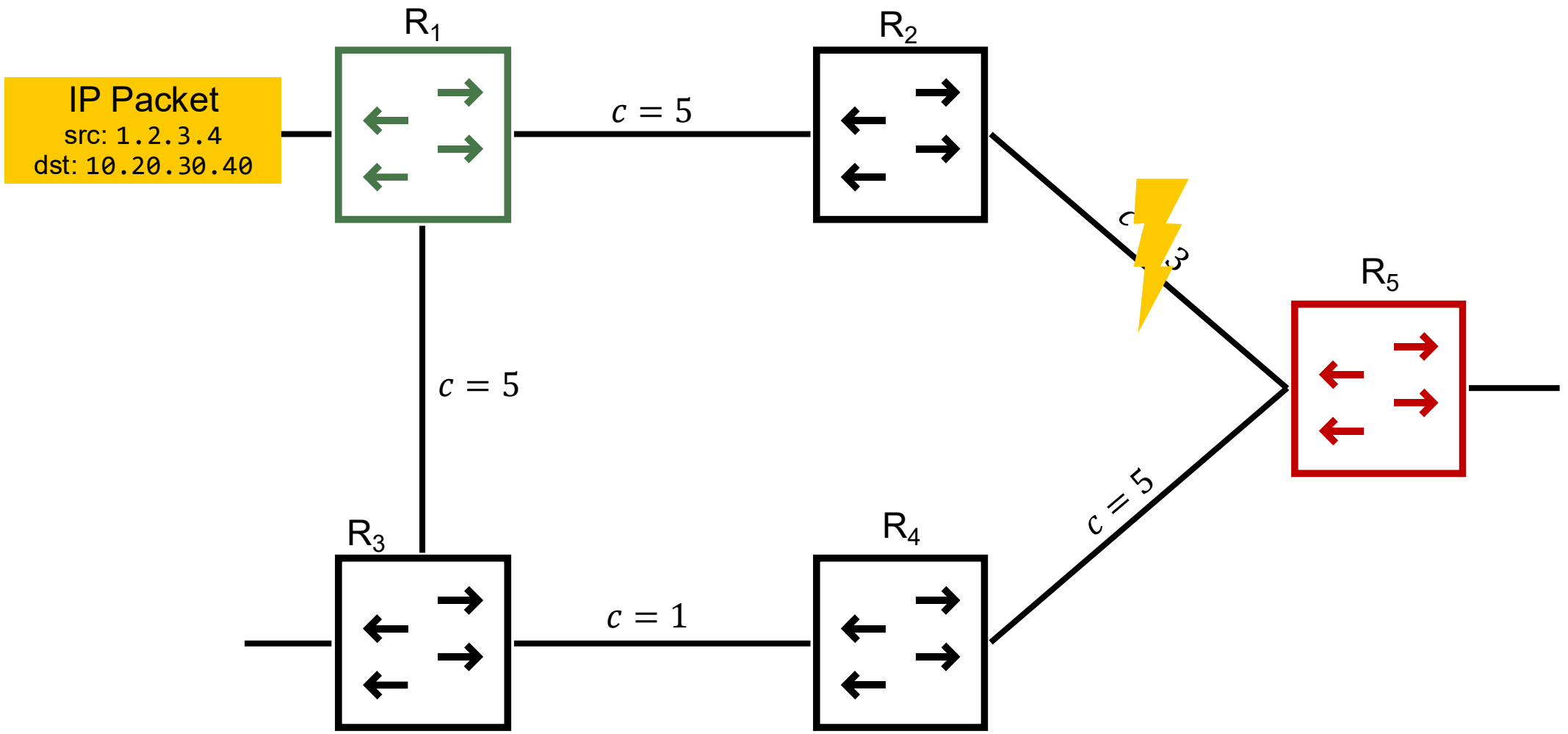
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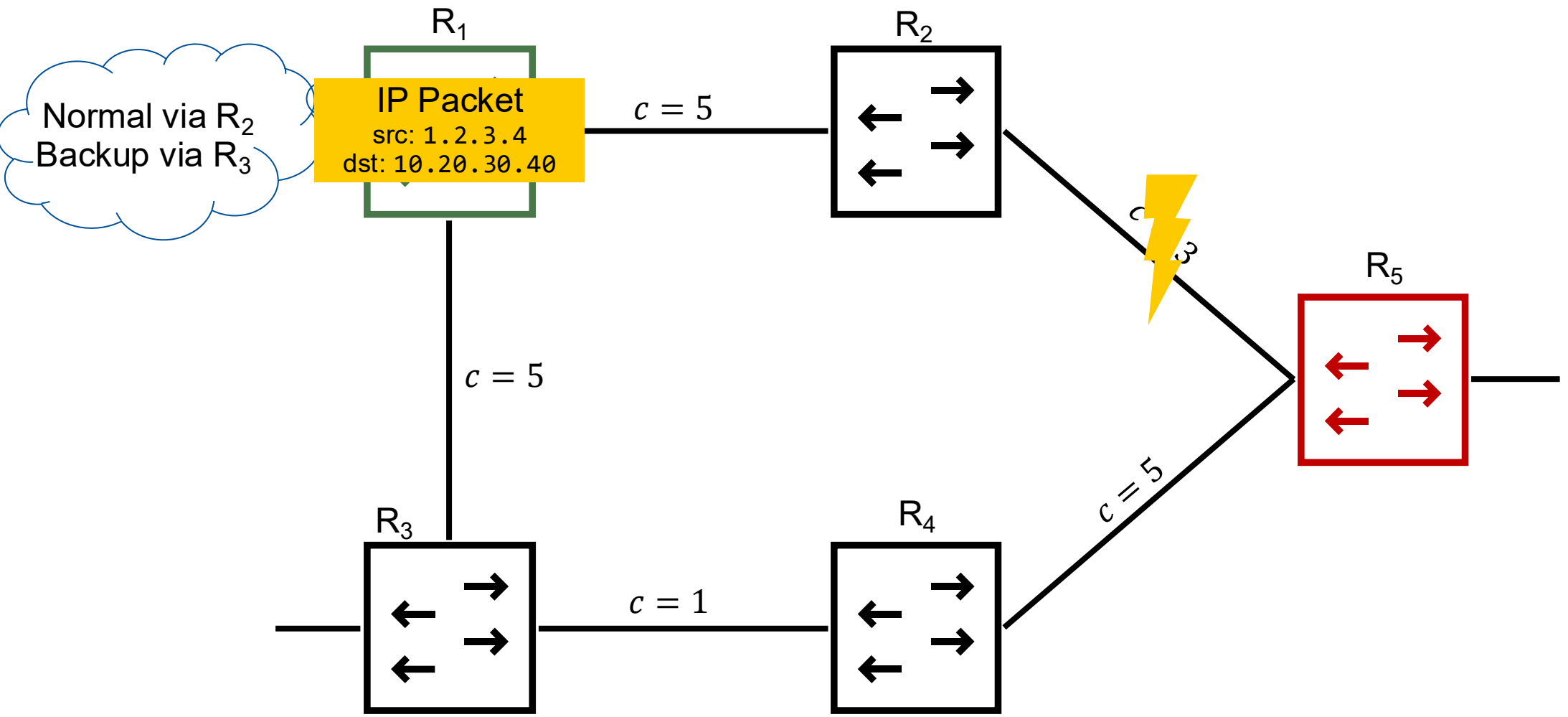
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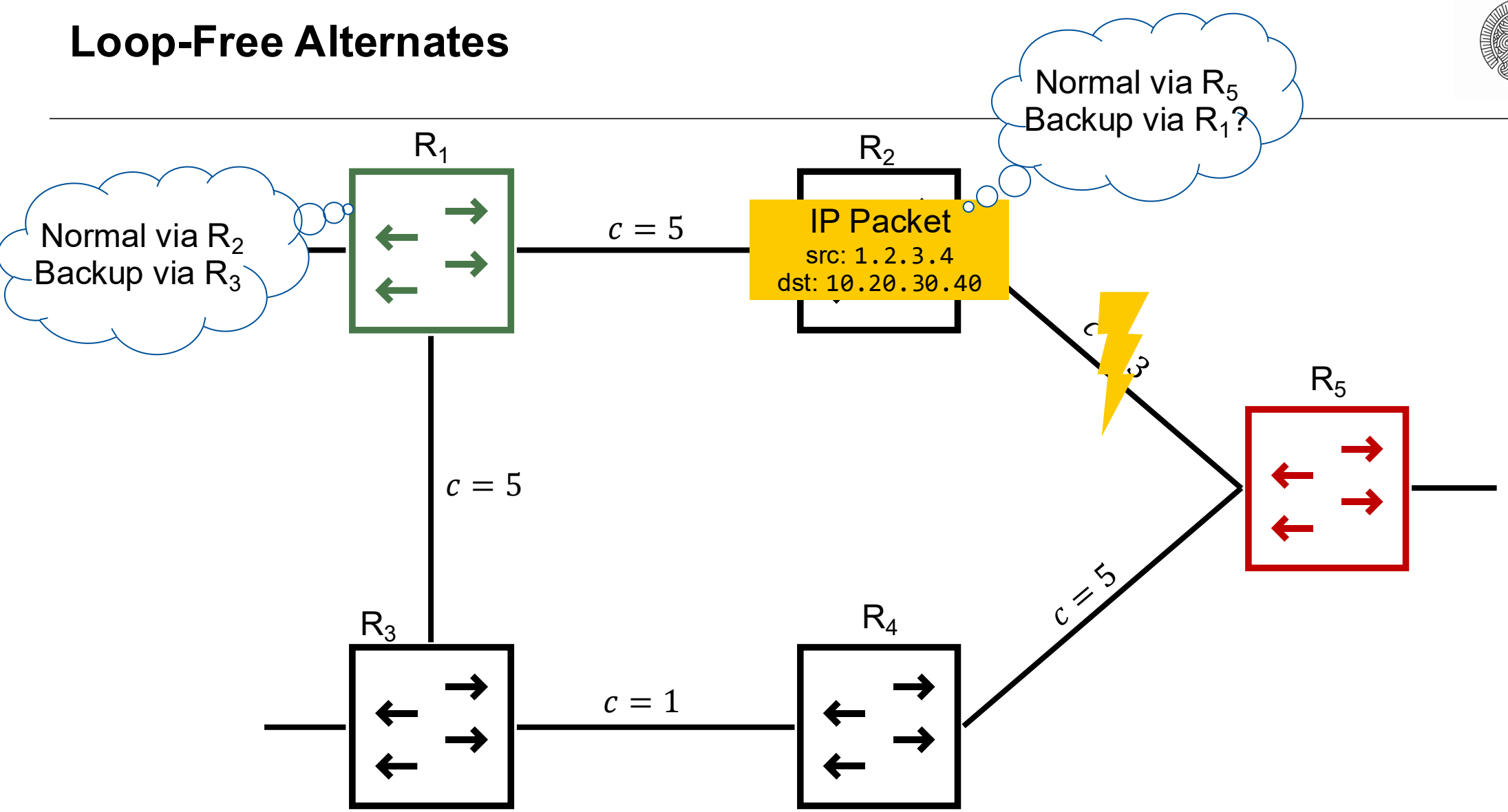
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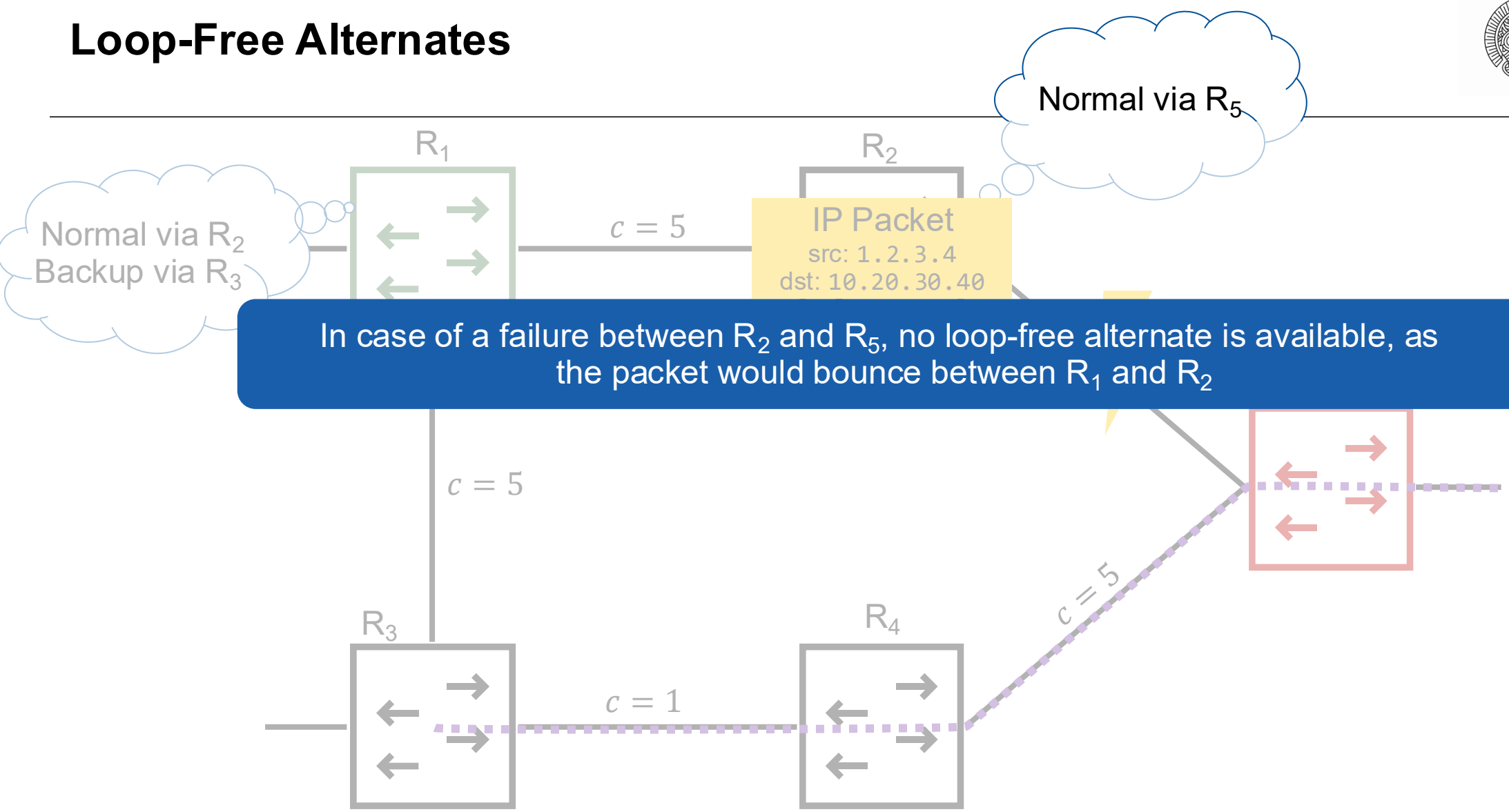
Loop-Free Alternates



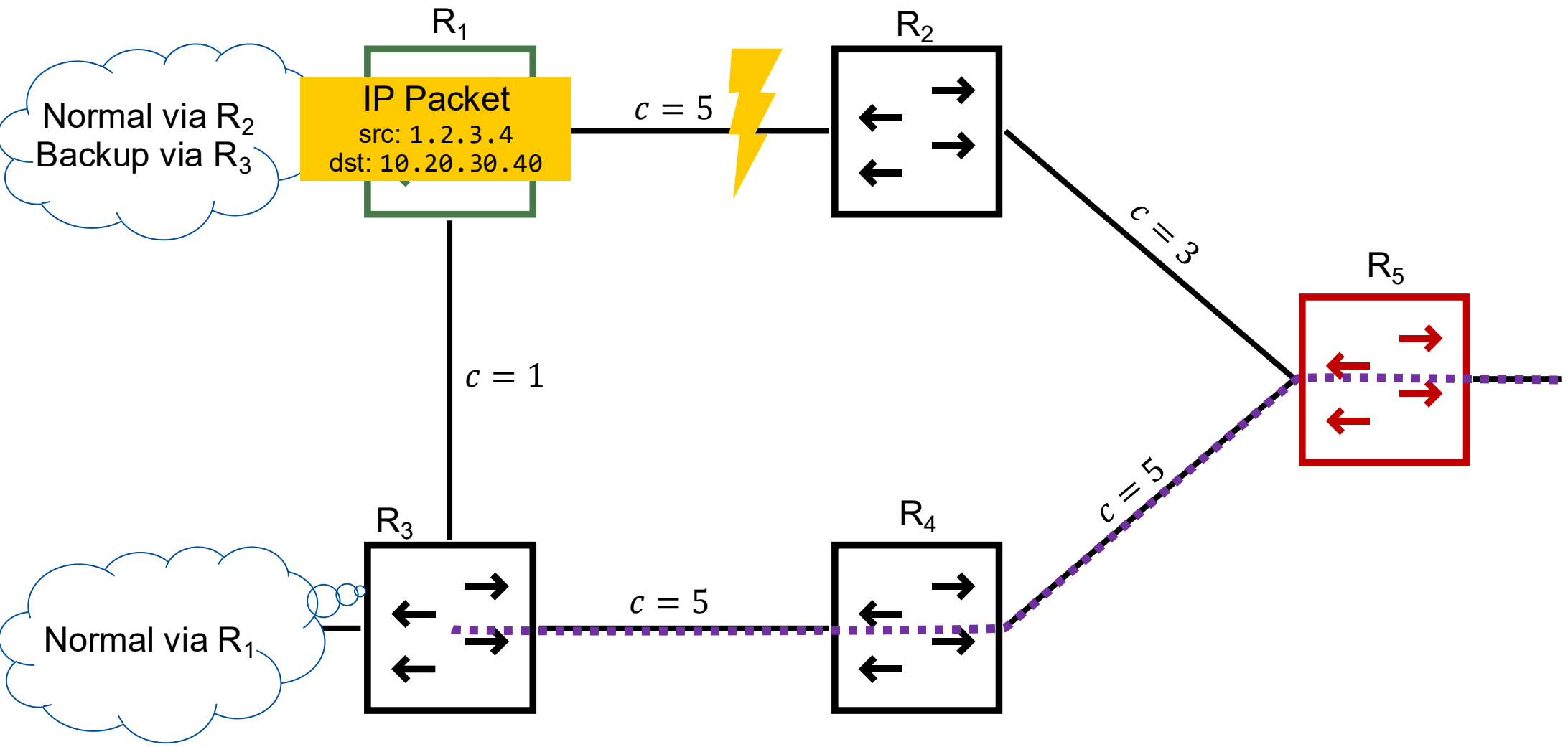
Loop-Free Alternates



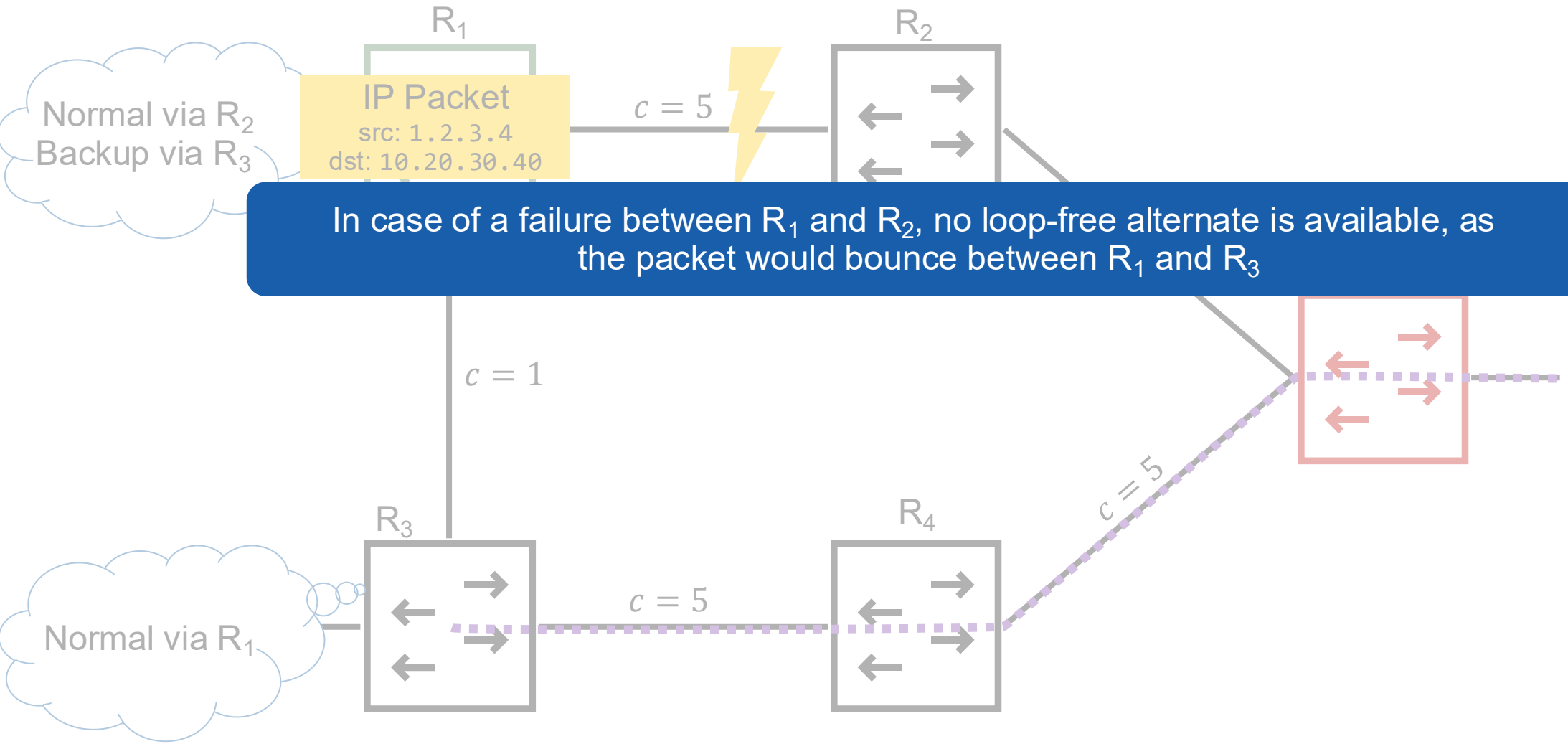
Loop-Free Alternates



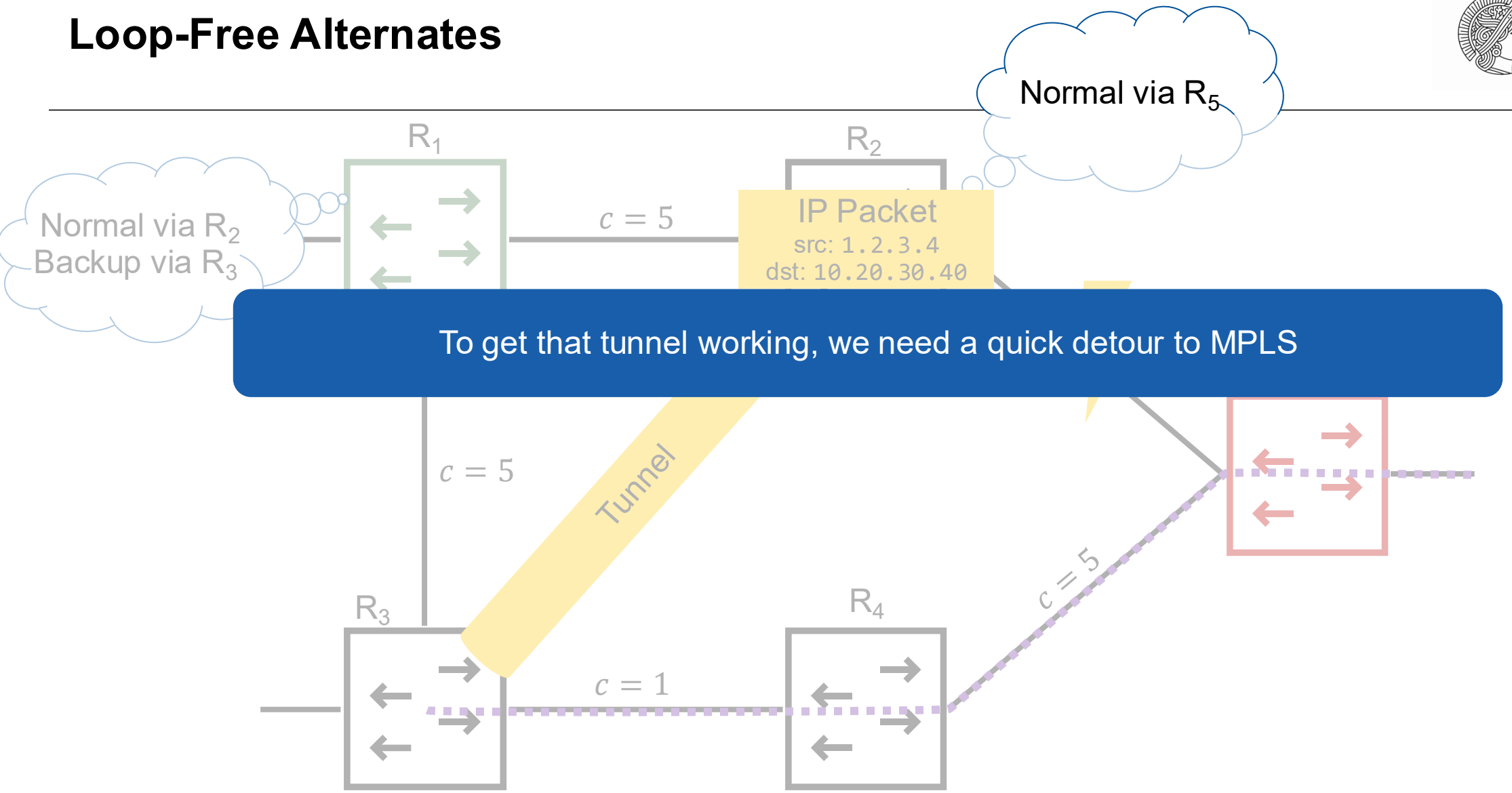
Loop-Free Alternates



Loop-Free Alternates



Loop-Free Alternates



Multiprotocol Label Switching (MPLS)

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Instead of addressing nodes by their IP-address, MPLS addresses paths

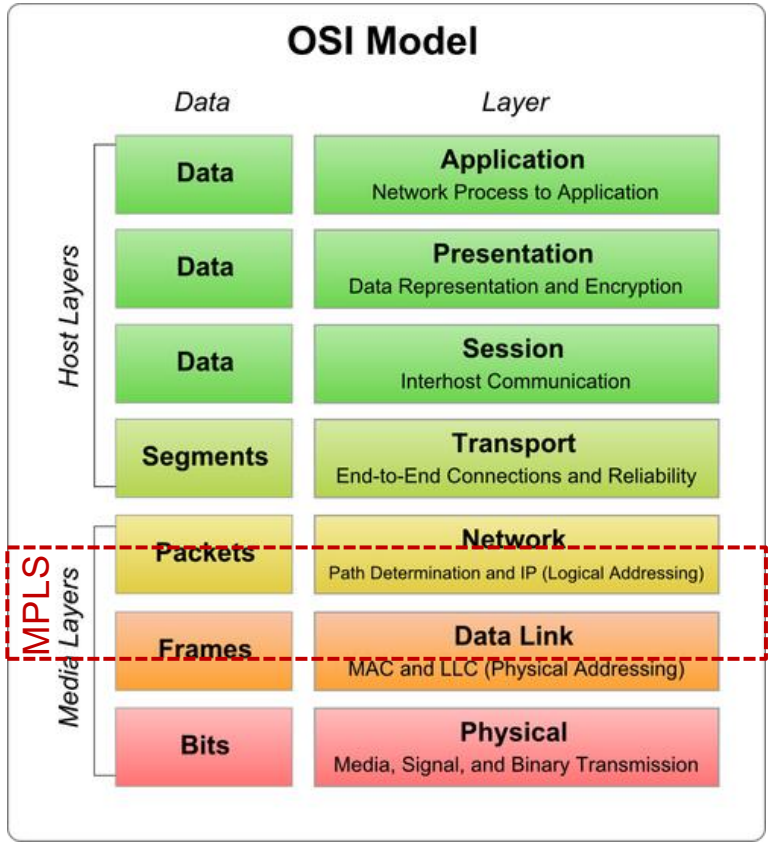
- Encapsulates the packets of classical network protocols

General idea: Assign labels to packets that determine their future path

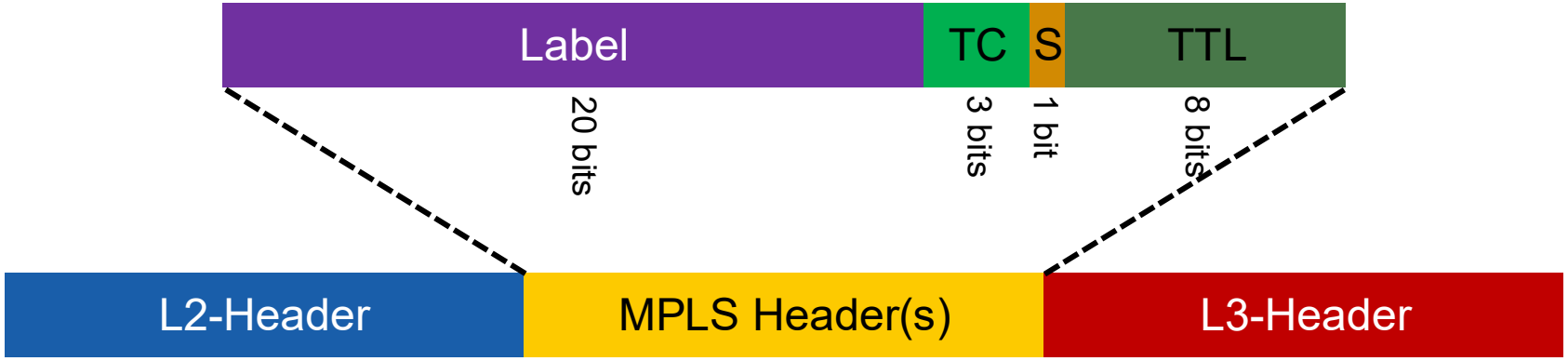
MPLS improves the efficiency of transport networks

- Switching based on labels is less complex compared to longest-prefix-matching (standard of IP)
- Label-Switched Paths (LSPs) are determined once entering the MPLS network, which packets will follow while traversing the network

A Label Edge Router assigns a label to each incoming packet



MPLS Header Format



The MPLS Header is located between the L2 and L3 header

- In Ethernet, its presence is communicated using the Ethernet type `0x8846` (unicast) and `0x8847` (multicast)
- It consists of a label value, a traffic class (TC), a flag indicating if it is the last MPLS Label, and a Time-to-Live (TTL)
- There can be multiple MPLS Header at the same time, but each router will only inspect the topmost label

Routing in MPLS

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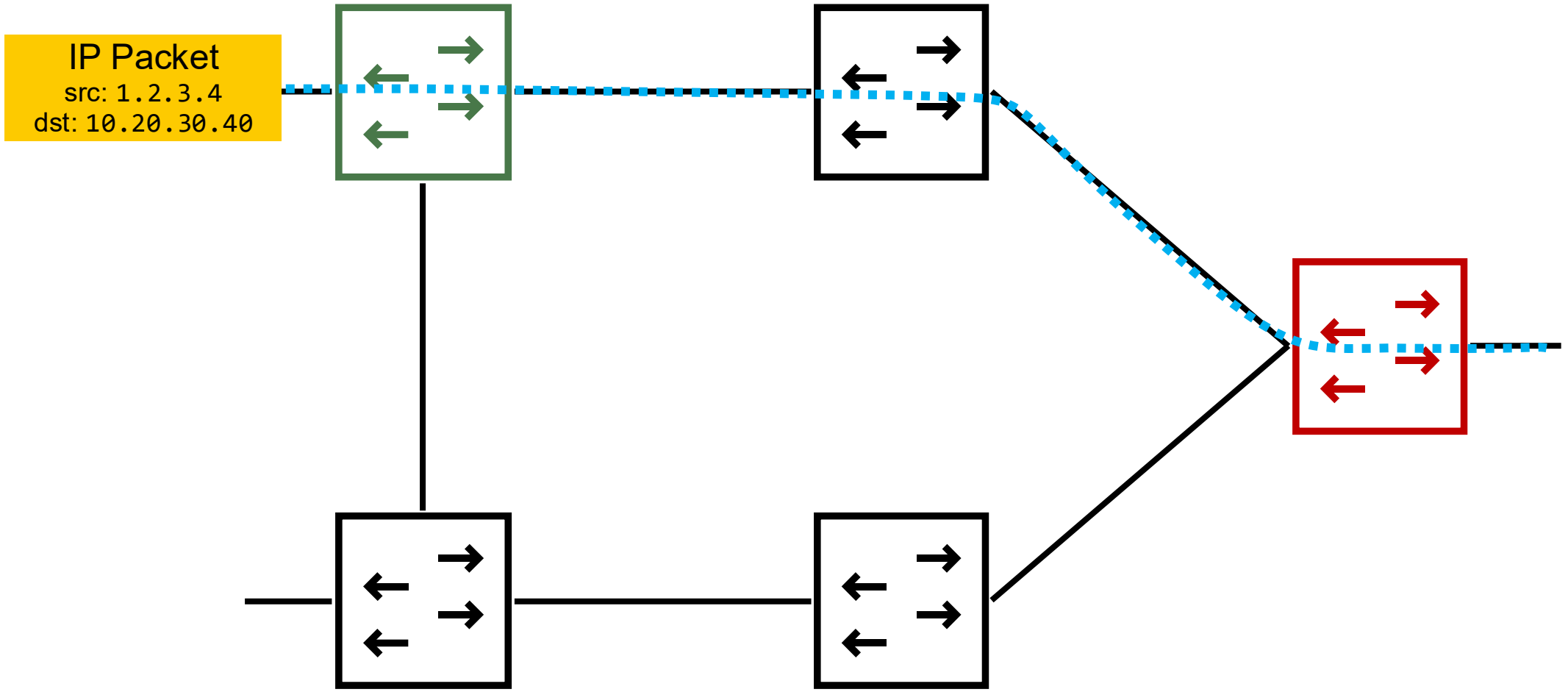
Paths are predetermined

- Based on pre-calculated label-switched path (LSP) (often called MPLS tunnel)
  - Initiated using the Label Distribution Protocol (LDP) or the Resource Reservation Protocol (RSVP-TE)
  - Can find paths that are maximally diverse, i.e. share as little hops with other paths as possible
- Enables traffic engineering and QoS-guarantees

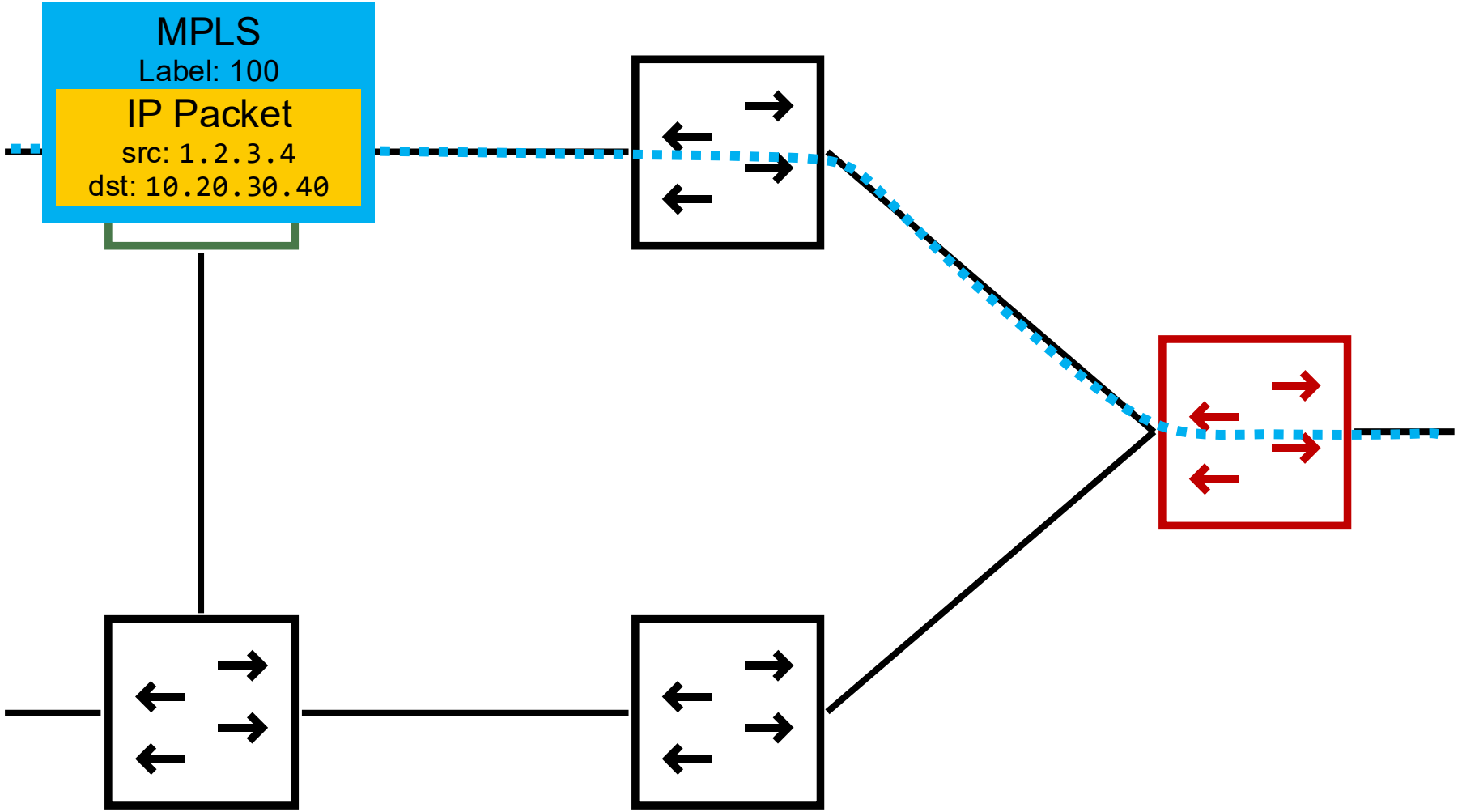
Decision-making hop-by-hop

- Simple lookup based on the topmost label
- Each router can either
  - **Swap** the label (to set the next hop of the path)
  - **Push** a new topmost label (create a tunnel within the tunnel)
  - **Pop** the topmost label (terminate the tunnel)
- The first tunnel is always initiated at the ingress node and (often) terminated at the egress node

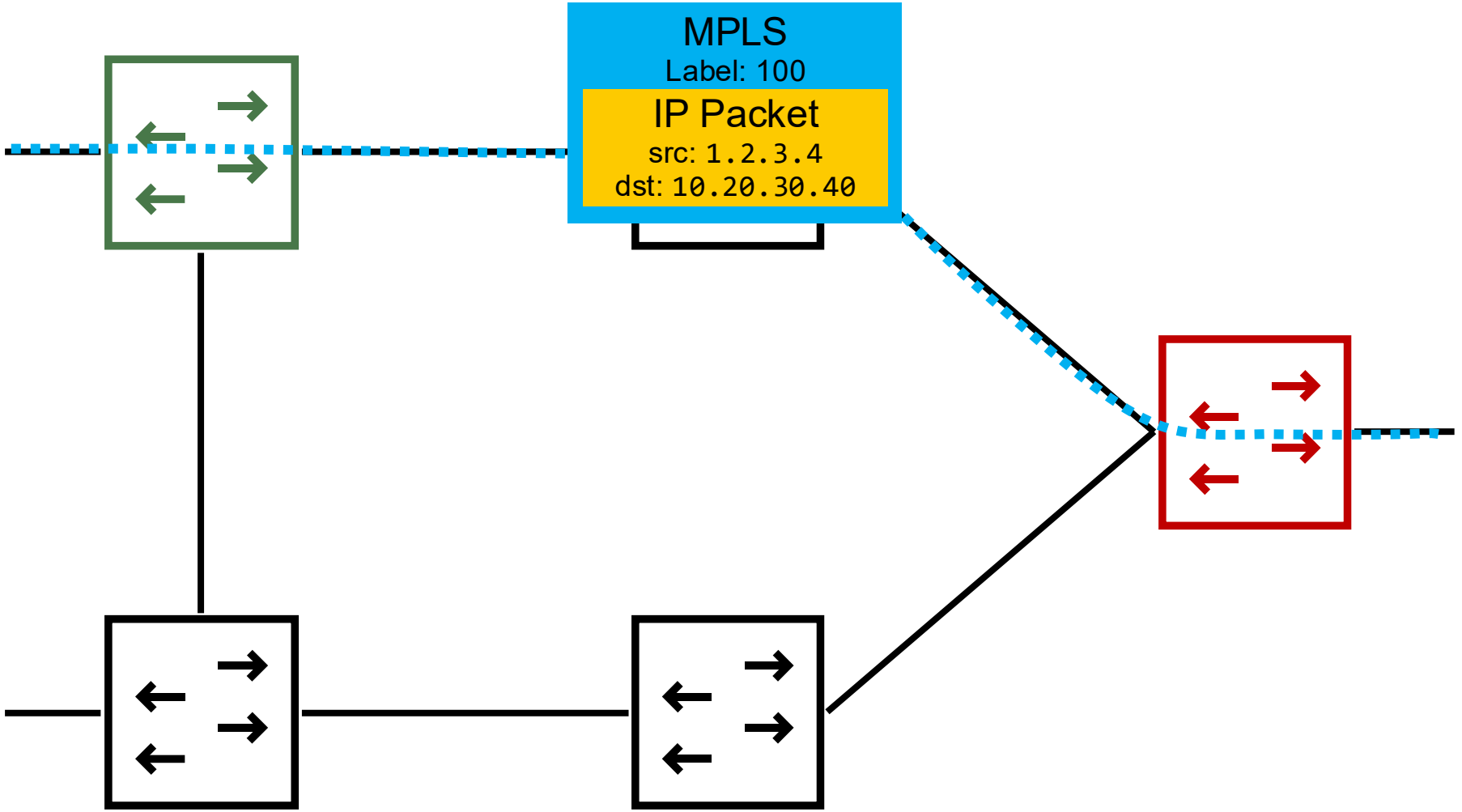
MPLS Routing



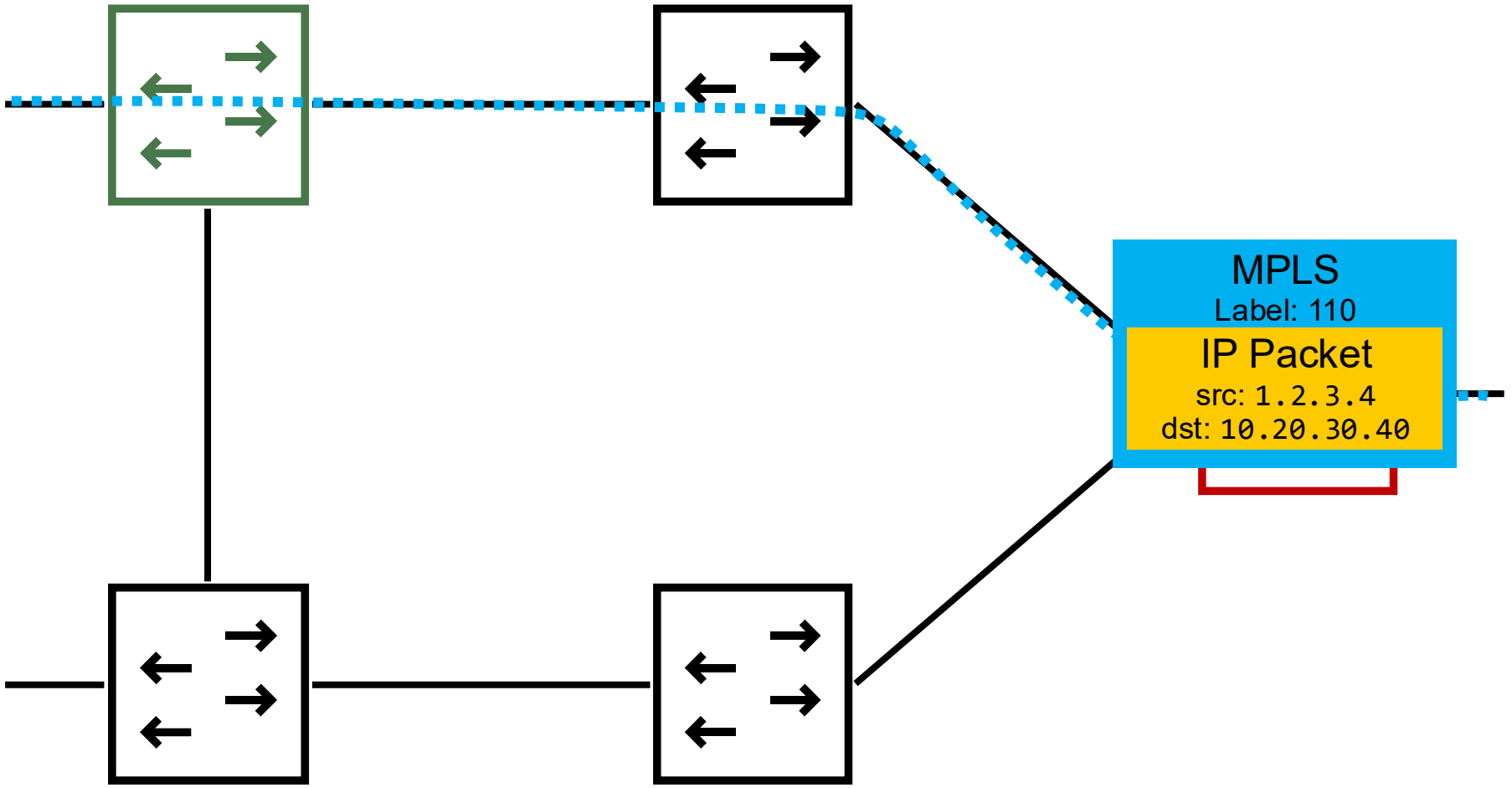
MPLS Routing



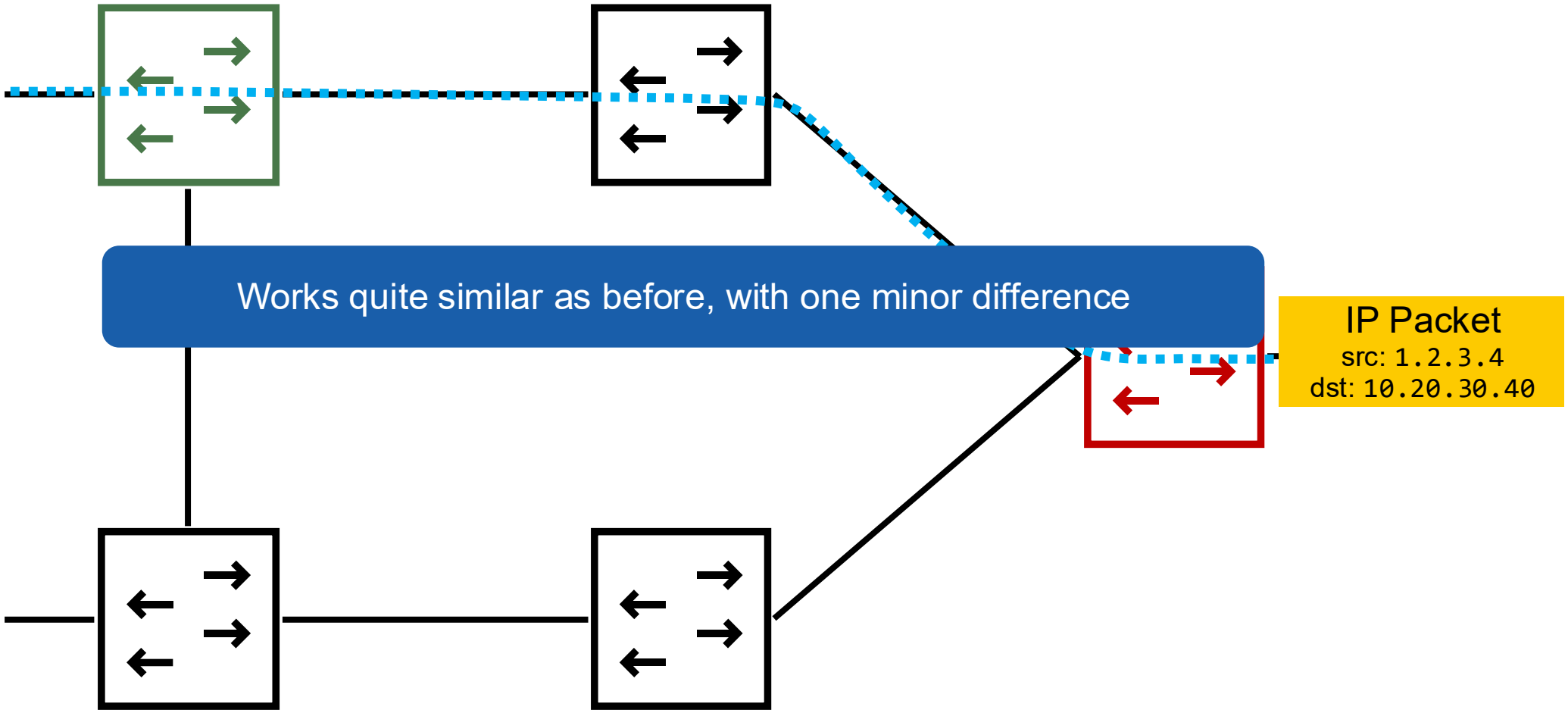
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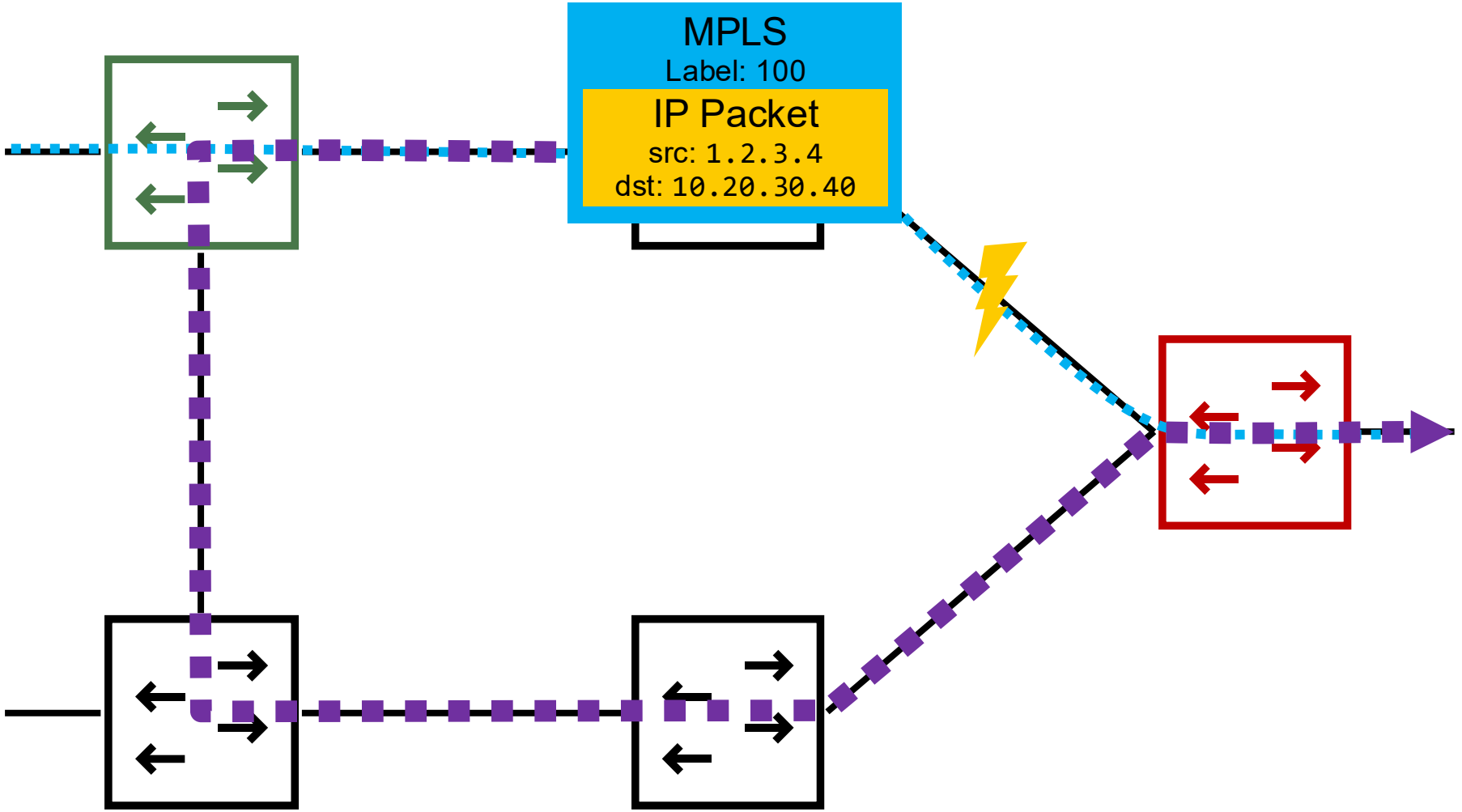
MPLS Routing



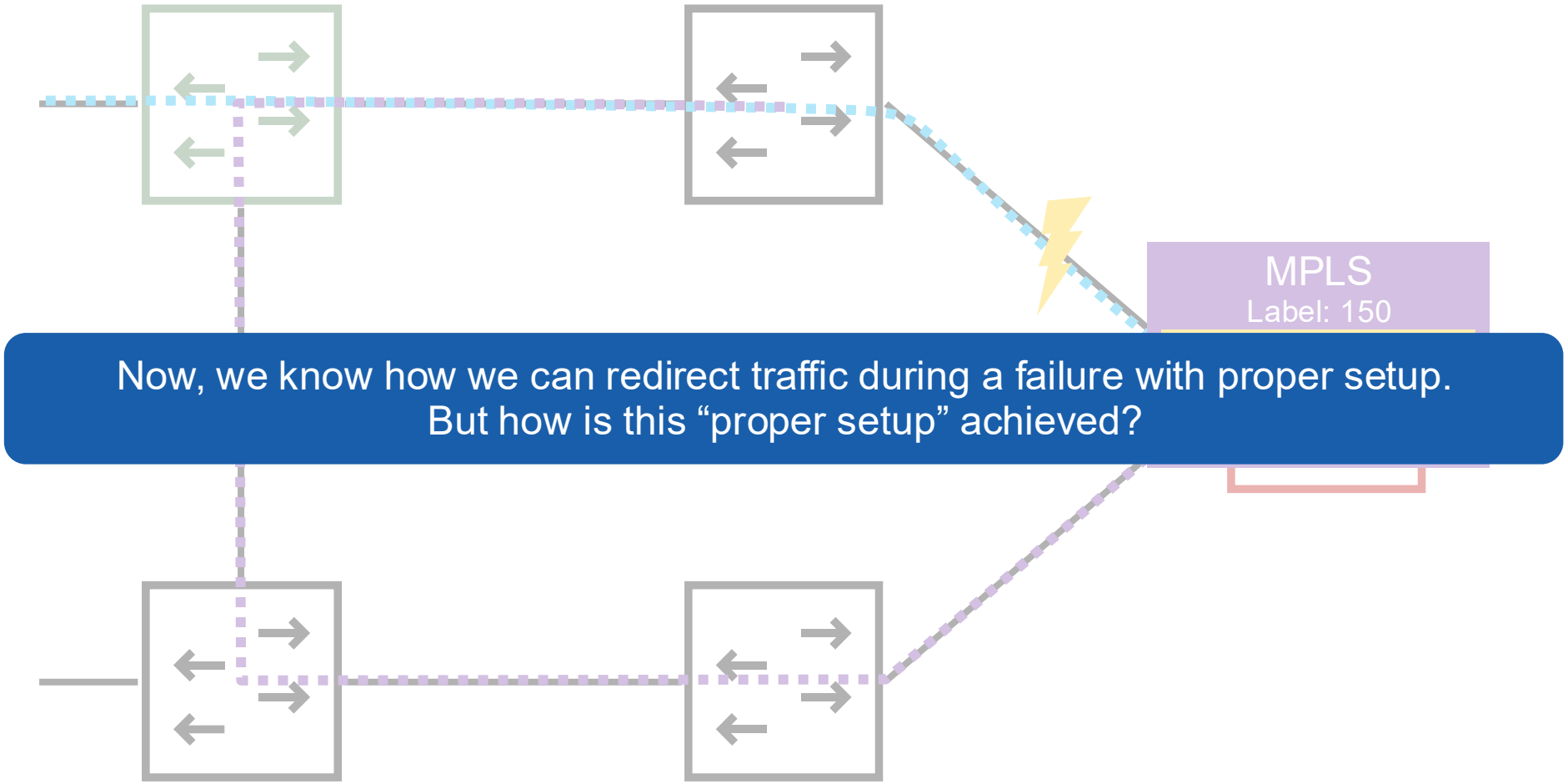
MPLS Routing



MPLS Fast Reroute – Basic Scenario



MPLS Fast Reroute – Basic Scenario

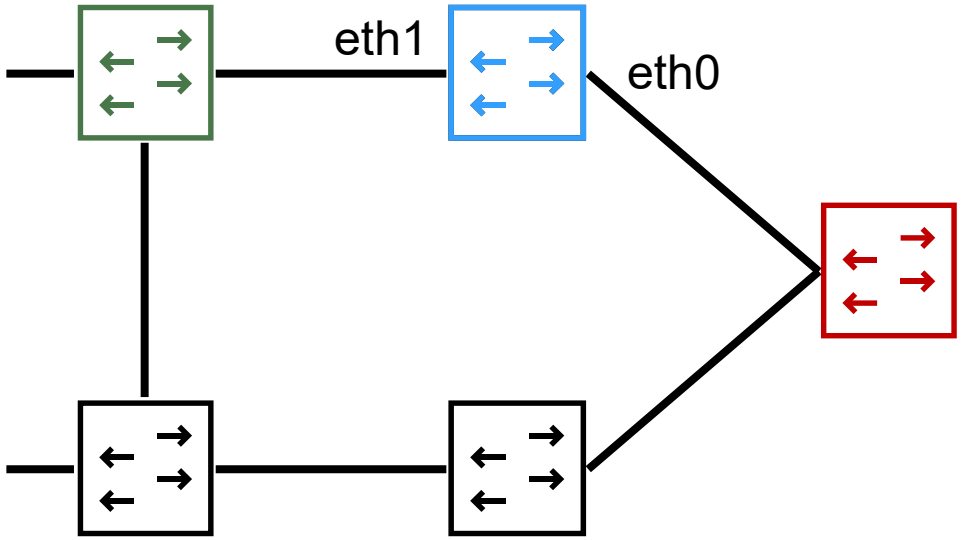


Label Forwarding Information Base (LFIB)

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The LFIB contains routing information (similar like the IP routing table)

- Based on labels (MP***L***S) instead of IPs
- Decides forwarding based on incoming label
- Contains information about
  - The action to perform (push, pop, swap)
  - The outgoing label
  - The outgoing interface
  - Next-hop address



Example LFIB of blue router

Incoming Label	Operation	Outgoing Label	Outgoing Interface	Next-Hop Address
100	Swap	110 (primary)	eth0	(IP of red router)
100	Swap	120 (secondary)	eth1	(IP of green router)
…	…	…	…	…

Backup Paths using RSVP-TE in MPLS

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If a Label Switched Path (LSP) should be protected, a new path is calculated that intersects with the protected path

- Backup paths need to be pre-established
- More backup paths can lead to scalability issues

One-to-One Backup

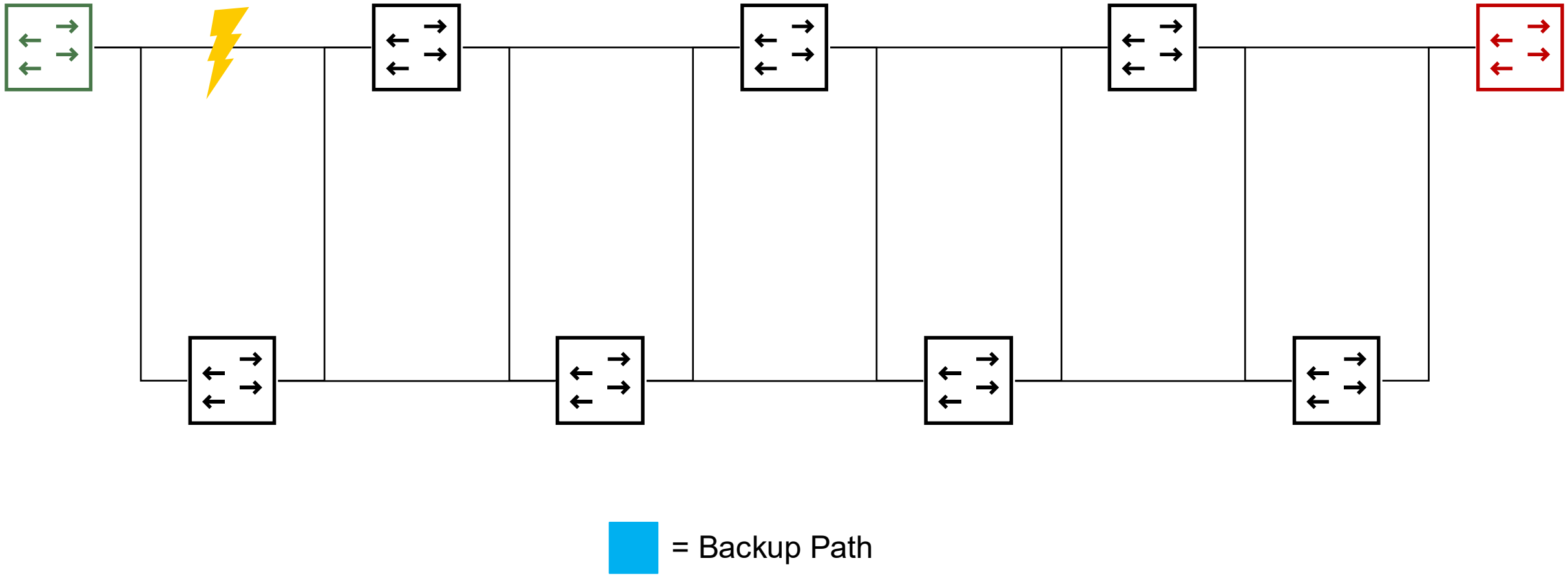
- One backup path per protected entity and LSP
- Very easy to setup, but can lead to scalability issues

Facility Backup

- Instead of creating one LSP for each failure, a MPLS tunnel is used to handle multiple failures at once
- More efficient compared to One-to-One Backup
- An MPLS tunnel is used for the detour

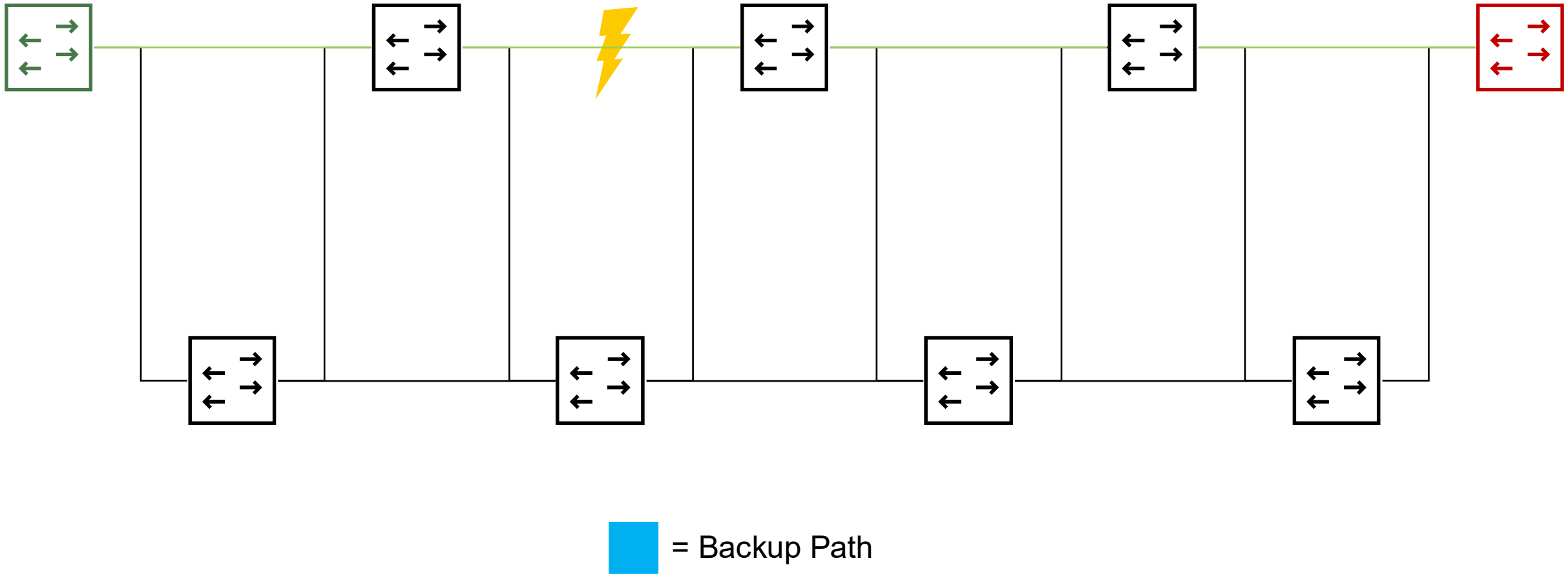
One-to-One Backup

Example backup paths



One-to-One Backup

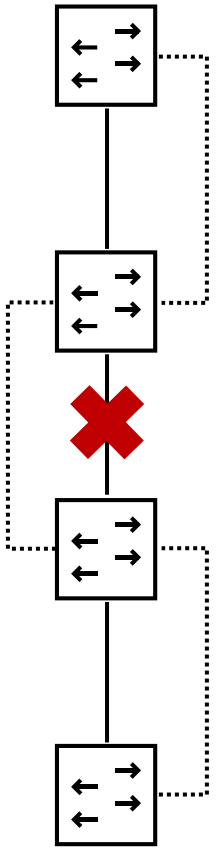
Example backup paths



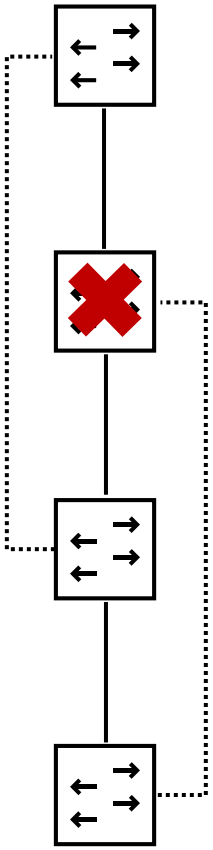
Failure protection modes

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Link protection

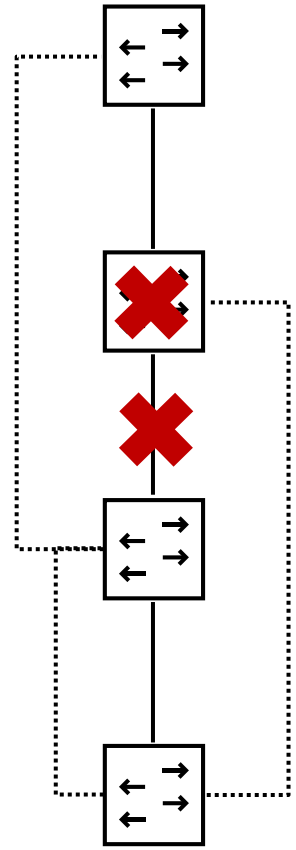


Node protection

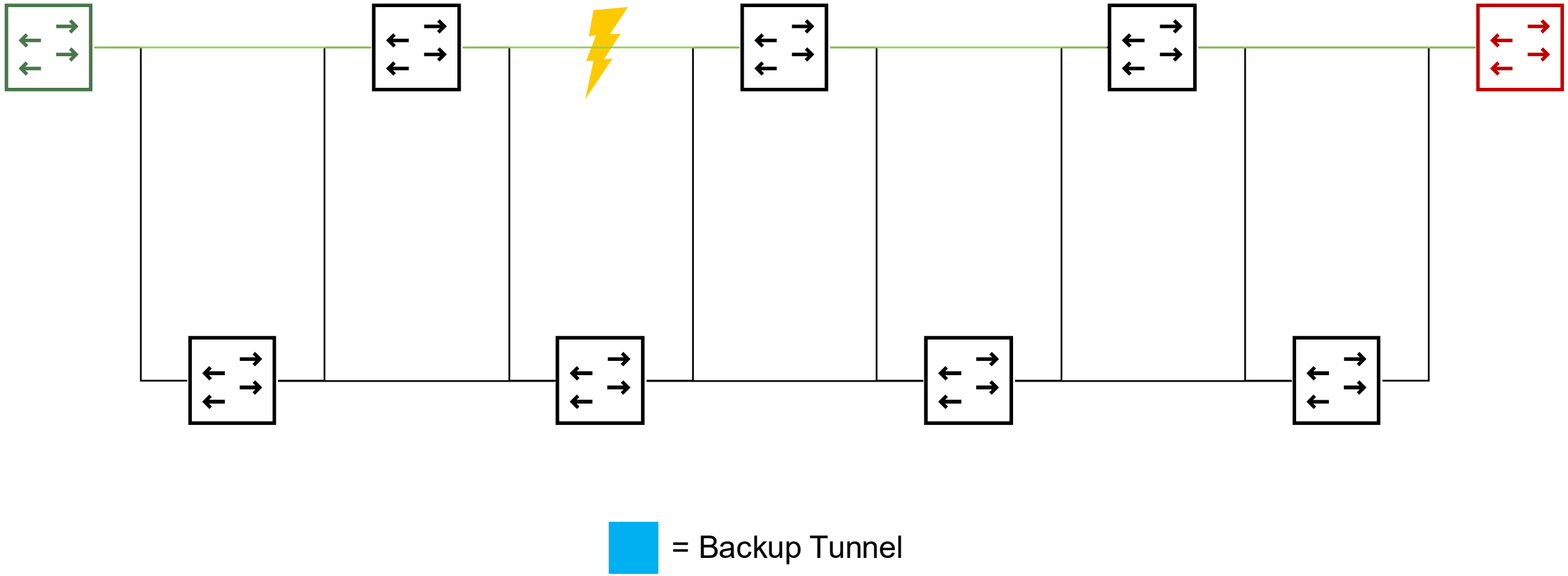


Element protection

(Link + Node)

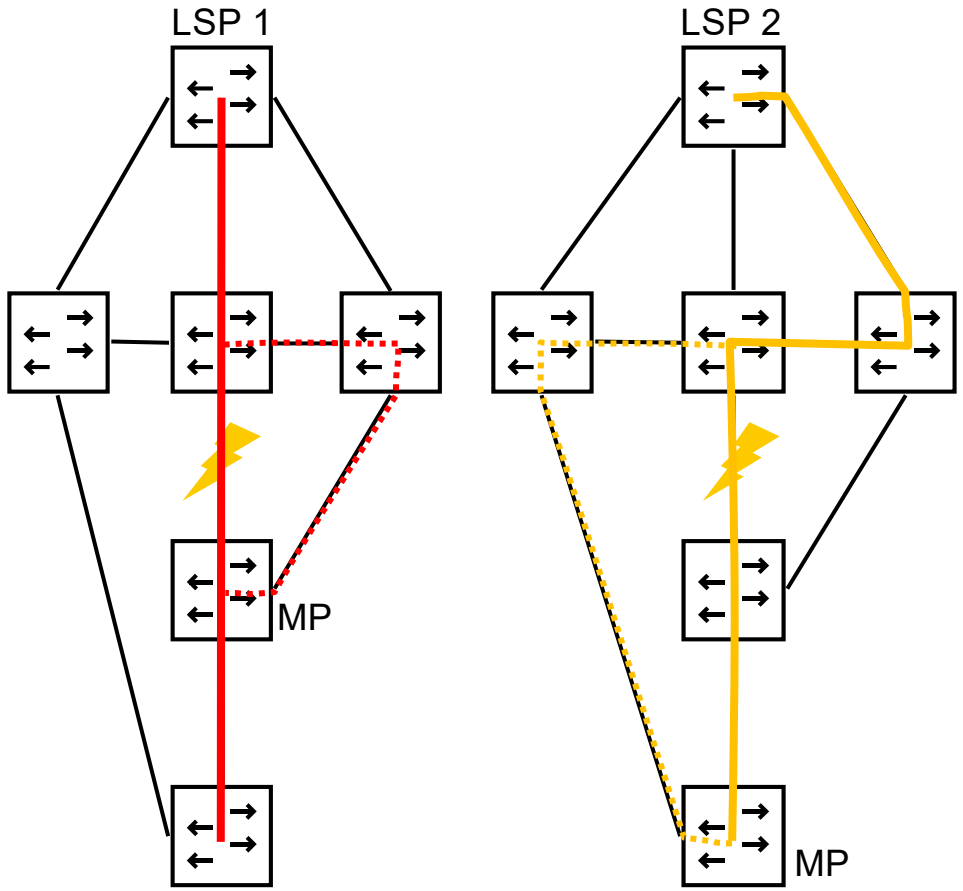


Facility Backup
Example backup paths



One-to-One Backup vs. Facility Backup

One-to-One Backup



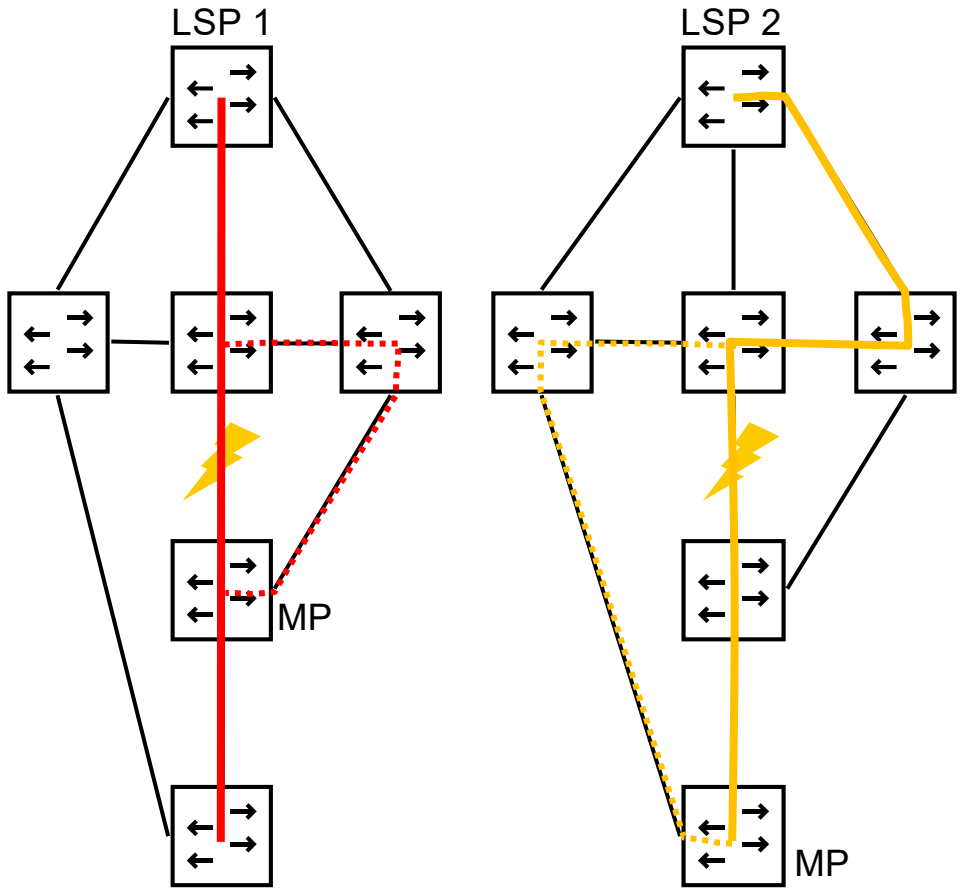
Facility Backup

MP = Merge Point

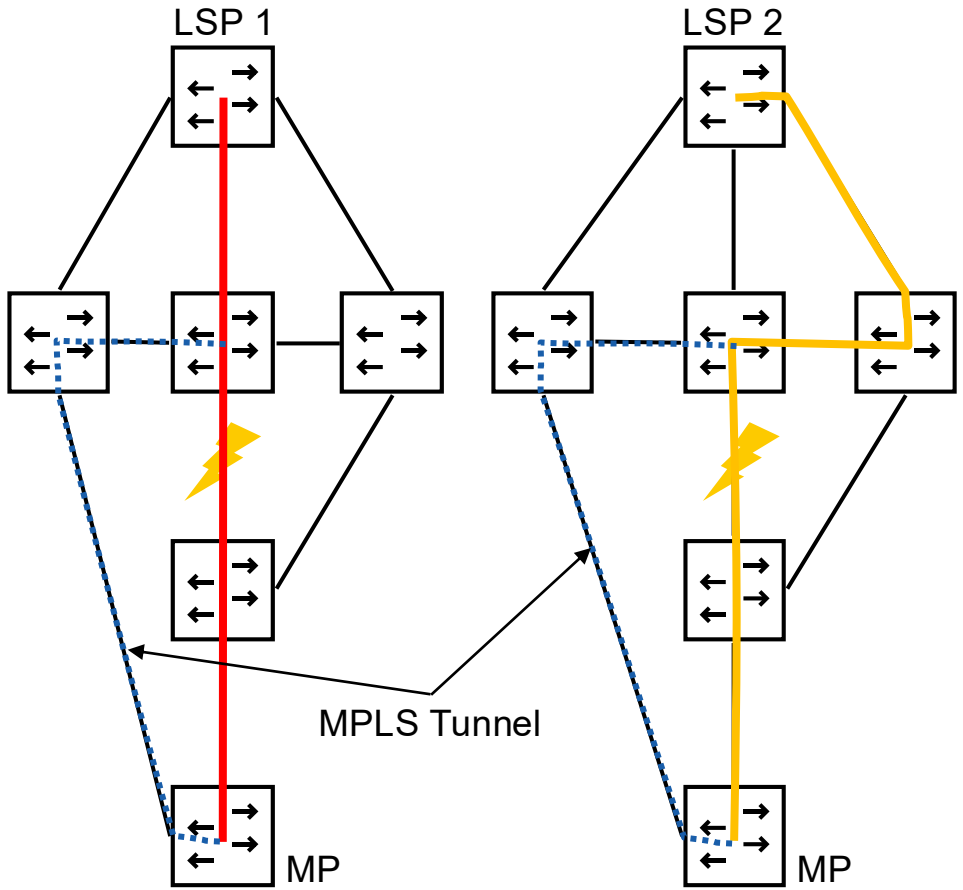
One-to-One Backup vs. Facility Backup

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One-to-One Backup



Facility Backup

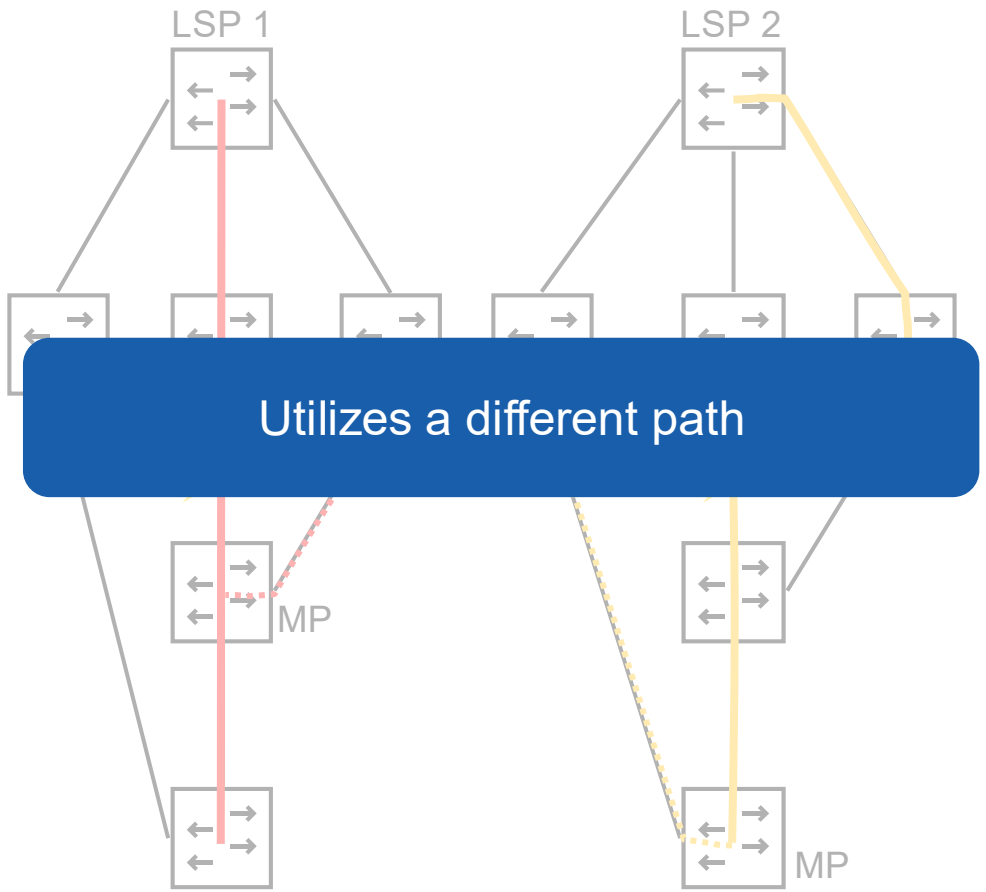


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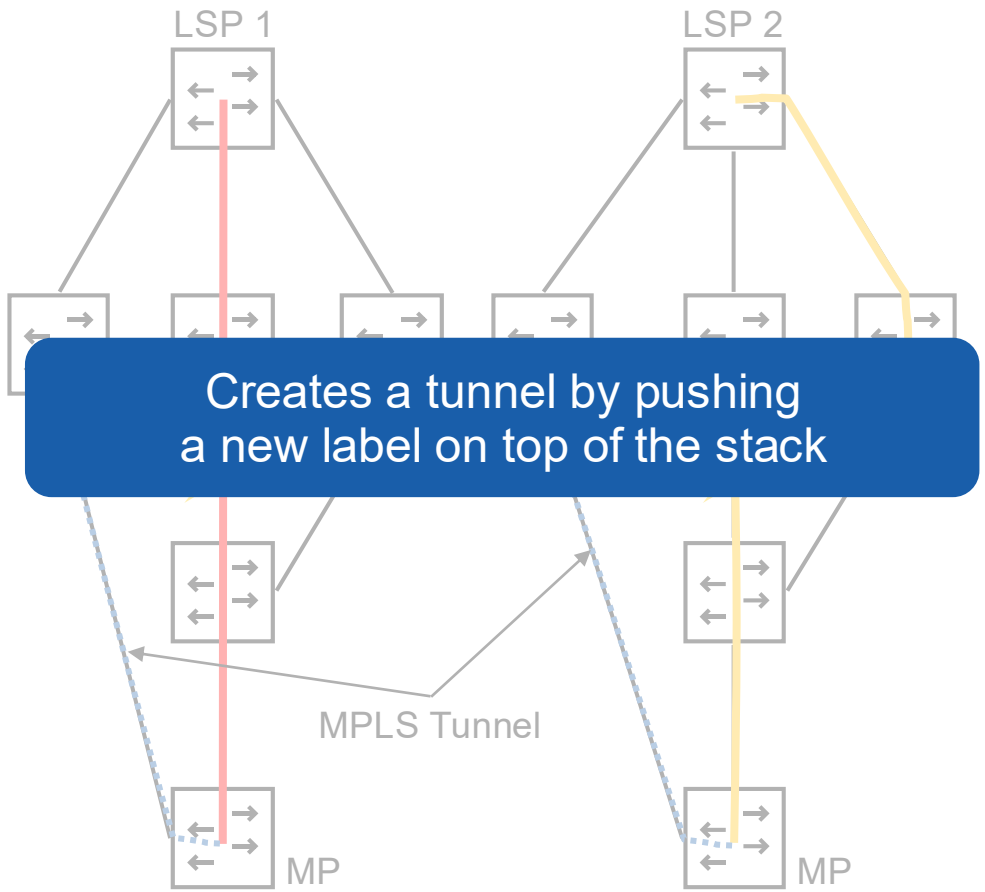
One-to-One Backup vs. Facility Backup

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One-to-One Backup



Facility Backup



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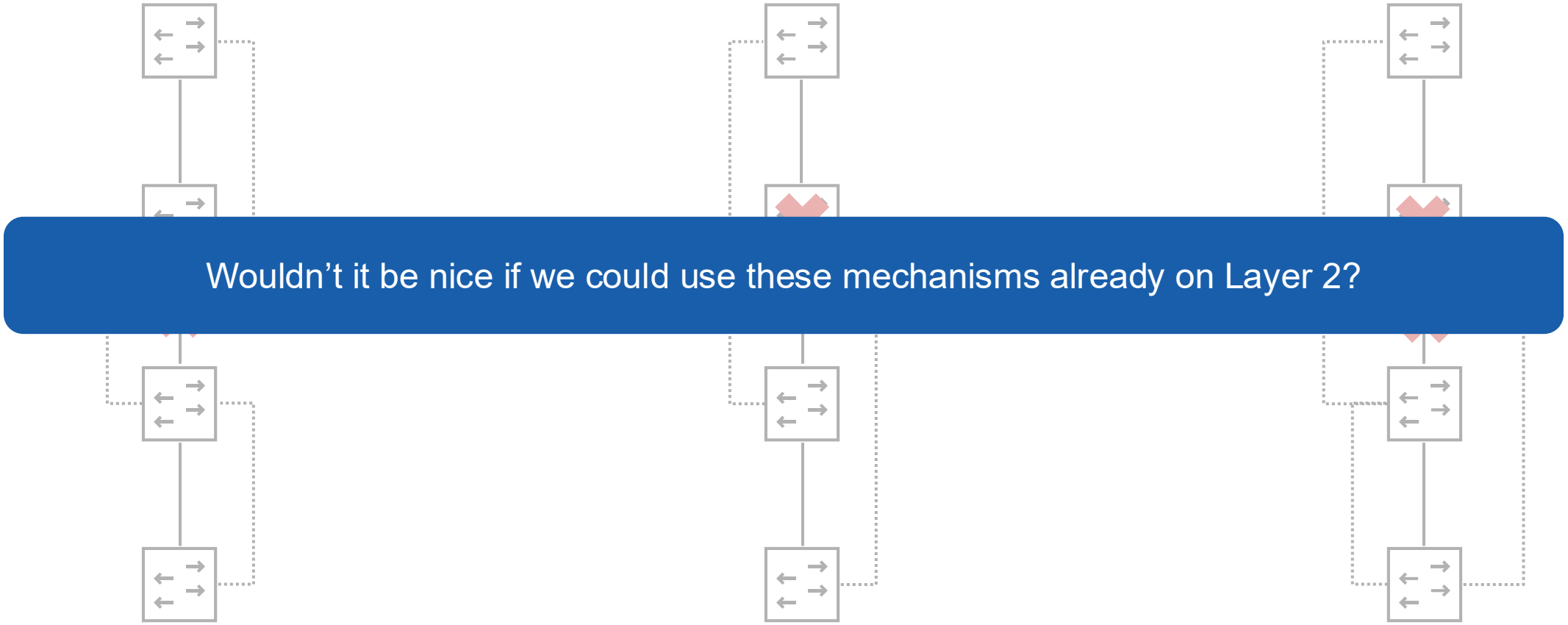
Failure protection modes

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Link protection

Node protection

Element protection
(Link + Node)



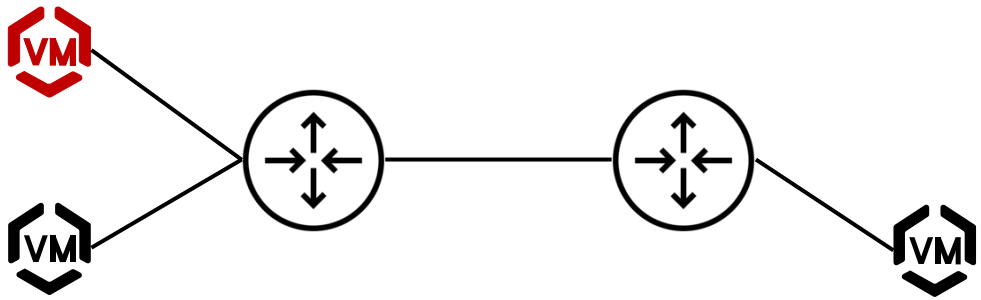
Wouldn’t it be nice if we could use these mechanisms already on Layer 2?

Ethernet Virtual Private Network (EVPN)

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Ethernet is a global standard

- Changes are hard or impossible to make at scale
- Building on top of networks is a common approach, and you can put into the payload whatever you want



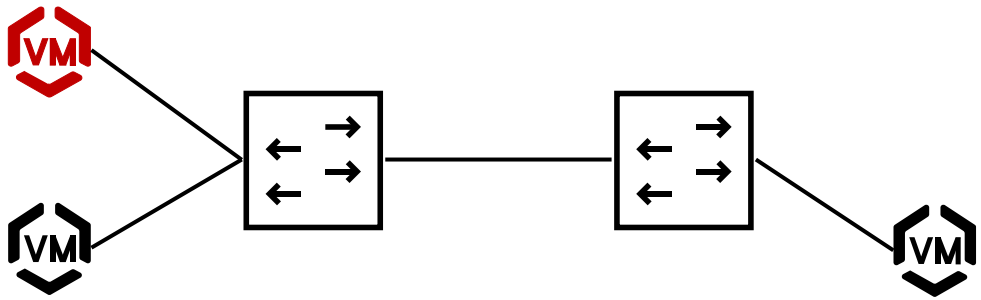


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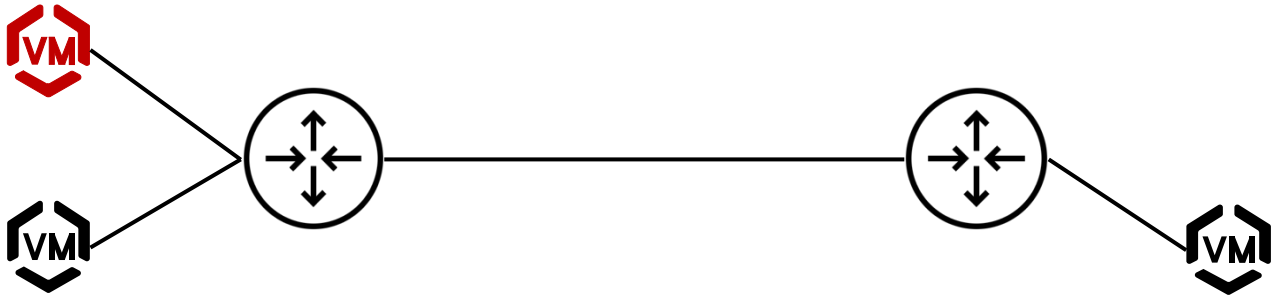
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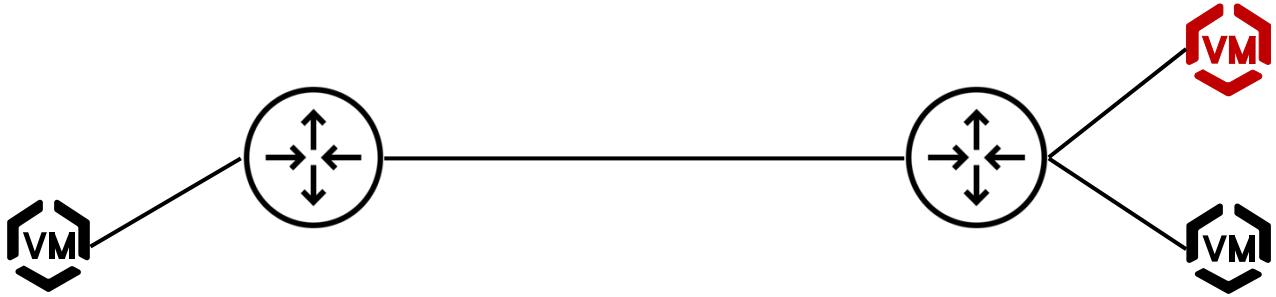
Instead of using Ethernet directly, we build Ethernet networks on top of VXLAN or MPLS



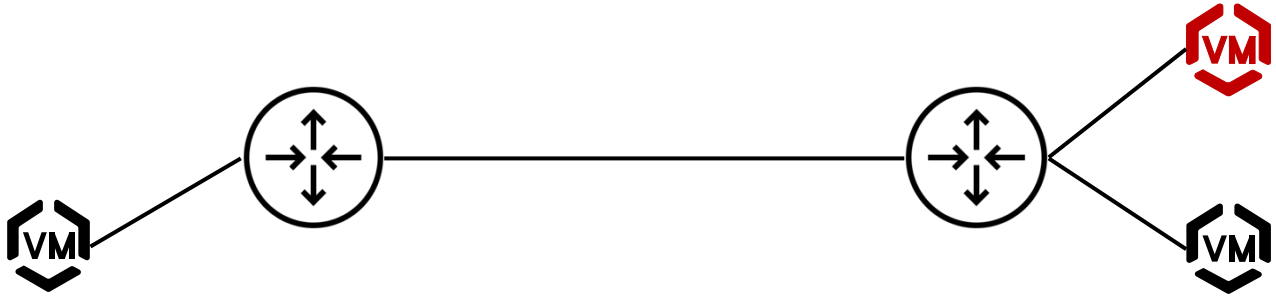
High Churn and Number of Devices in Today's Networks



High Churn and Number of Devices in Today's Networks

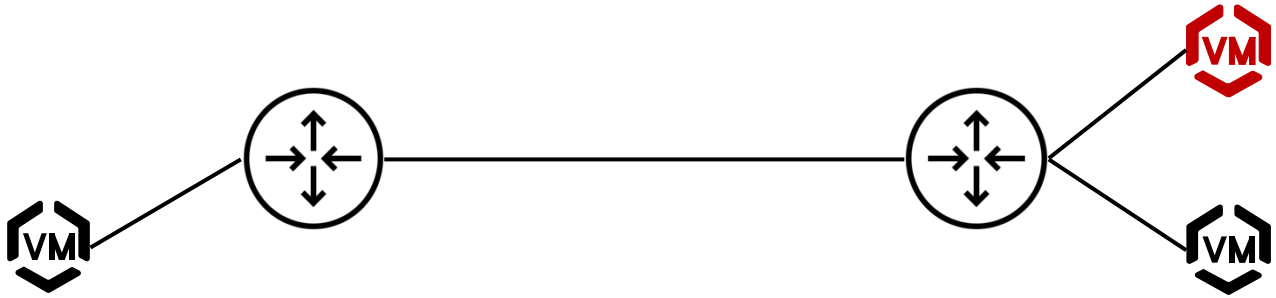


High Churn and Number of Devices in Today's Networks



Quick Disclamer: VMs are not physically moved, but transferred from one server to another

High Churn and Number of Devices in Today's Networks



In earlier times (when Shortest Path Bridging was developed), only few changes happened in the network

- SPB is still rather inefficient as MAC addresses need to be advertised using flooding in the network
- If MAC addresses change frequently and/or the number of devices is high, this can put a significant strain on the network

We need to get rid of all the broadcast traffic in the network!

Idea: Improve the scalability of L2-networks through tunneling

EVPN
Basic Functionality

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EVPN can be split into a control plane part and a data plane part

- Data plane is for forwarding of packets
- Control plane is for the control of the network (like Spanning Tree Protocol (STP), or Shortest Path Bridging (SPB))

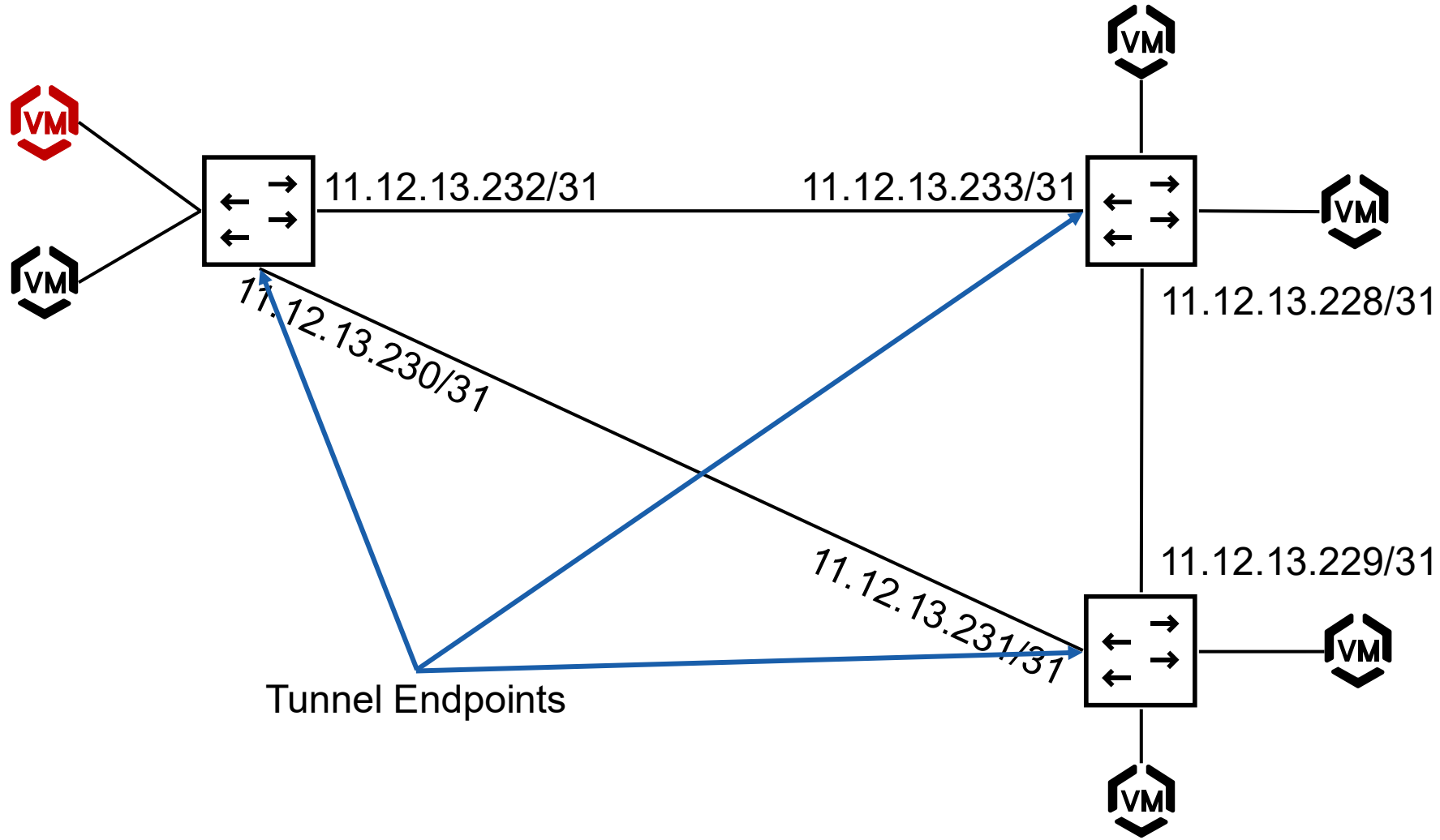
EVPN control plane is based on Multiprotocol Border Gateway Protocol (MP-BGP)

- Active exchange of MAC-addresses instead of flooding
- Planning of routes in the network

EVPN encapsulates Ethernet packets and can use multiple protocols on the data plane

- Multi-Protocol Label Switching (MPLS)
- Virtual eXtensible LAN (VXLAN)

EVPN basic setup
IP-based



EVPN Control Plane

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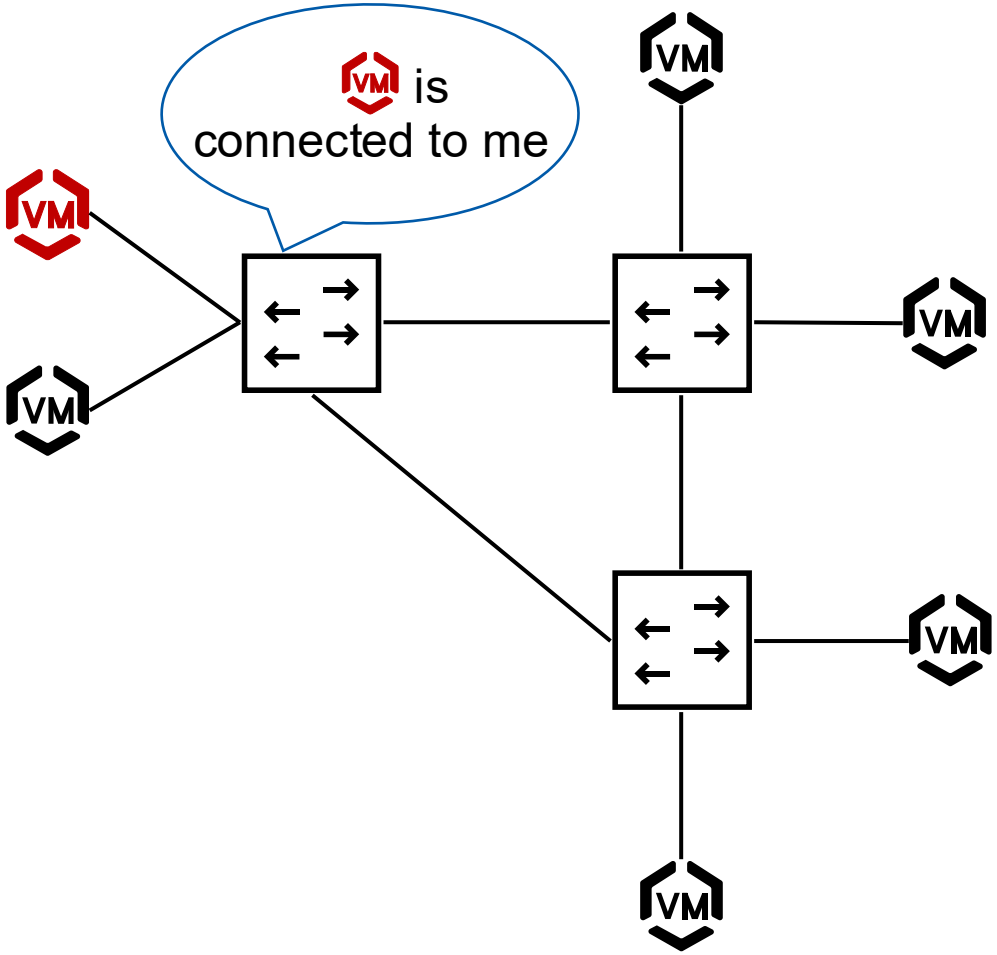
Multiprotocol Border Gateway Protocol (MP-BGP)

- MAC-connectivity is exchanged via control messages (announced as type 2 routes)
- Routers learn MAC addresses similarly to switches, but do not forward the ARP any further if the requested MAC is already known (if MAC suppression is enabled)

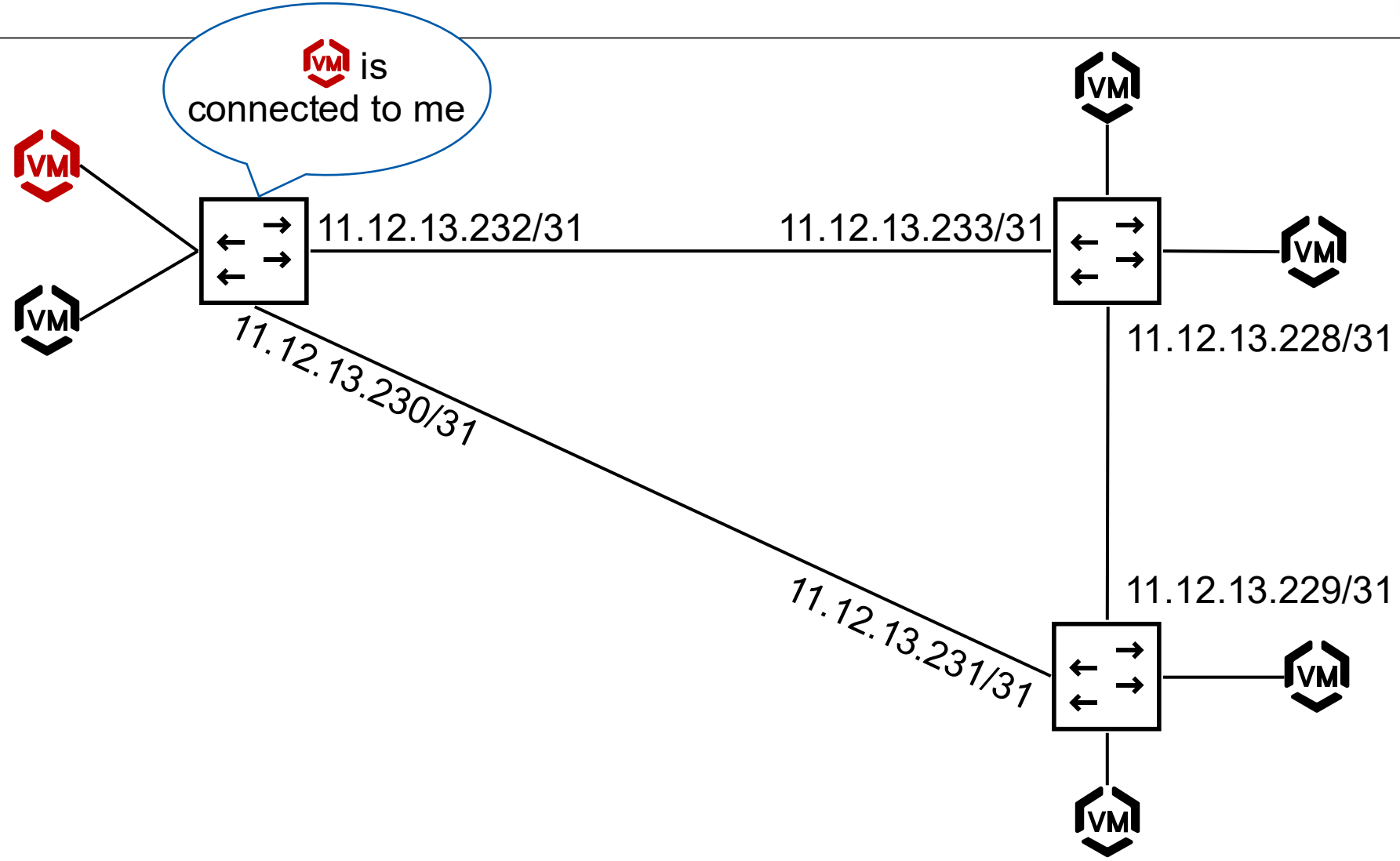
Once a router detects a new device connecting

- It publishes route a route to all other routers in the network

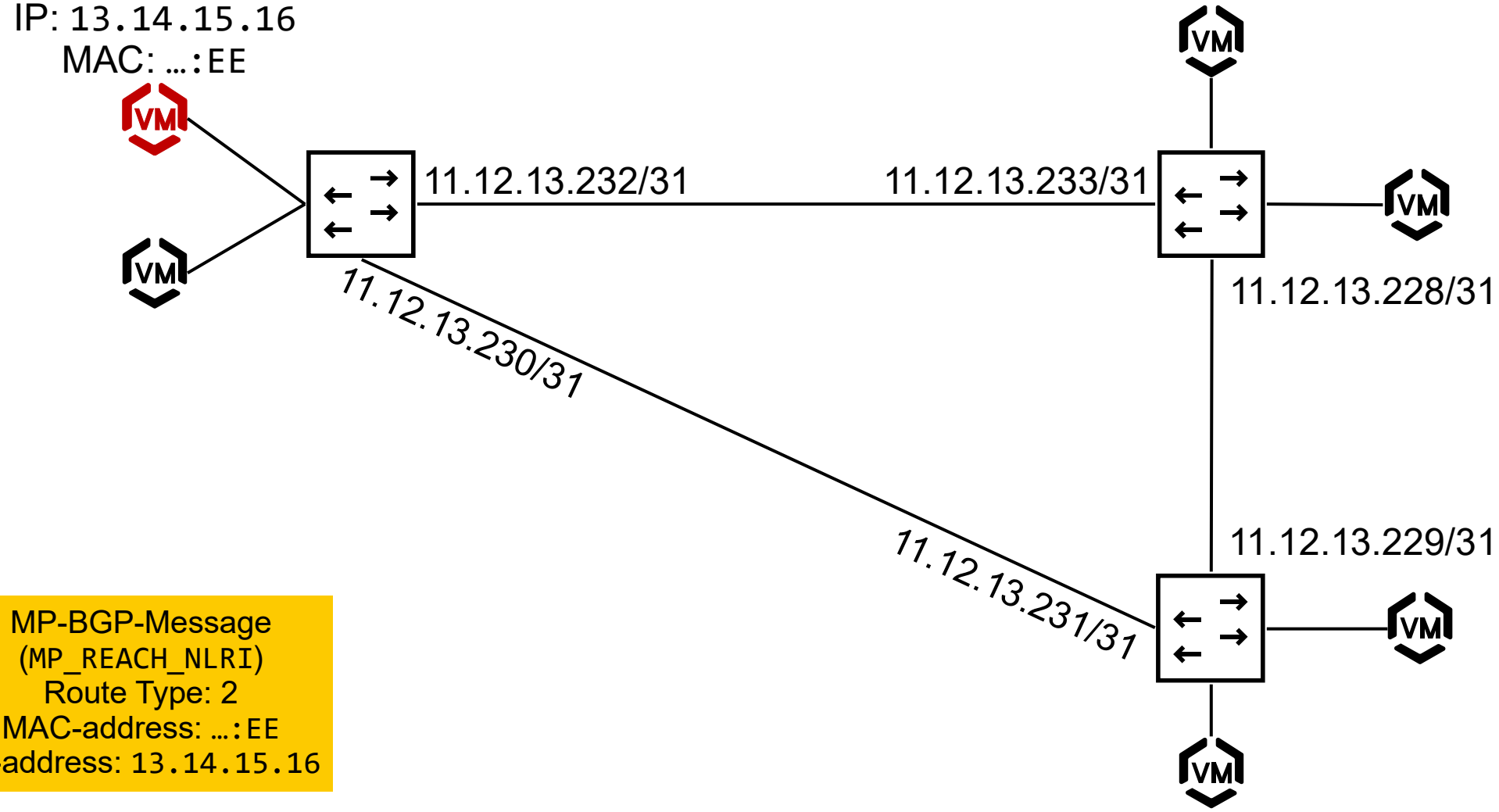
MP-BGP route announcements are only sent on change!



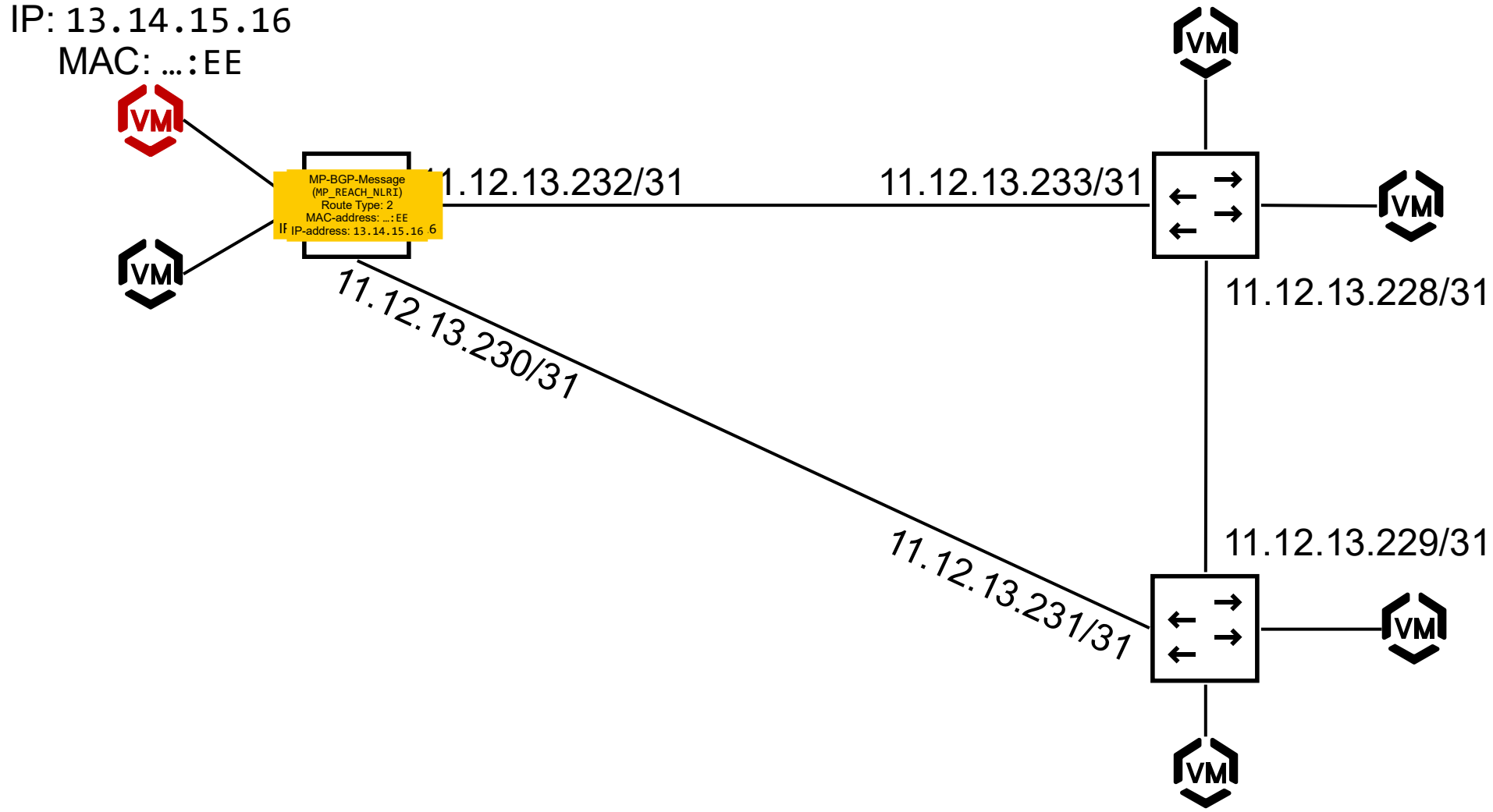
EVPN Control Plane



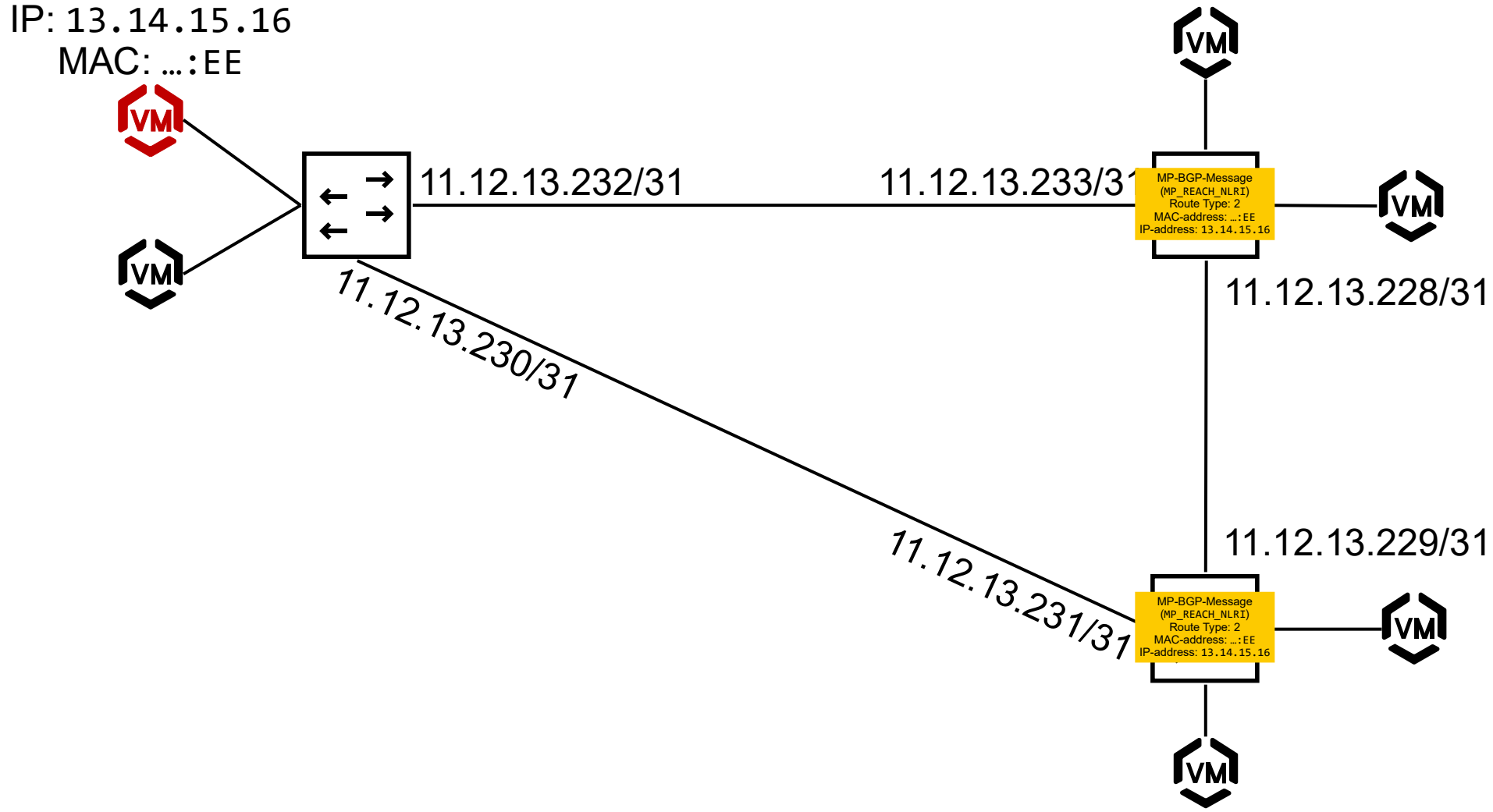
EVPN Control Plane
Network Layer Reachability Information (NLRI)



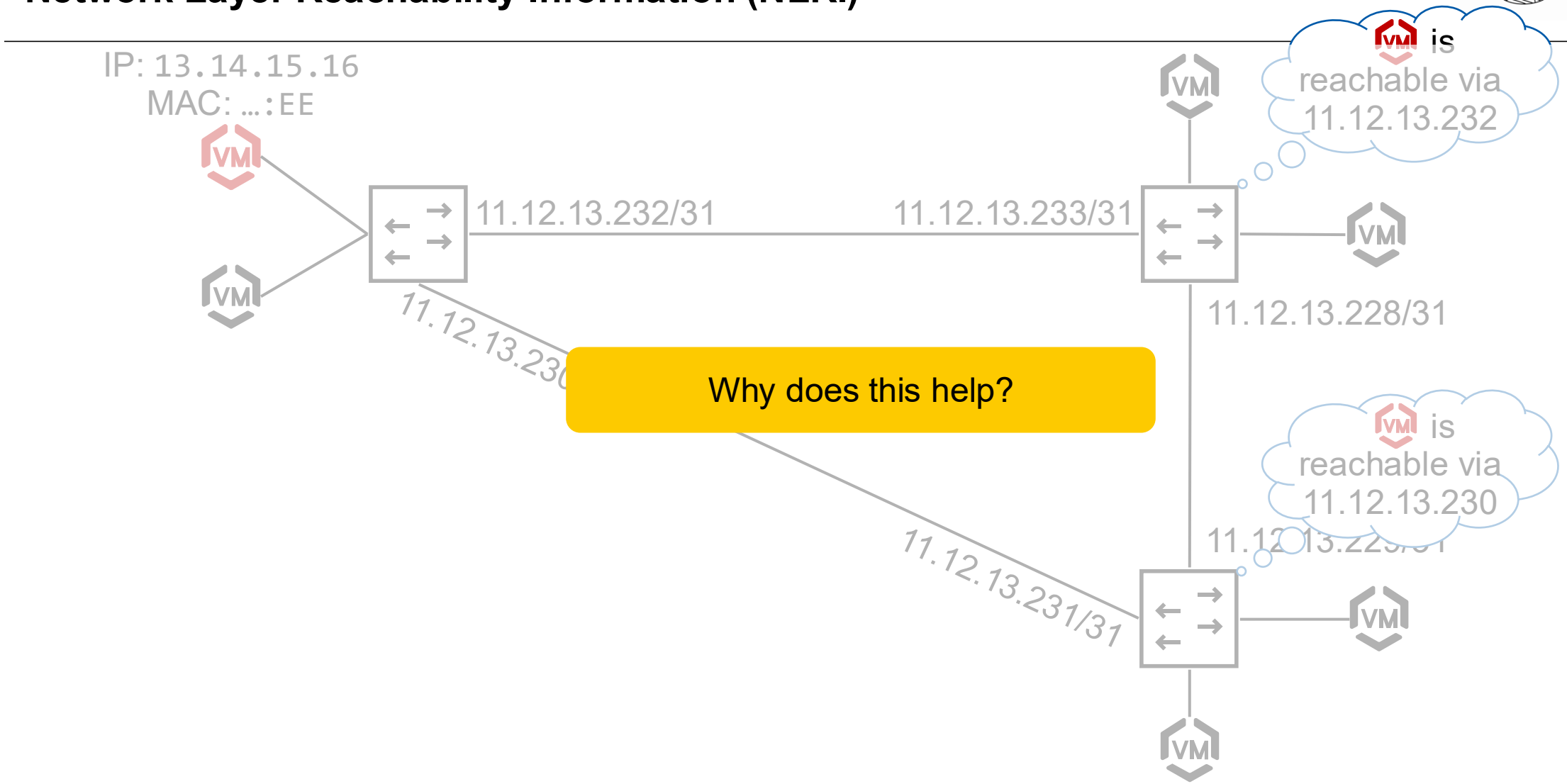
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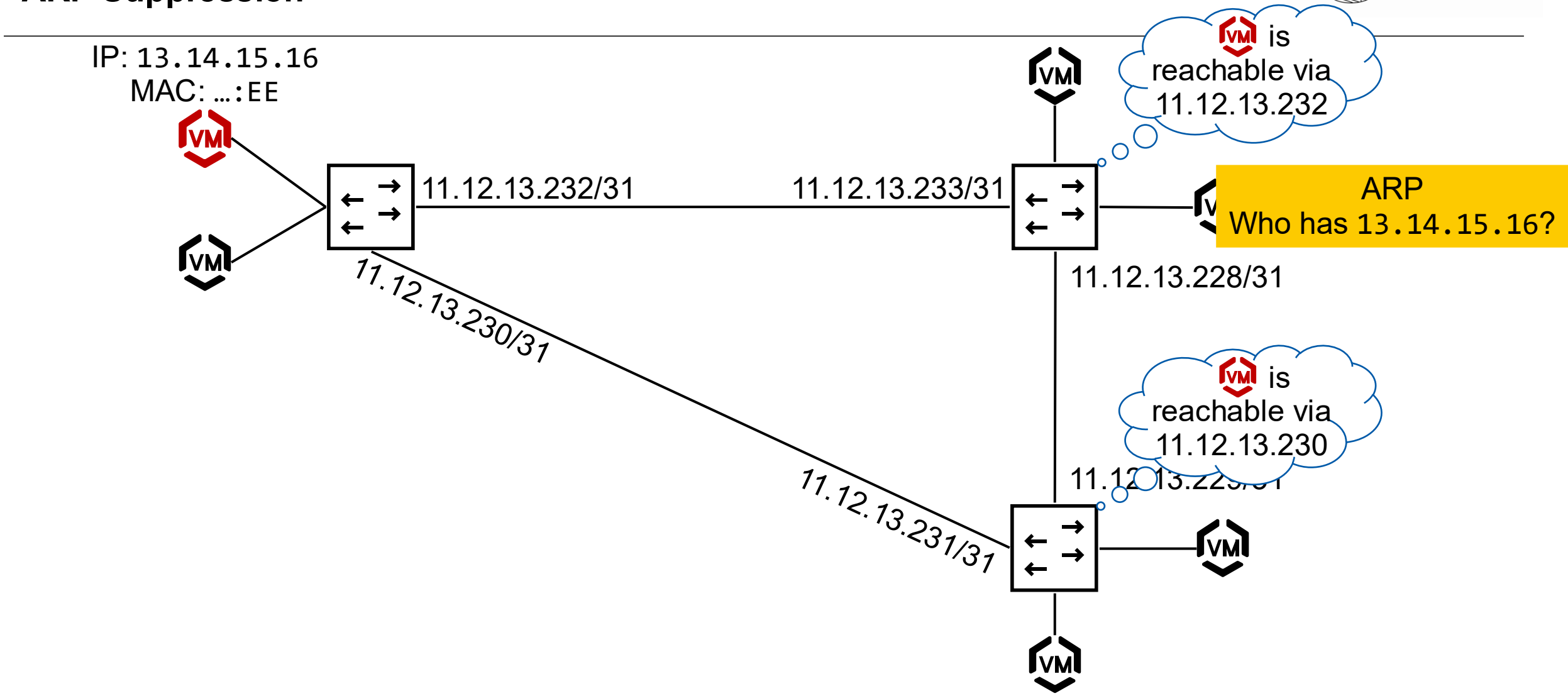
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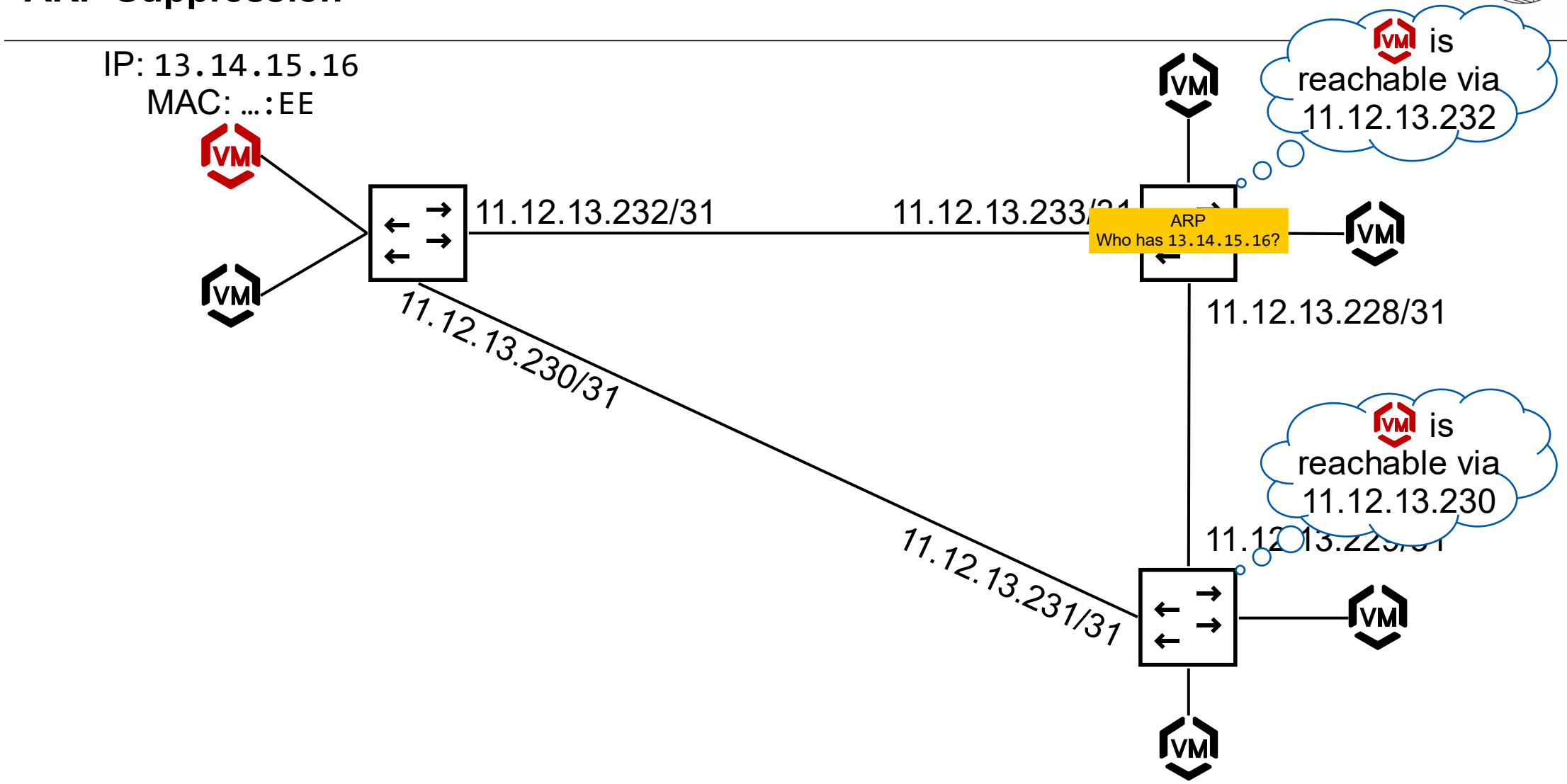
EVPN Control Plane
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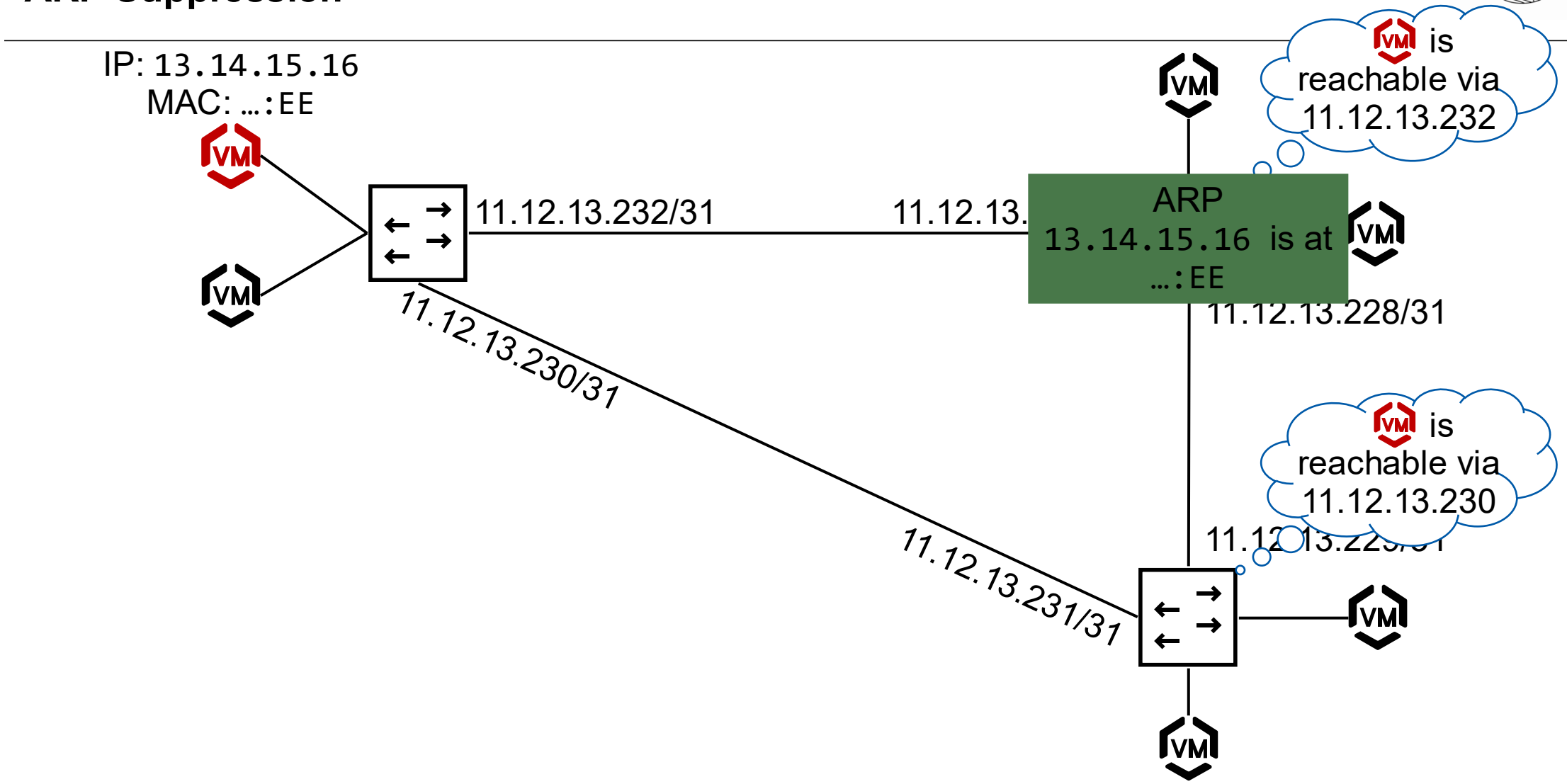
EVPN Control Plane
ARP Suppression



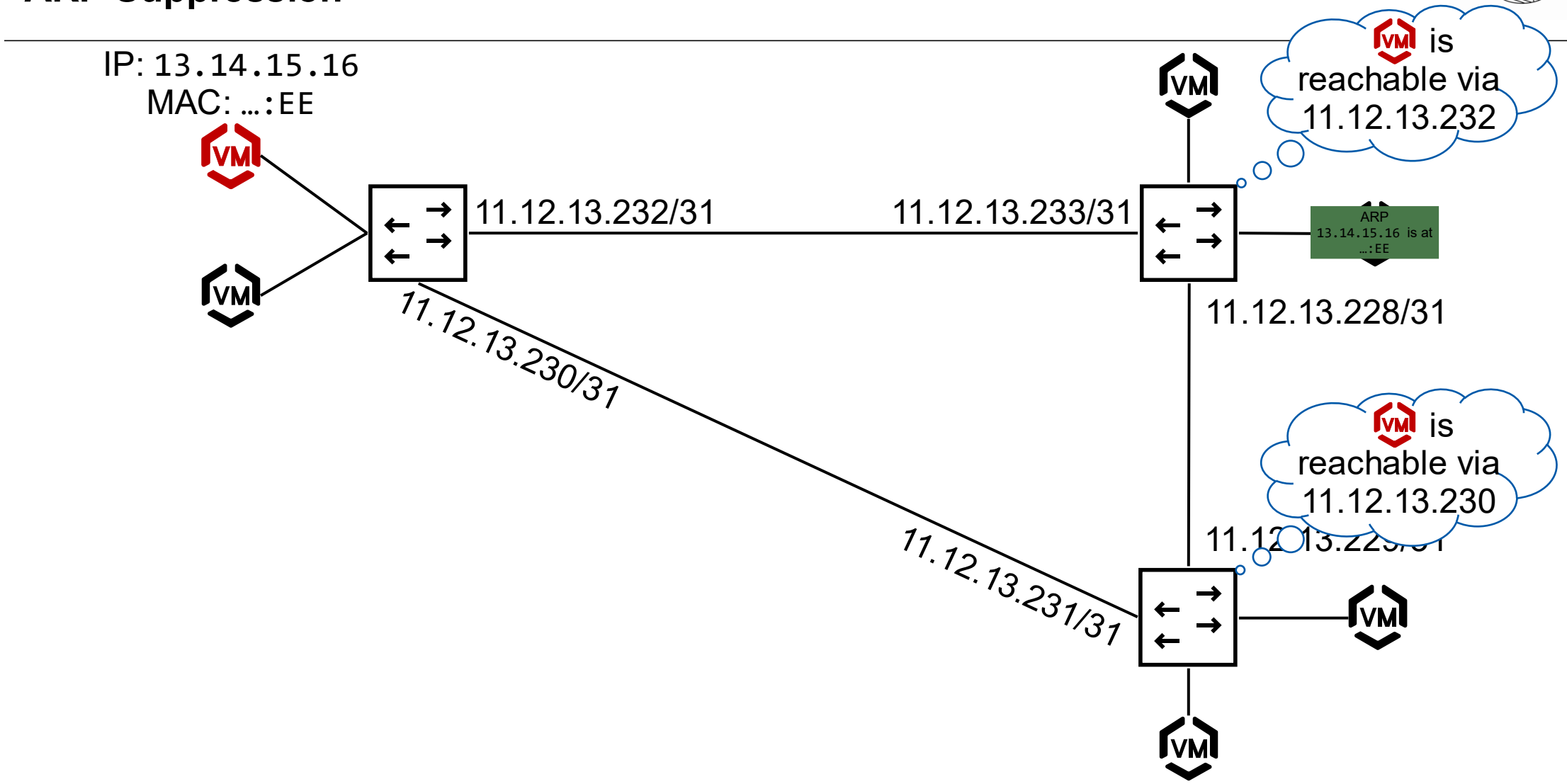
EVPN Control Plane
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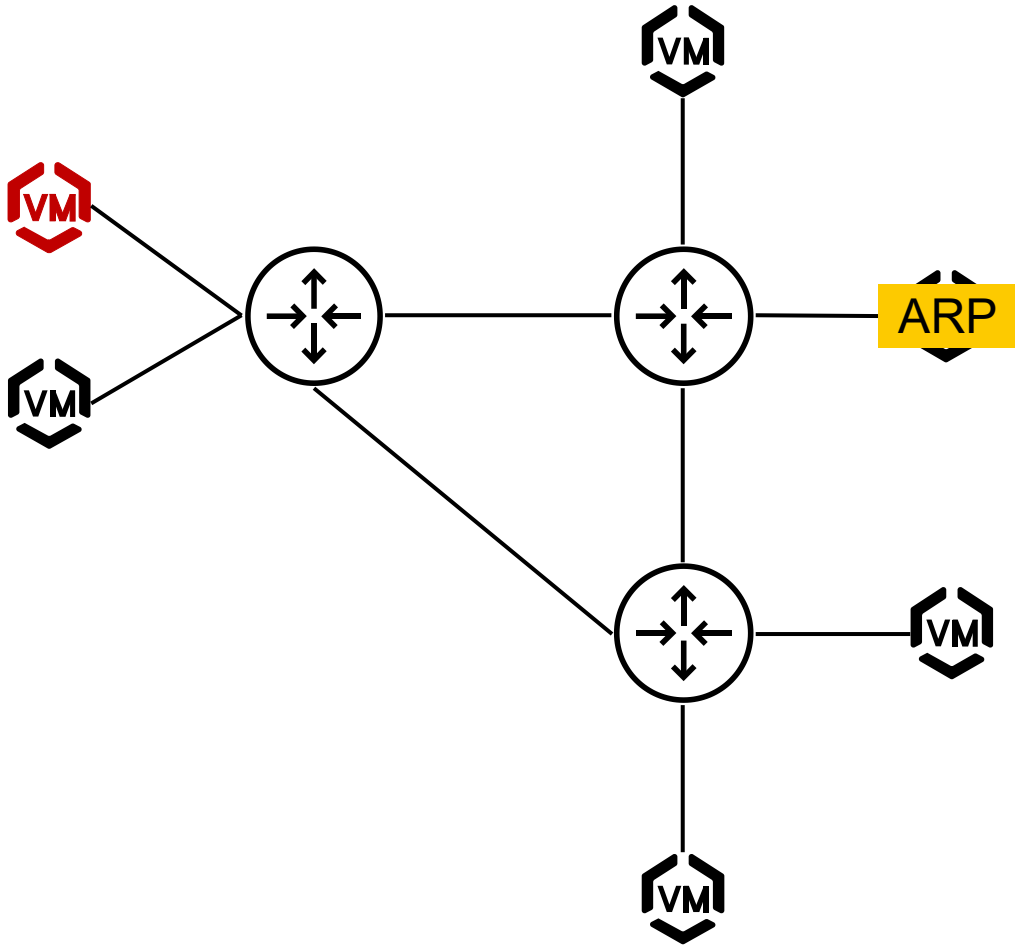
EVPN Control Plane
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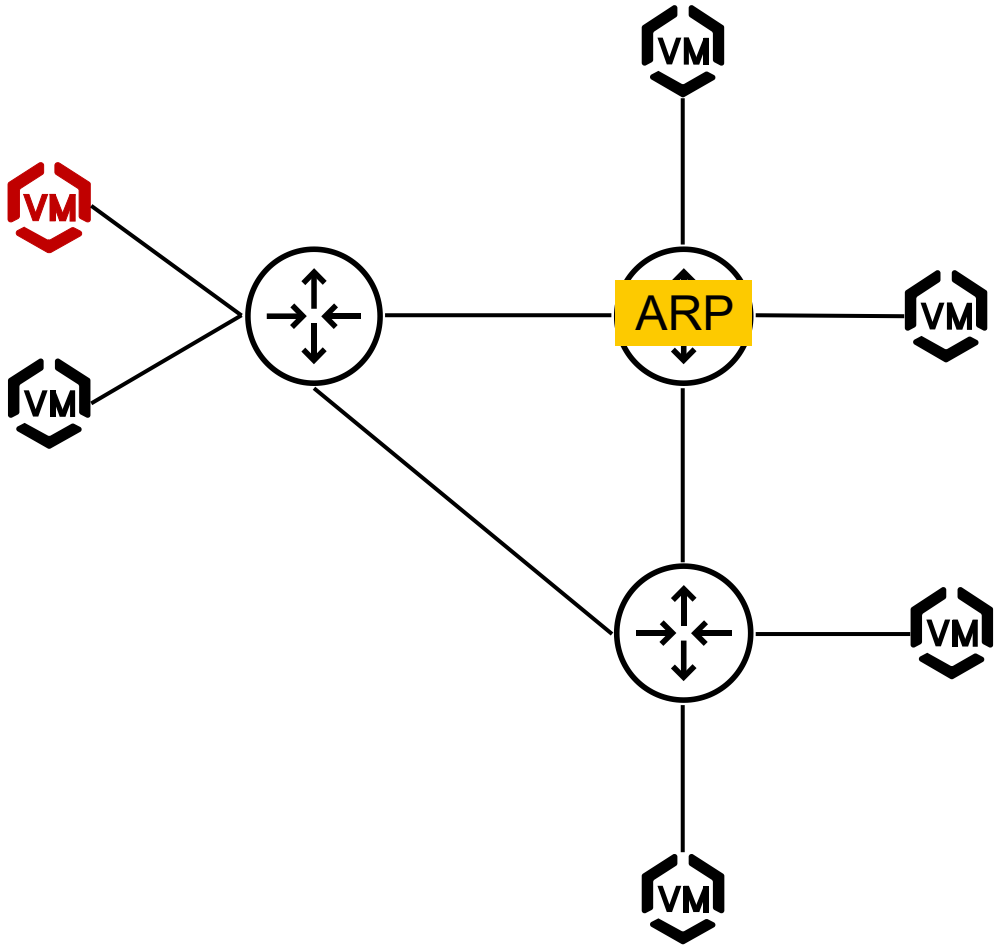
EVPN Control Plane
ARP Suppression



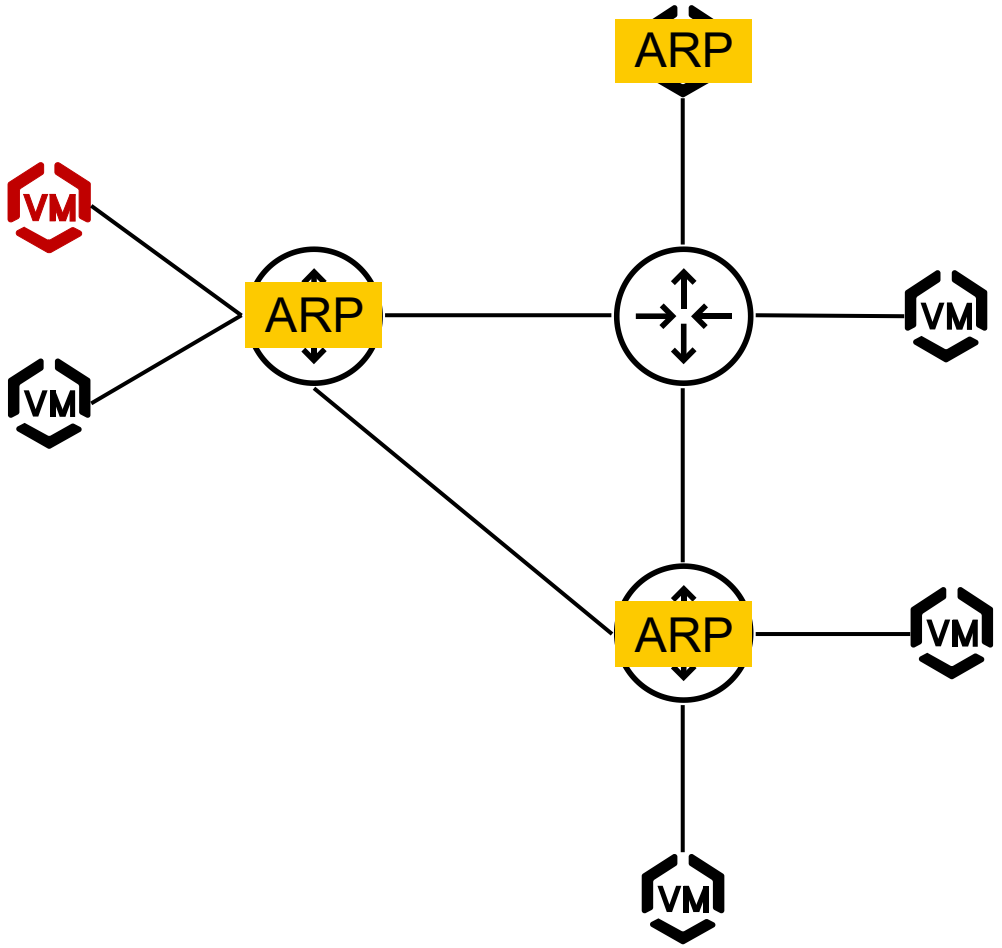
ARPs in Ethernet



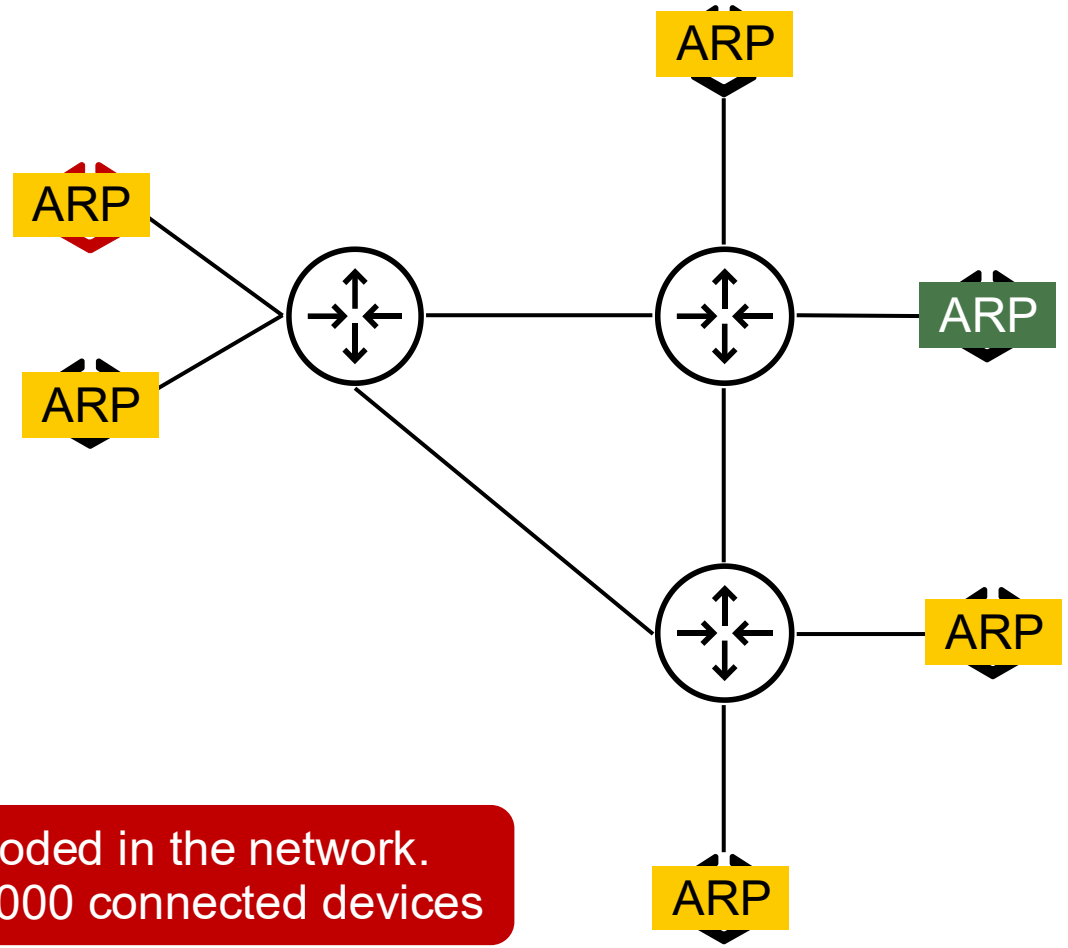
ARPs in Ethernet



ARPs in Ethernet

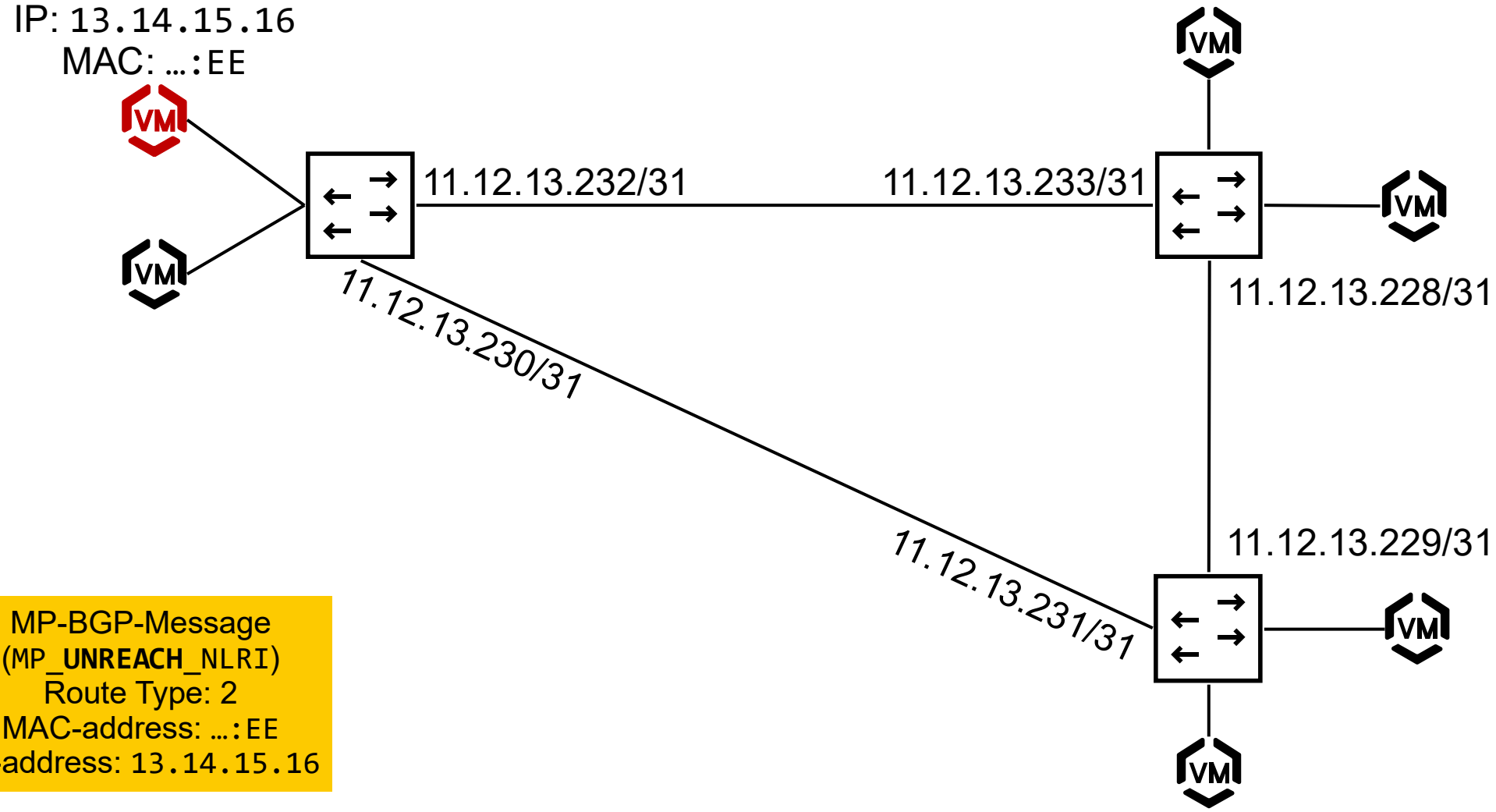


ARPs in Ethernet

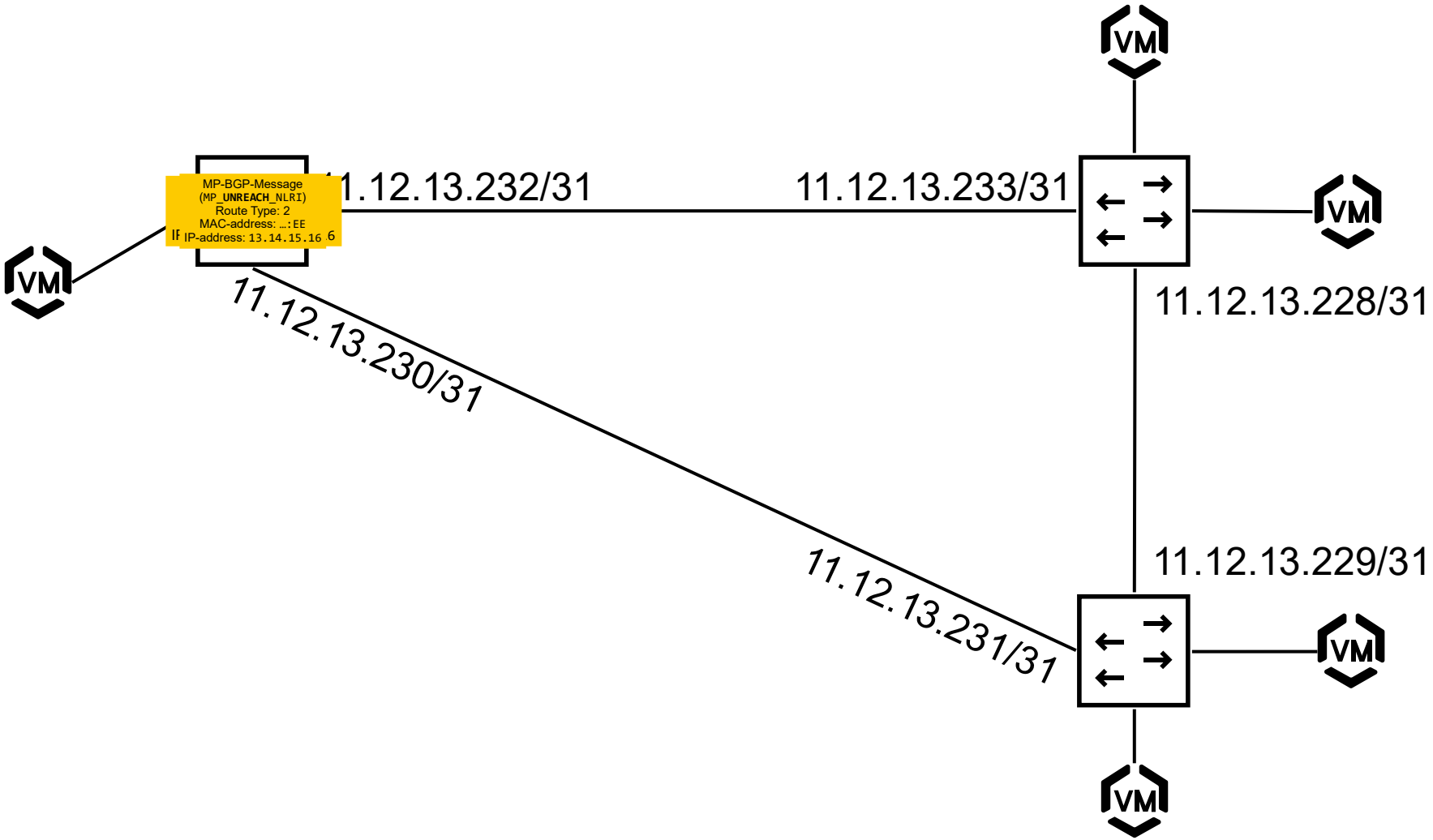


Every ARP Request gets flooded in the network.
Imagine this process with 200.000 connected devices

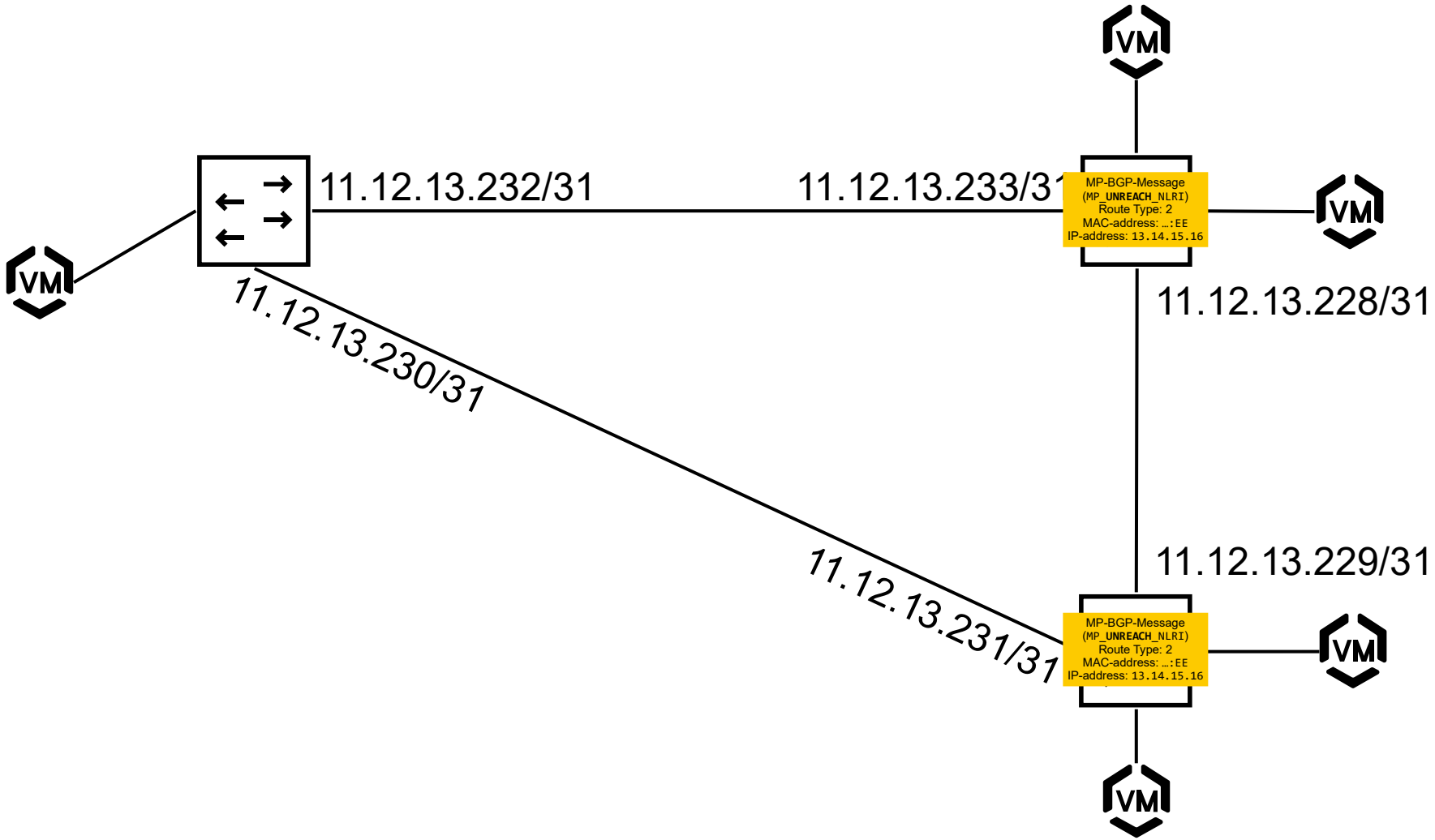
EVPN Control Plane



EVPN Control Plane
Network Layer Reachability Information (NLRI)



EVPN Control Plane
Network Layer Reachability Information (NLRI)



EVPN Data Plane

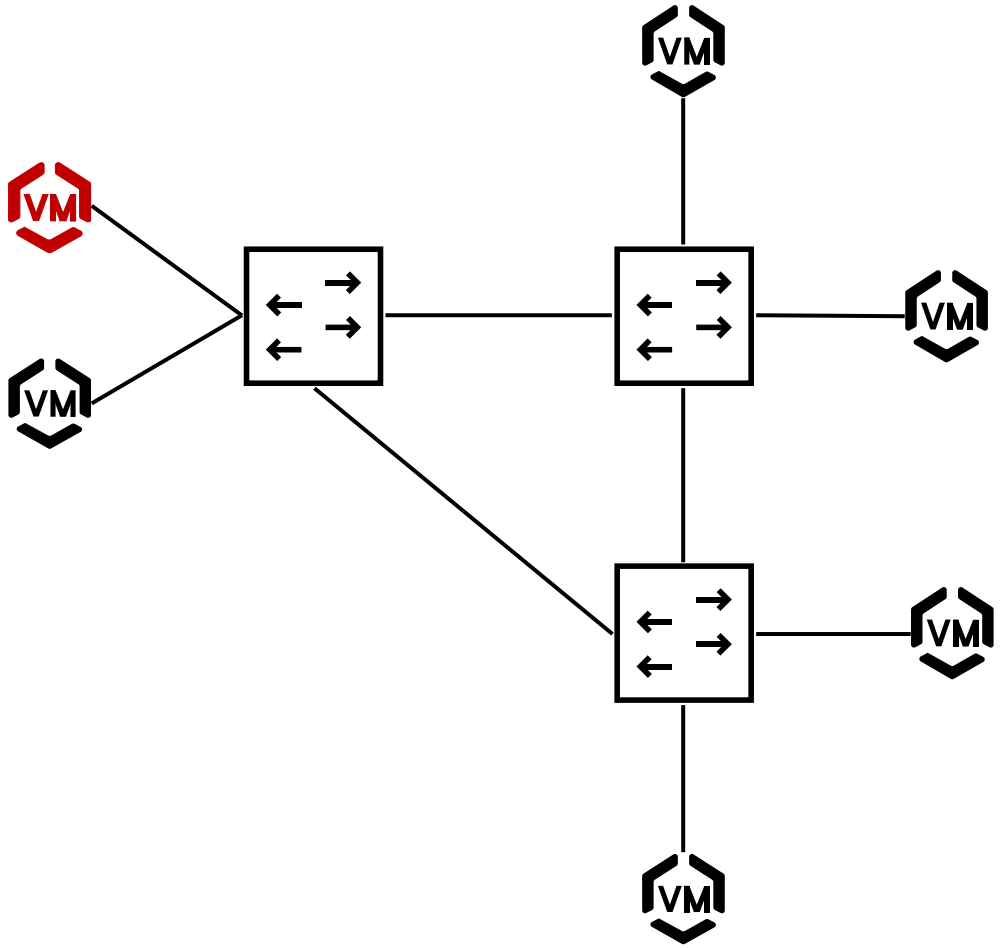
TECHNISCHE UNIVERSITÄT DARMSTADT

Tunnels Ethernet traffic often through

- Multi-Protocol Label Switching (MPLS)
- Virtual eXtensible LAN (VXLAN)

And can rely on their failure recovery mechanisms

- Can utilize built-in resilience approaches (e.g., Fast Reroute, Loop-Free Alternates, …)
- Can utilize multi-path and multi-homing approaches
- Anycast Gateways



EVPN Data Plane

TECHNISCHE UNIVERSITÄT DARMSTADT

Tunnels Ethernet traffic often through

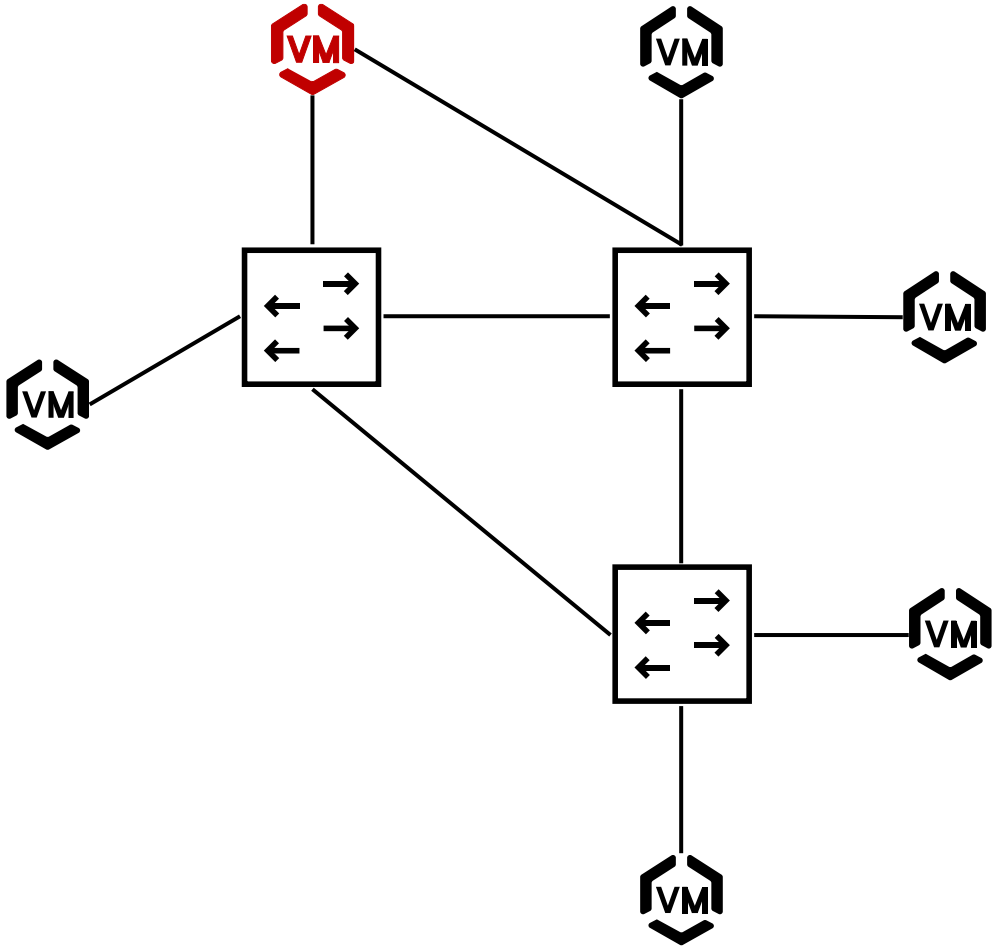
- Multi-Protocol Label Switching (MPLS)
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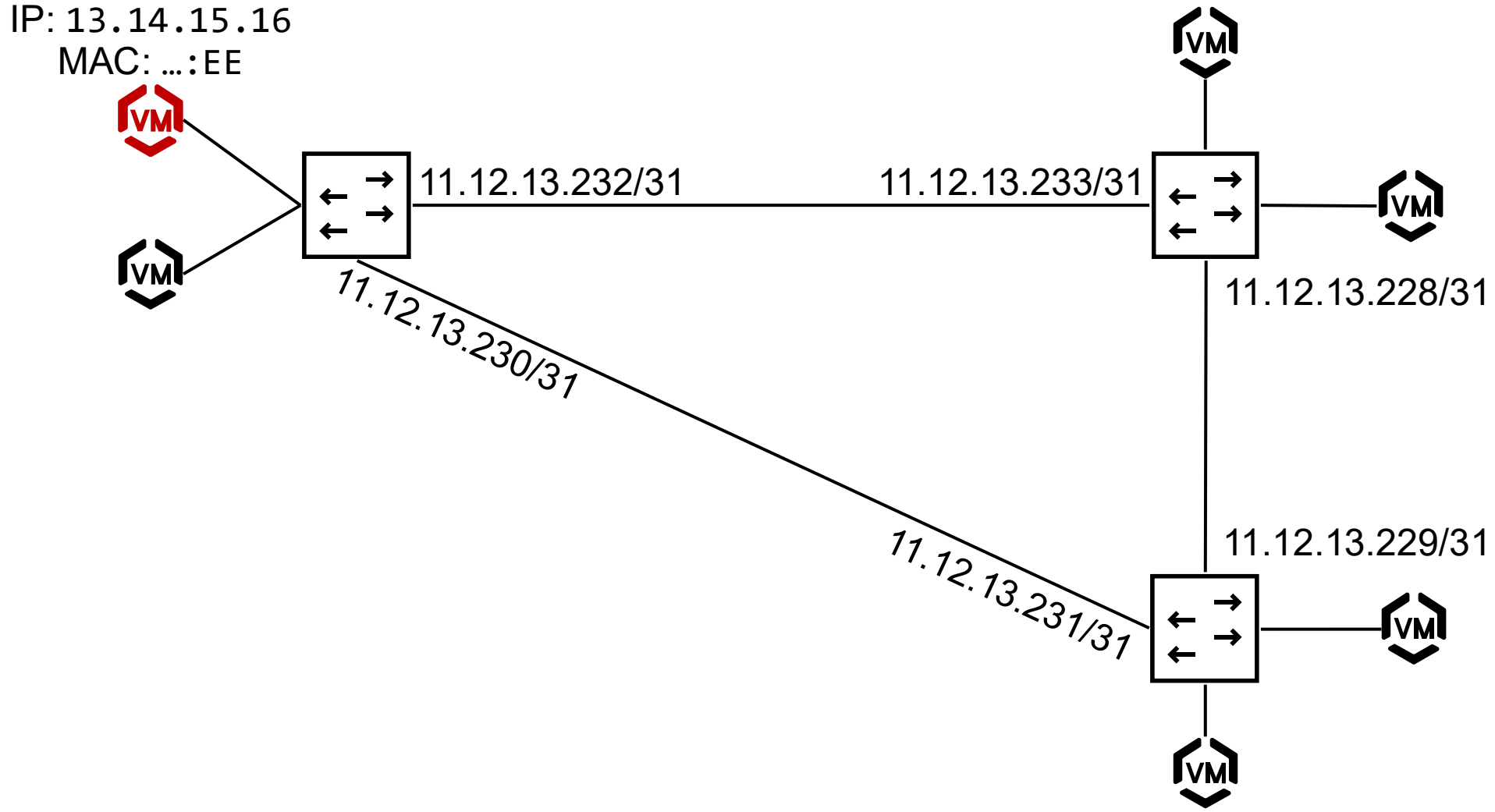
- Can utilize built-in resilience approaches (e.g., Fast Reroute, Loop-Free Alternates, …)
- Can utilize multi-path and multi-homing approaches
- Anycast Gateways

Can forward L2 and L3 traffic

- Also known as Integrated Routing and Bridging (IRB)

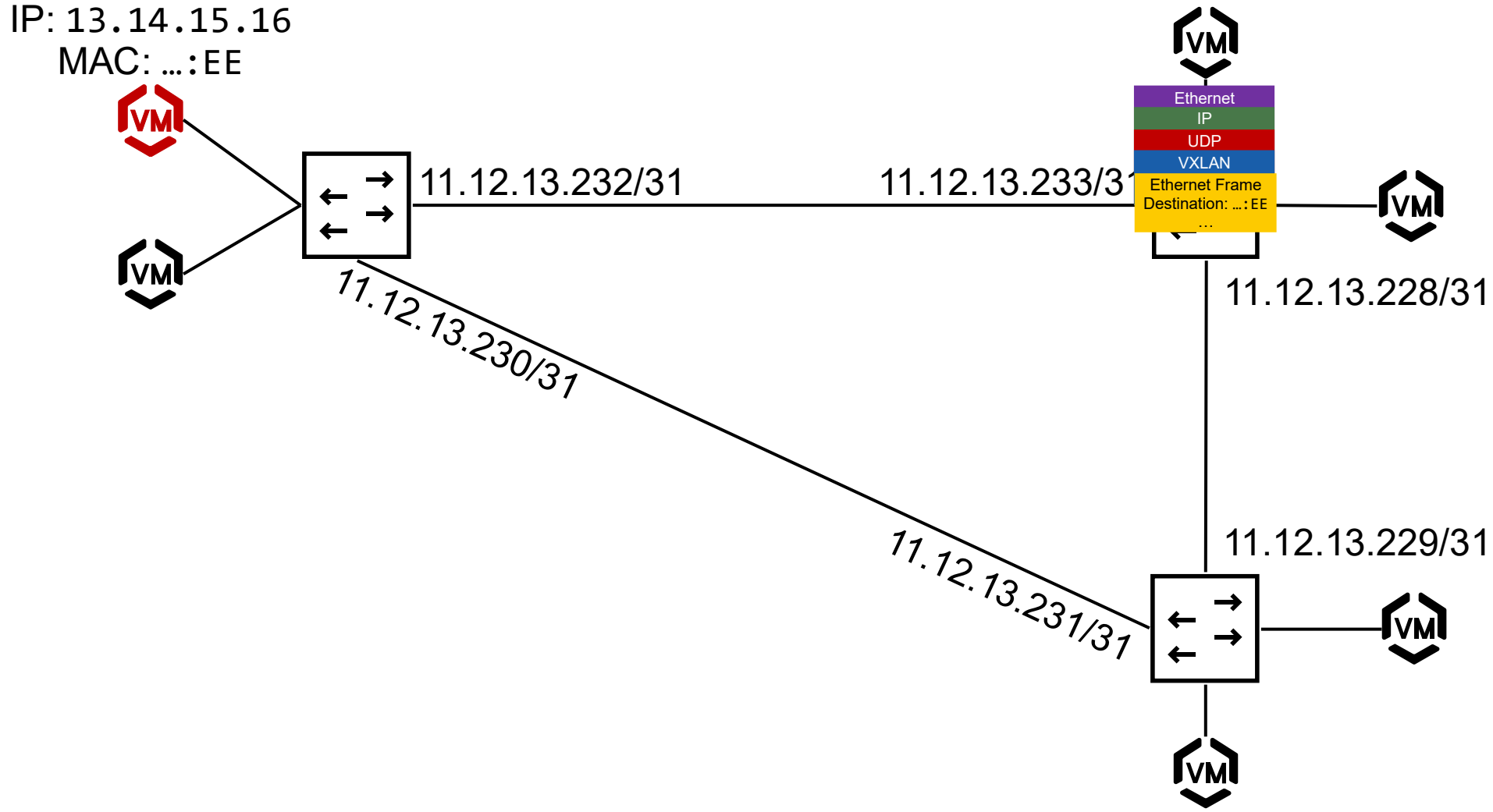


EVPN Data Plane
via VXLAN

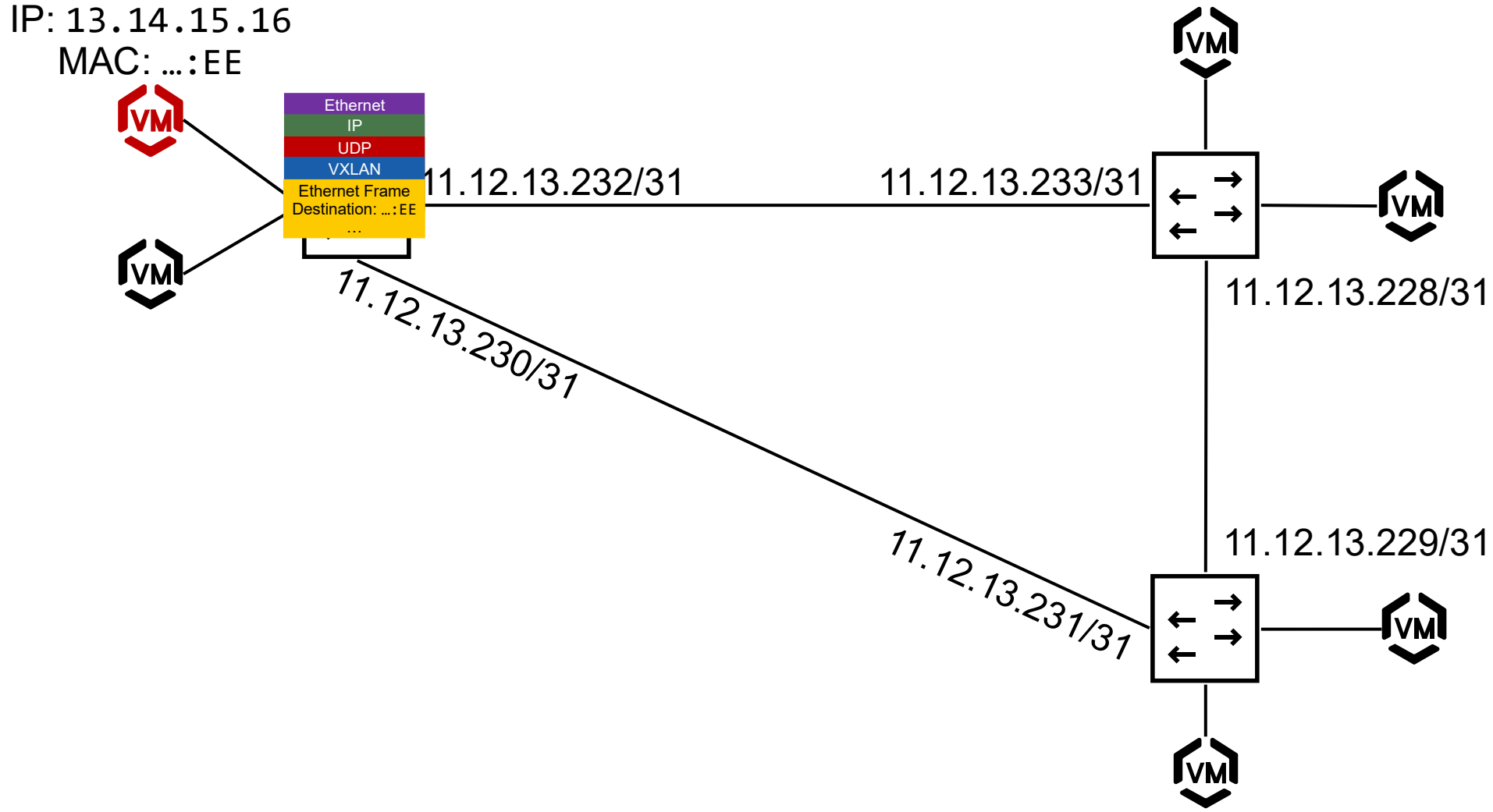


Ethernet Frame
Destination: …:EE
…

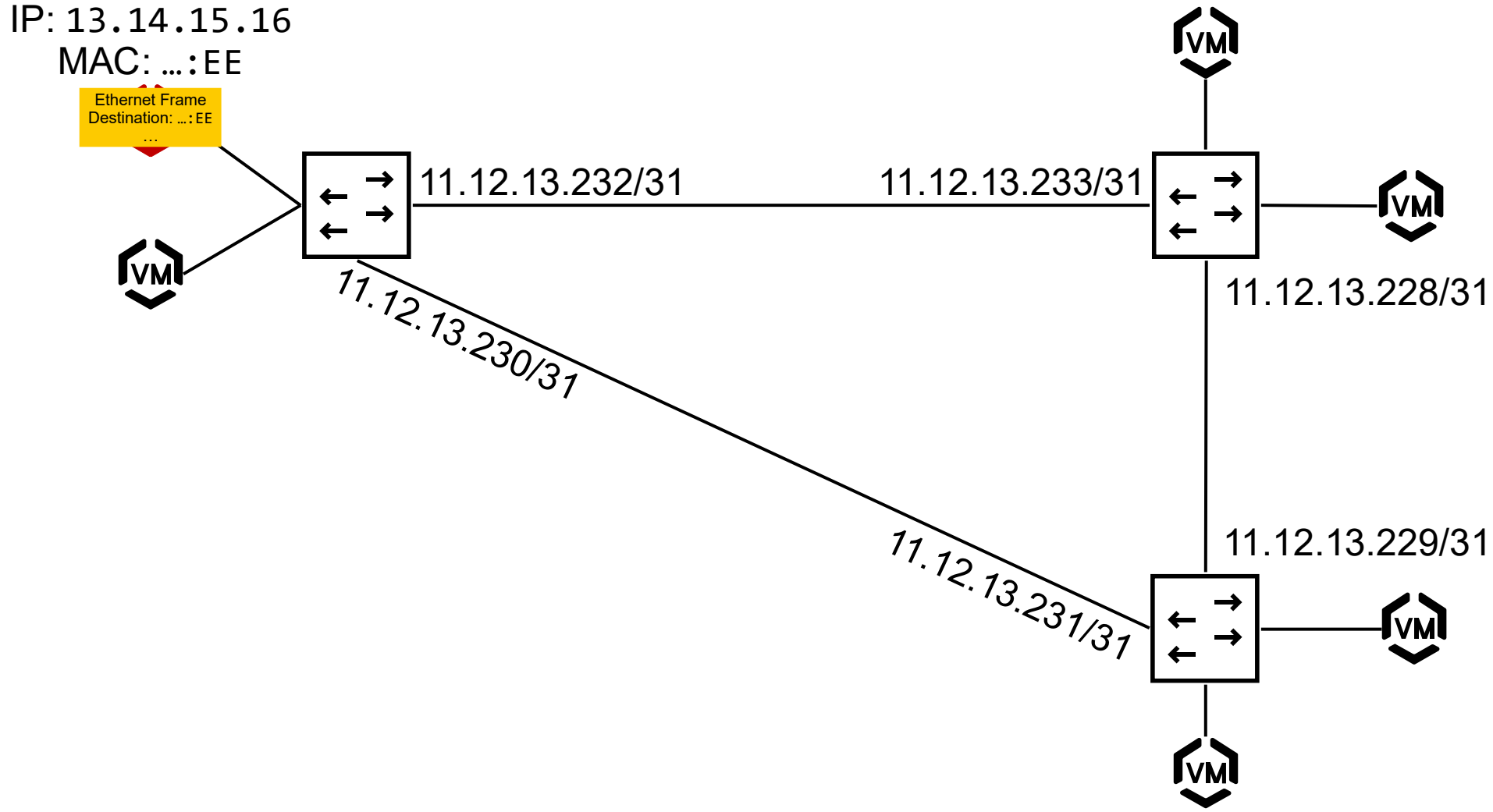
EVPN Data Plane
via VXLAN



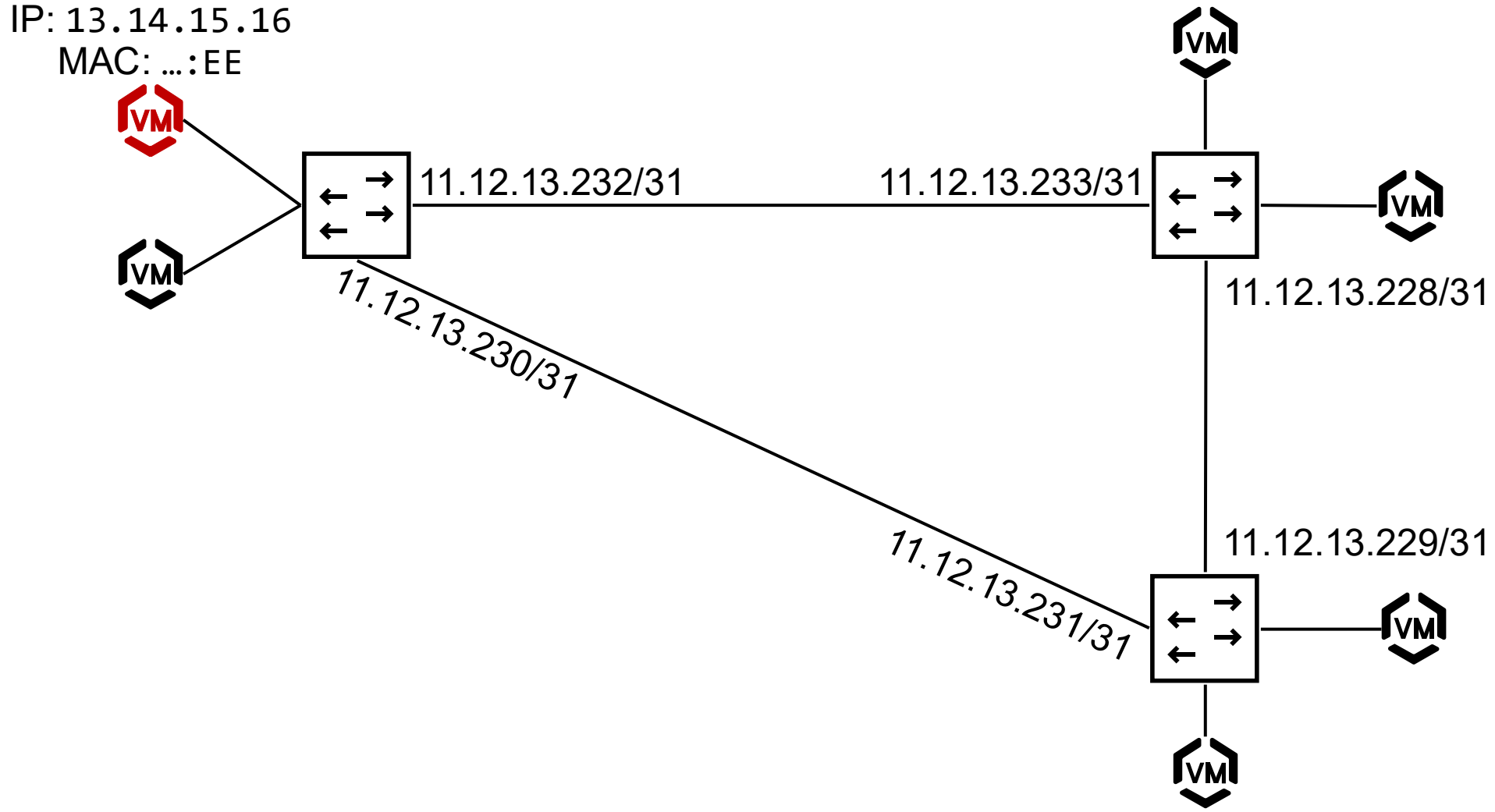
EVPN Data Plane
via VXLAN



EVPN Data Plane
via VXLAN

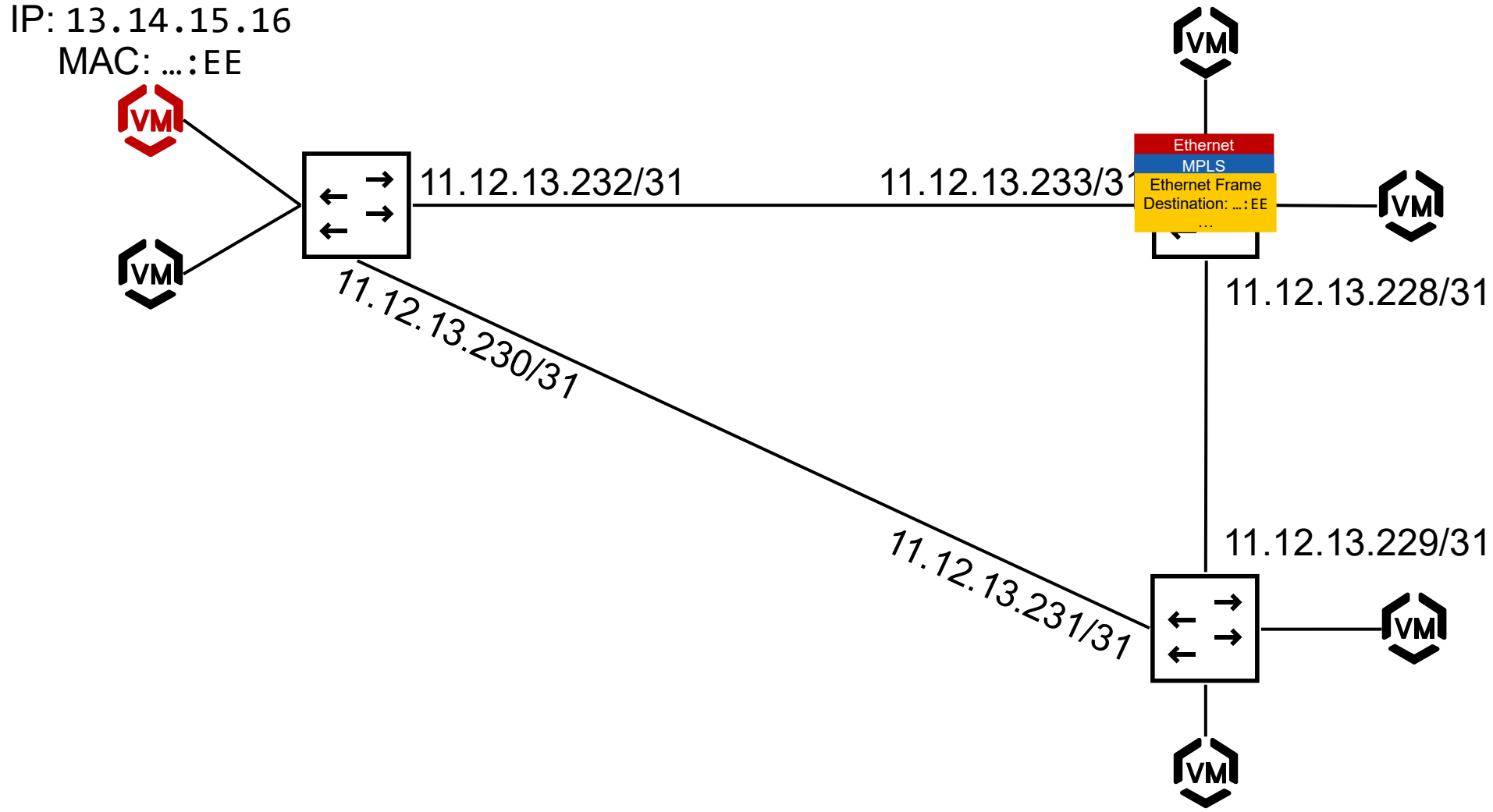


EVPN Data Plane
via MPLS

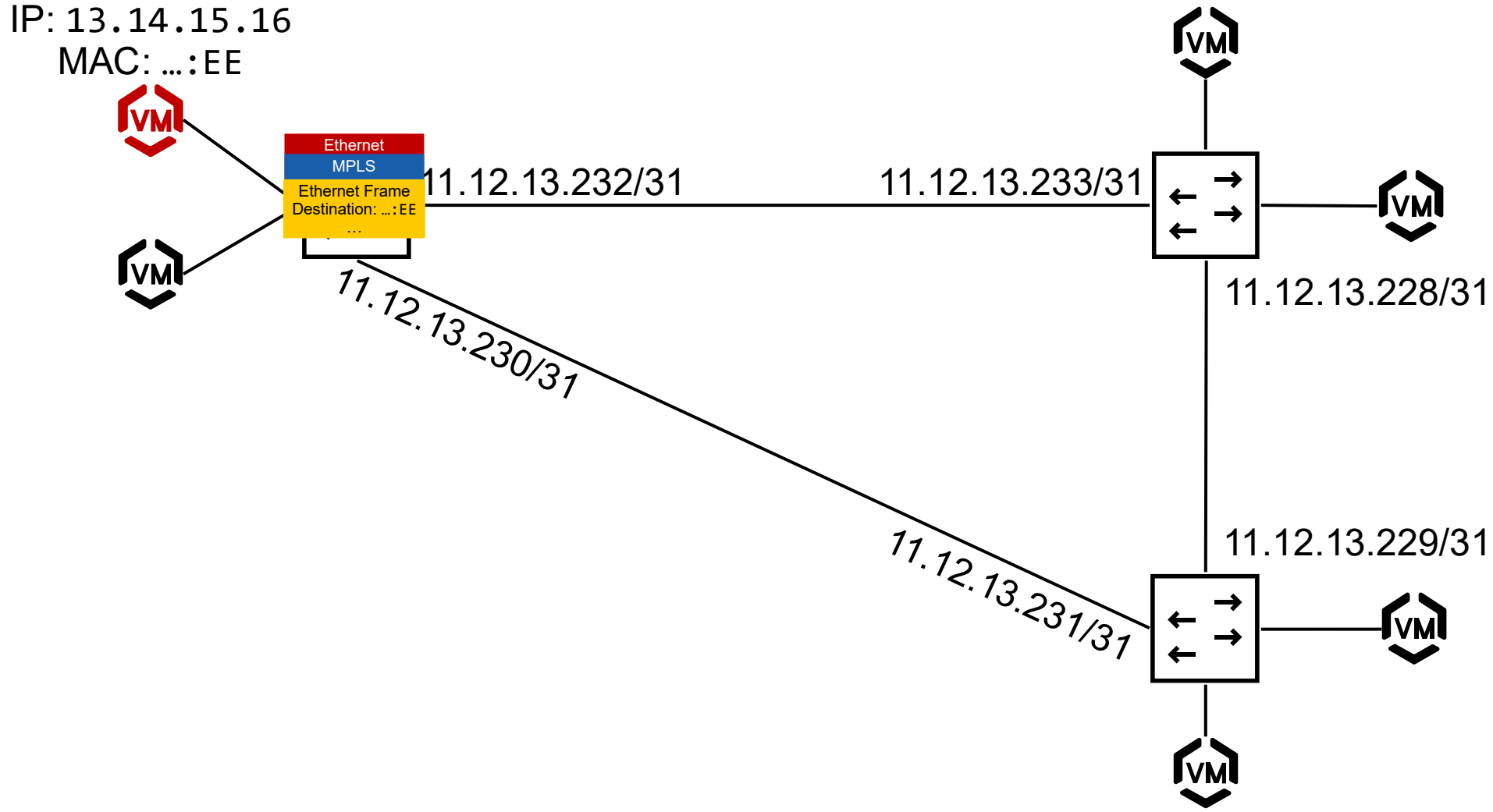


Ethernet Frame
Destination: …:EE
…

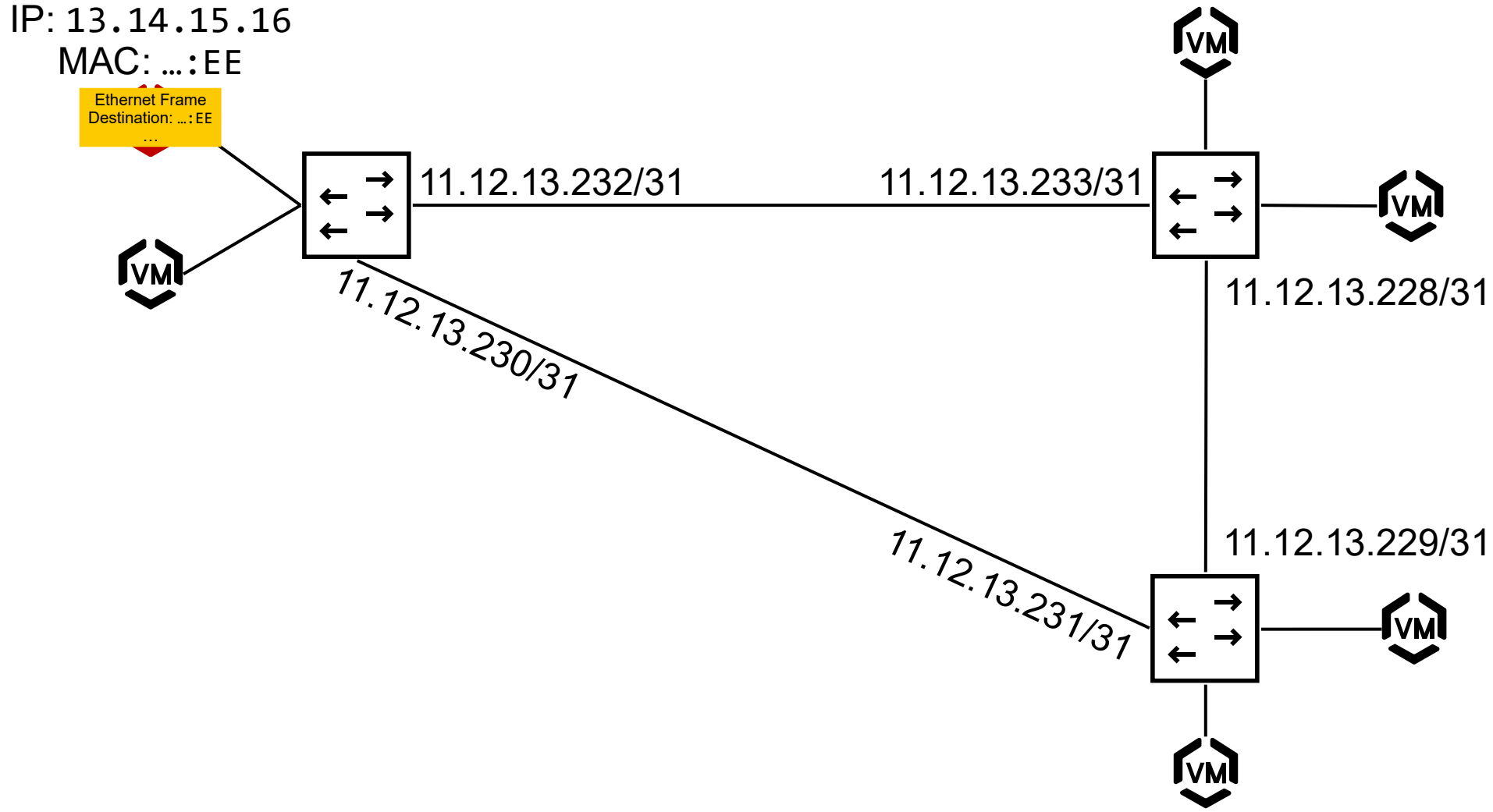
EVPN Data Plane
via MPLS



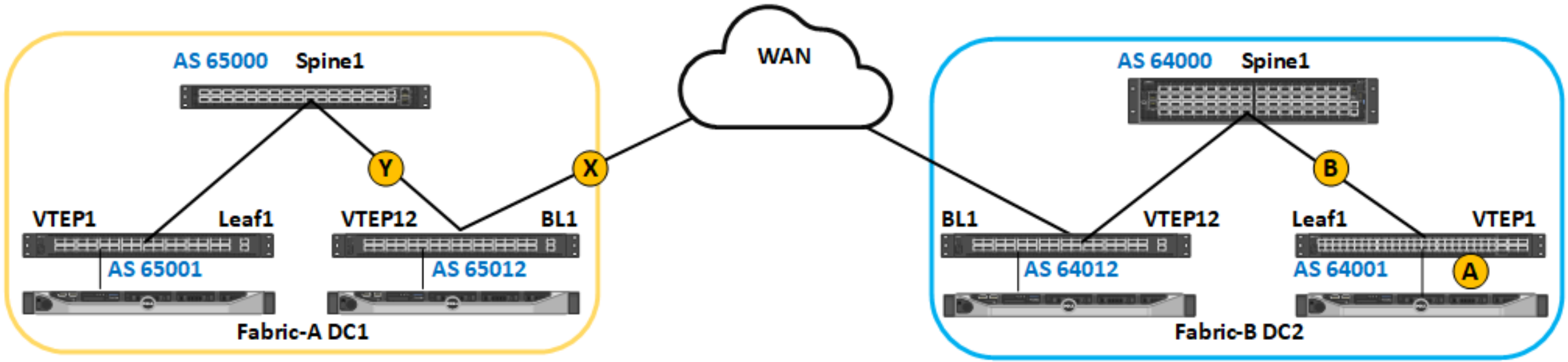
EVPN Data Plane
via MPLS



EVPN Data Plane
via MPLS

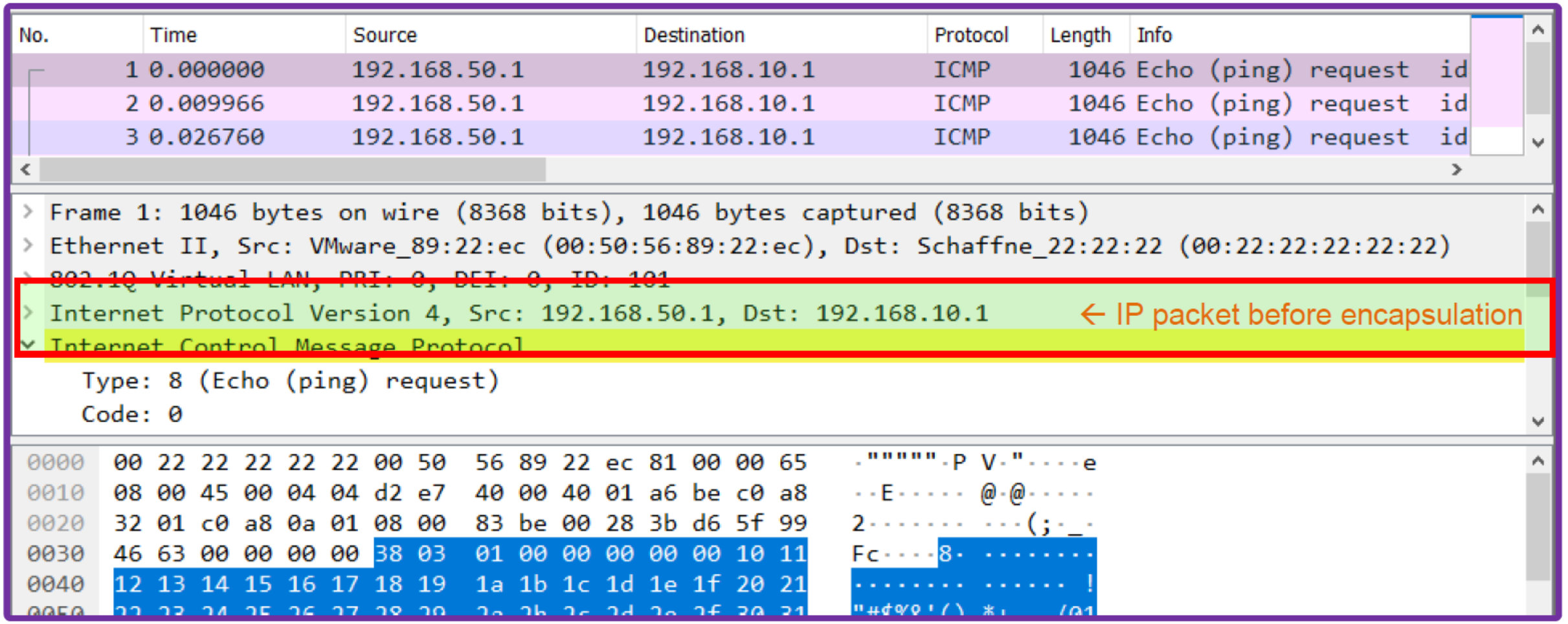


Ethernet Virtual Private Network (EVPN)
Example with VXLAN

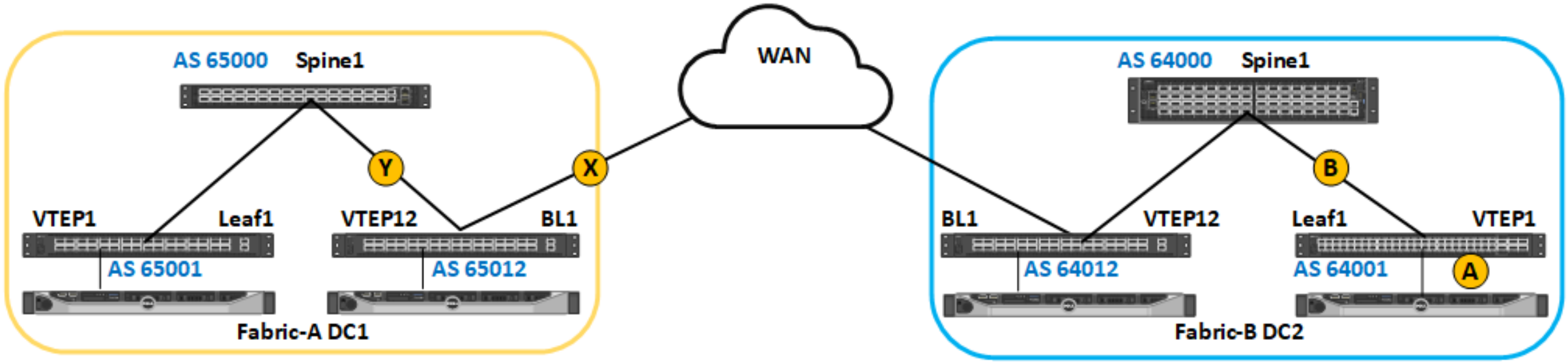


https://infohub.delltechnologies.com/en-us/l/evpn-vxlan-based-multisite-data-center-interconnect-dci-using-dell-enterprise-sonic/packet-capture-on-fabric-b-and-fabric-a/

Ethernet Virtual Private Network (EVPN)
Example with VXLAN at A

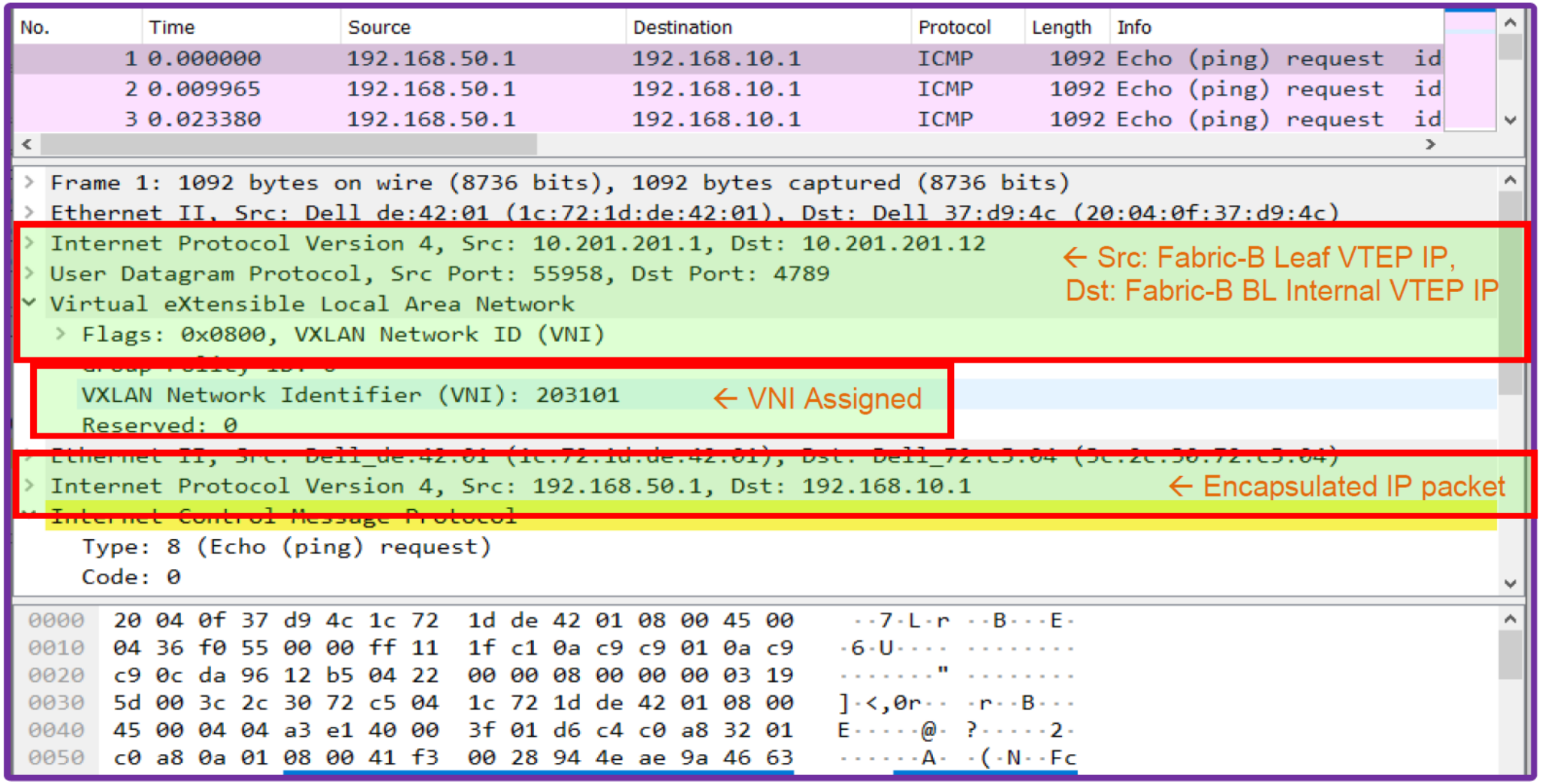


Ethernet Virtual Private Network (EVPN)
Example with VXLAN



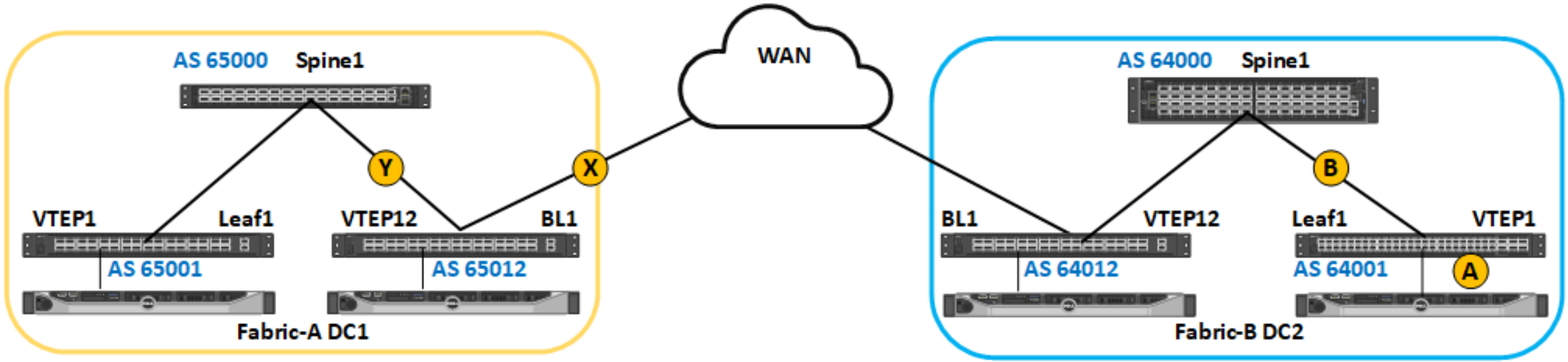
https://infohub.delltechnologies.com/en-us/l/evpn-vxlan-based-multisite-data-center-interconnect-dci-using-dell-enterprise-sonic/packet-capture-on-fabric-b-and-fabric-a/

Ethernet Virtual Private Network (EVPN)
Example with VXLAN at B



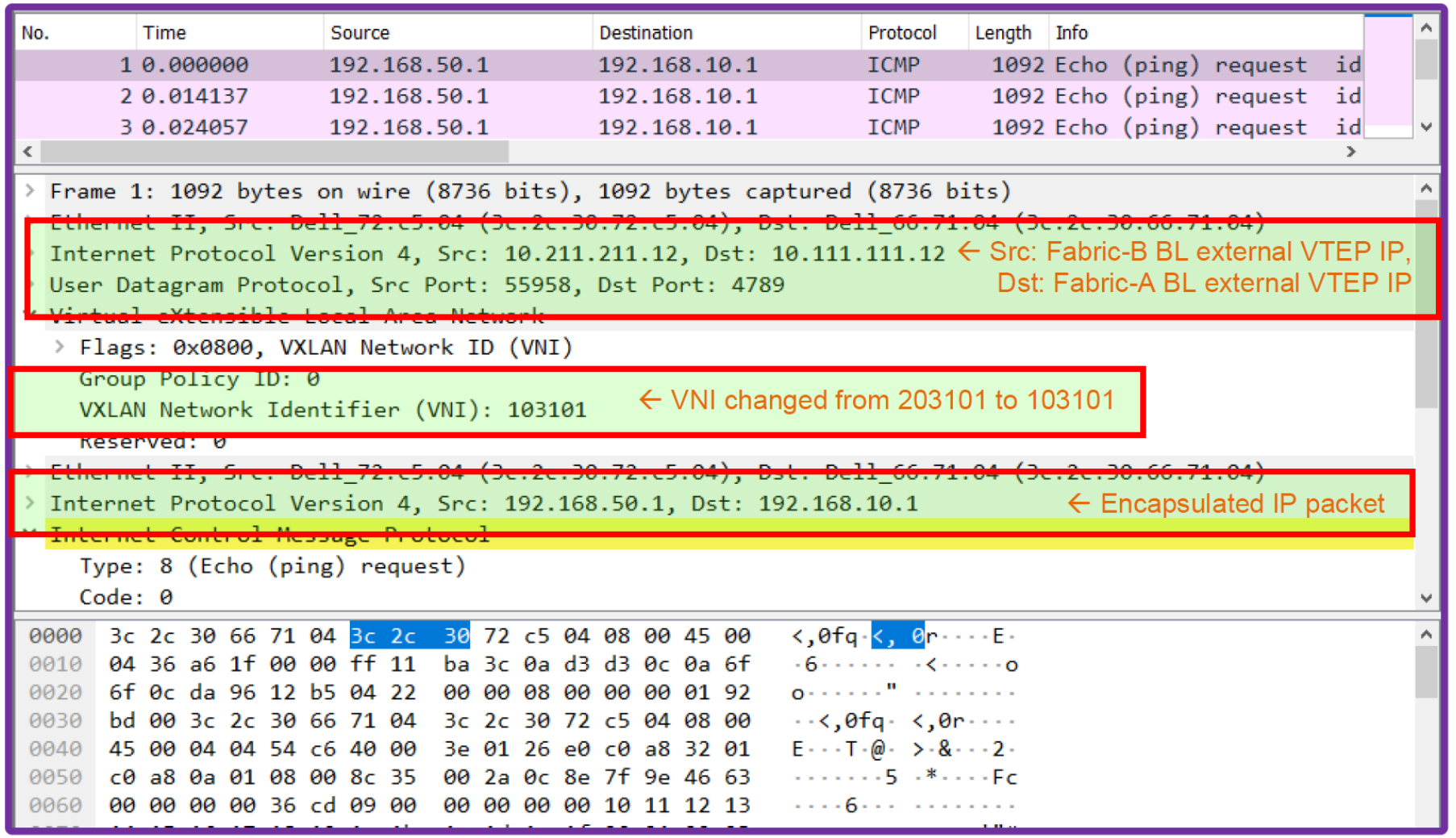
https://infohub.delltechnologies.com/en-us/l/evpn-vxlan-based-multisite-data-center-interconnect-dci-using-dell-enterprise-sonic/packet-capture-on-fabric-b-and-fabric-a/

Ethernet Virtual Private Network (EVPN)
Example with VXLAN

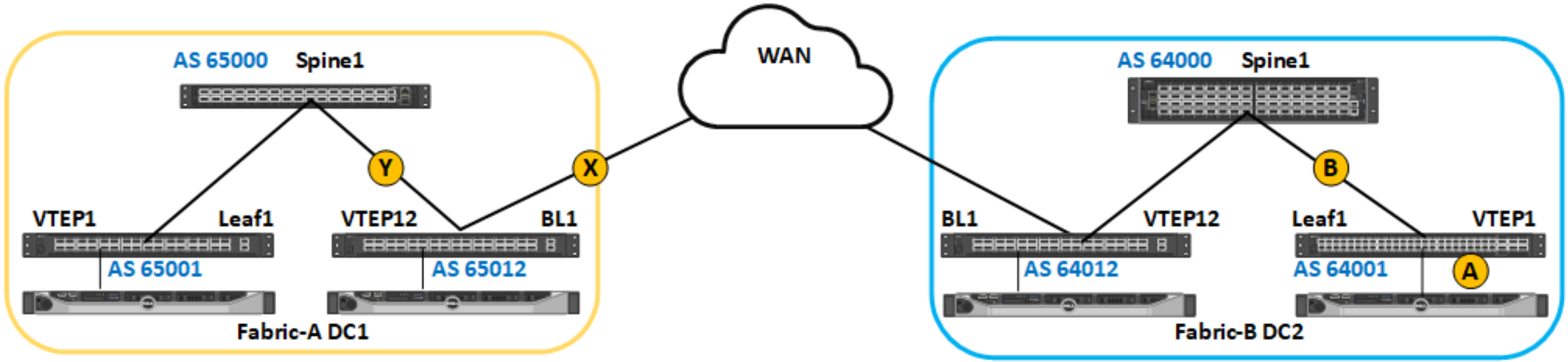


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Ethernet Virtual Private Network (EVPN)
Example with VXLAN at X

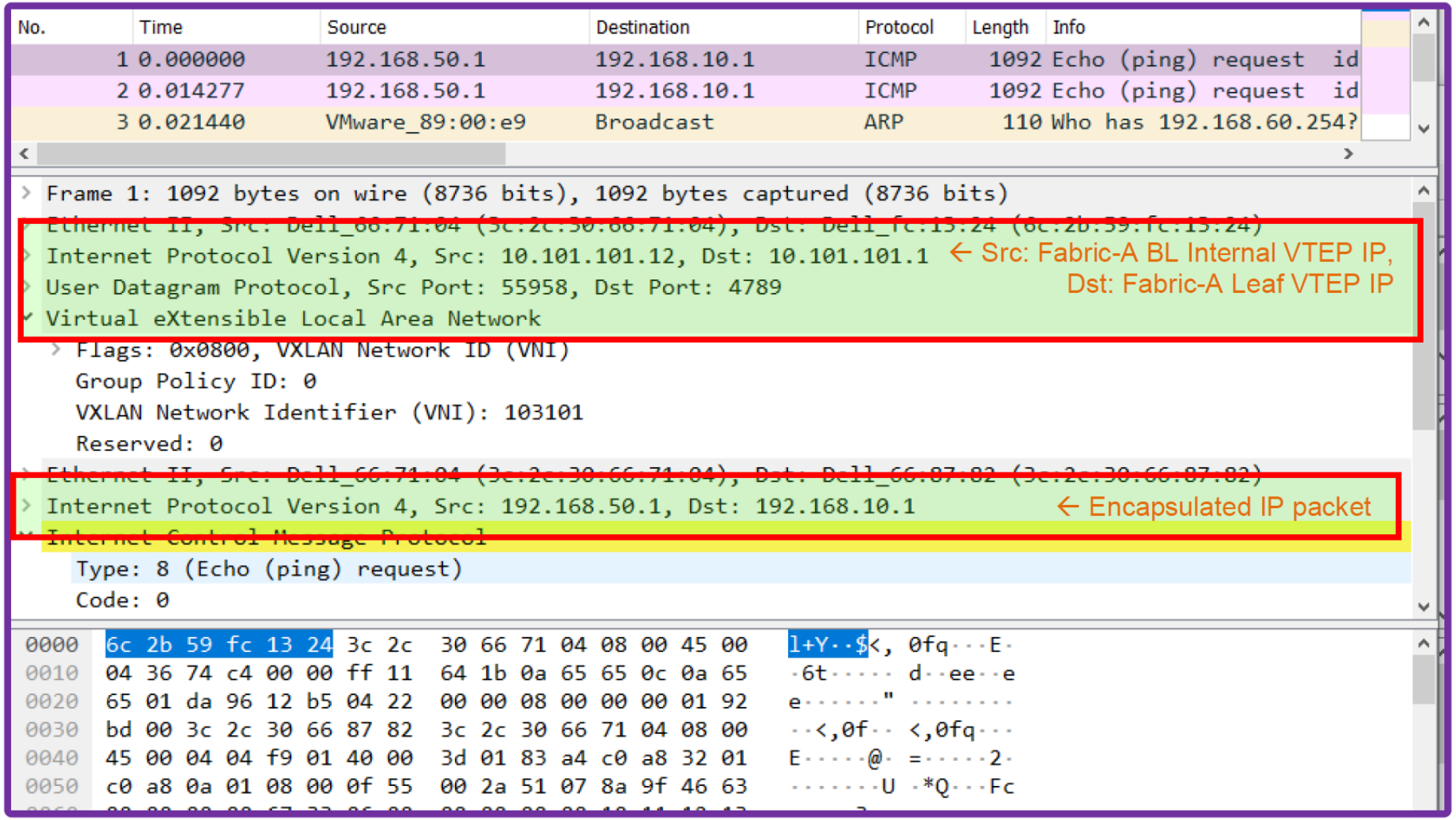


Ethernet Virtual Private Network (EVPN)
Example with VXLAN

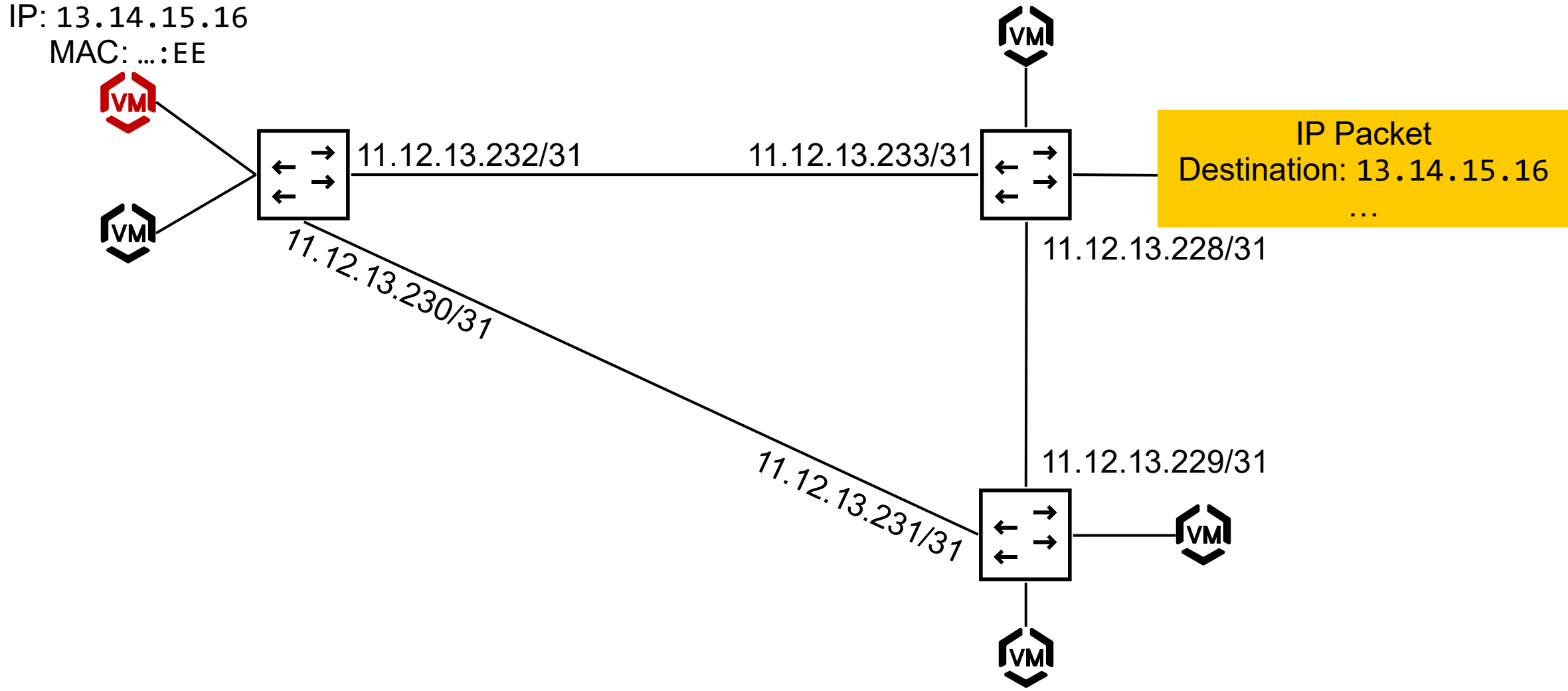


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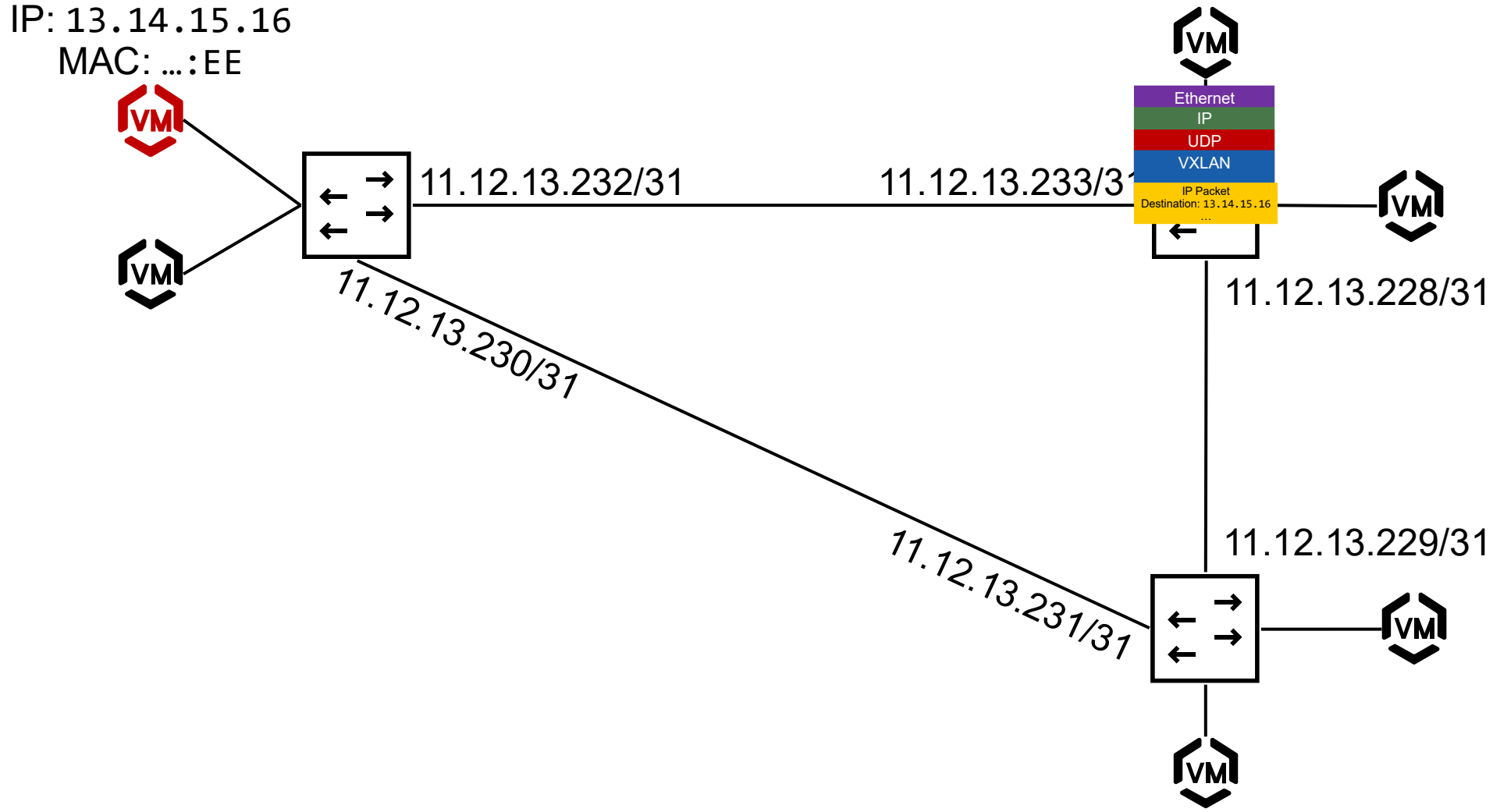
Ethernet Virtual Private Network (EVPN)
Example with VXLAN at Y



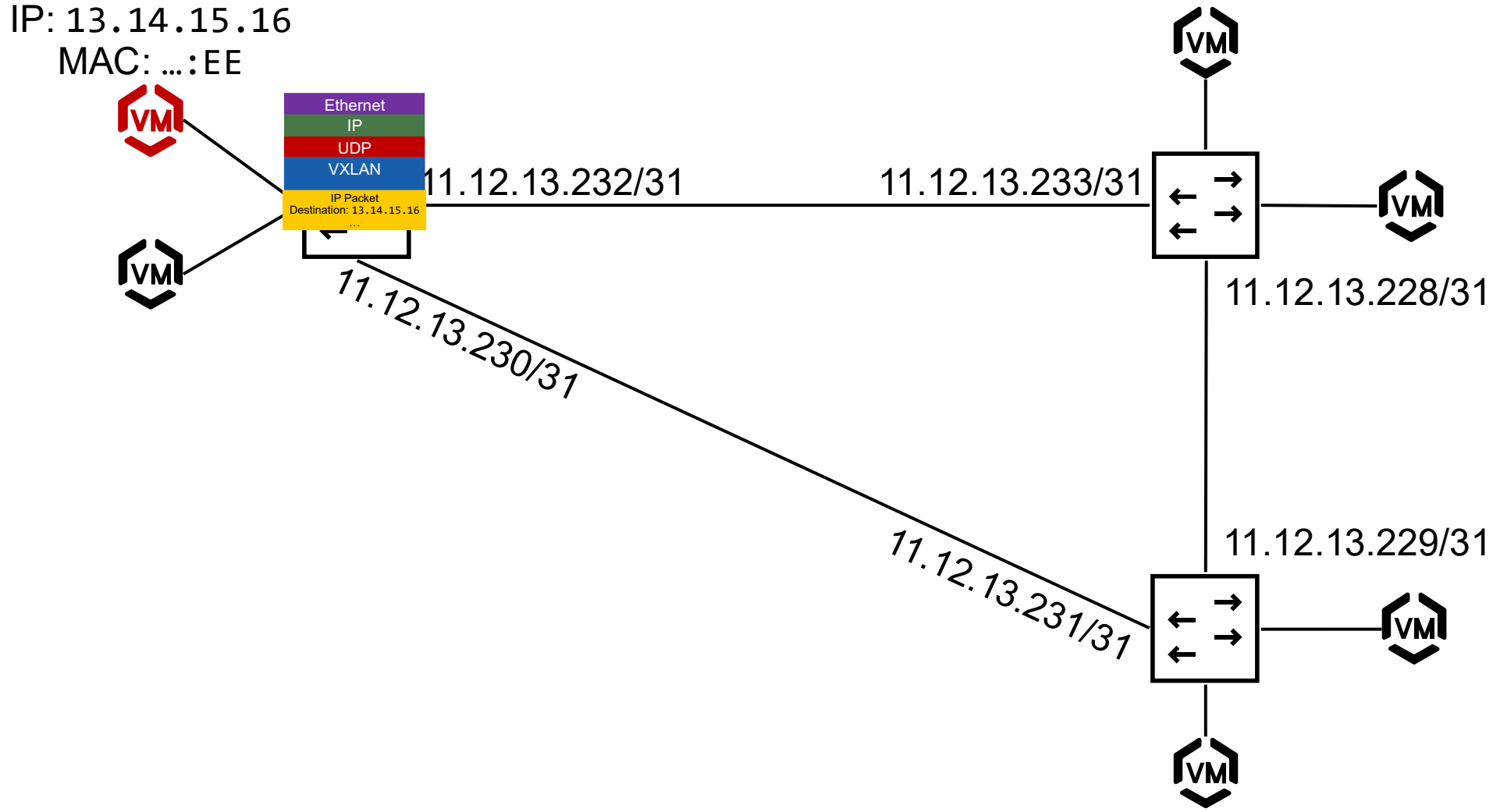
EVPN Data Plane – IP
via VXLAN



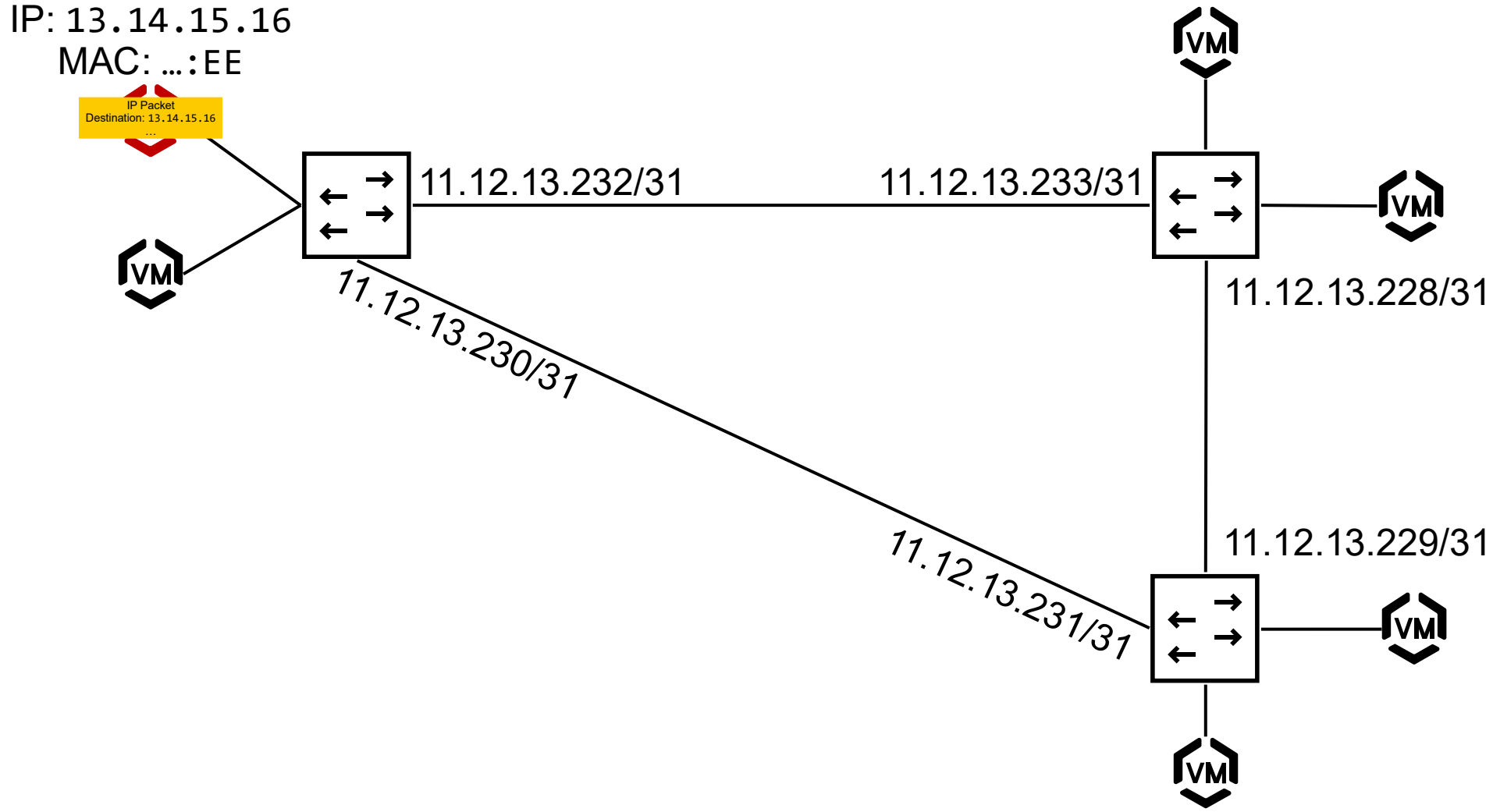
EVPN Data Plane – IP
via VXLAN



EVPN Data Plane – IP
via VXLAN



EVPN Data Plane – IP
via VXLAN

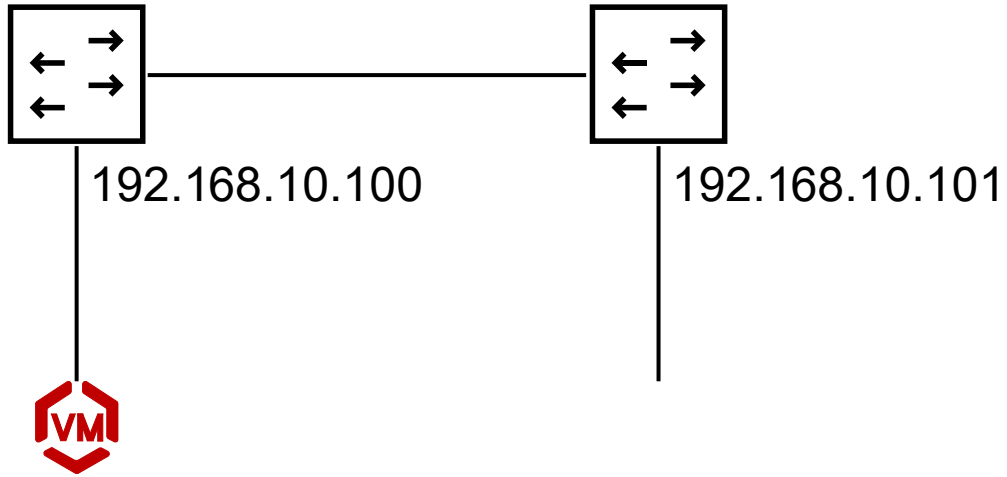


EVPN Data Plane – Anycast Gateways

TECHNISCHE UNIVERSITÄT DARMSTADT

When VMs that communicate on L3 move, their gateway would often change

- In this example, moving the VM to another switch would change its default gateway from 192.168.10.100 to 192.168.10.101
- Once a VM moves, it either requires reconfiguration, e.g., via DHCP, or uses inefficient paths in the network



Idea: Use the same gateway multiple times in your network

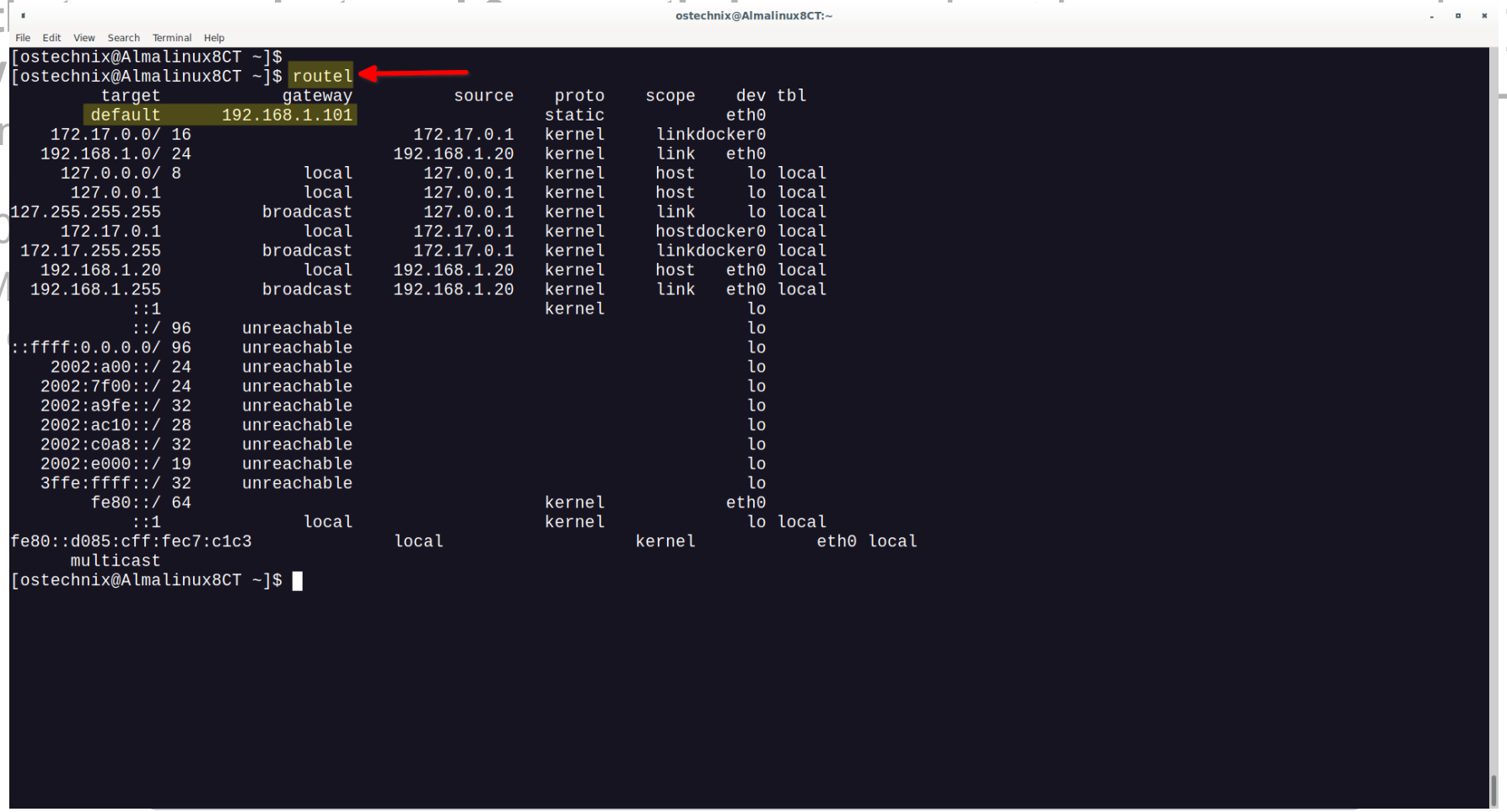
EVPN Data Plane – Anycast Gateways

TECHNISCHE UNIVERSITÄT DARMSTADT

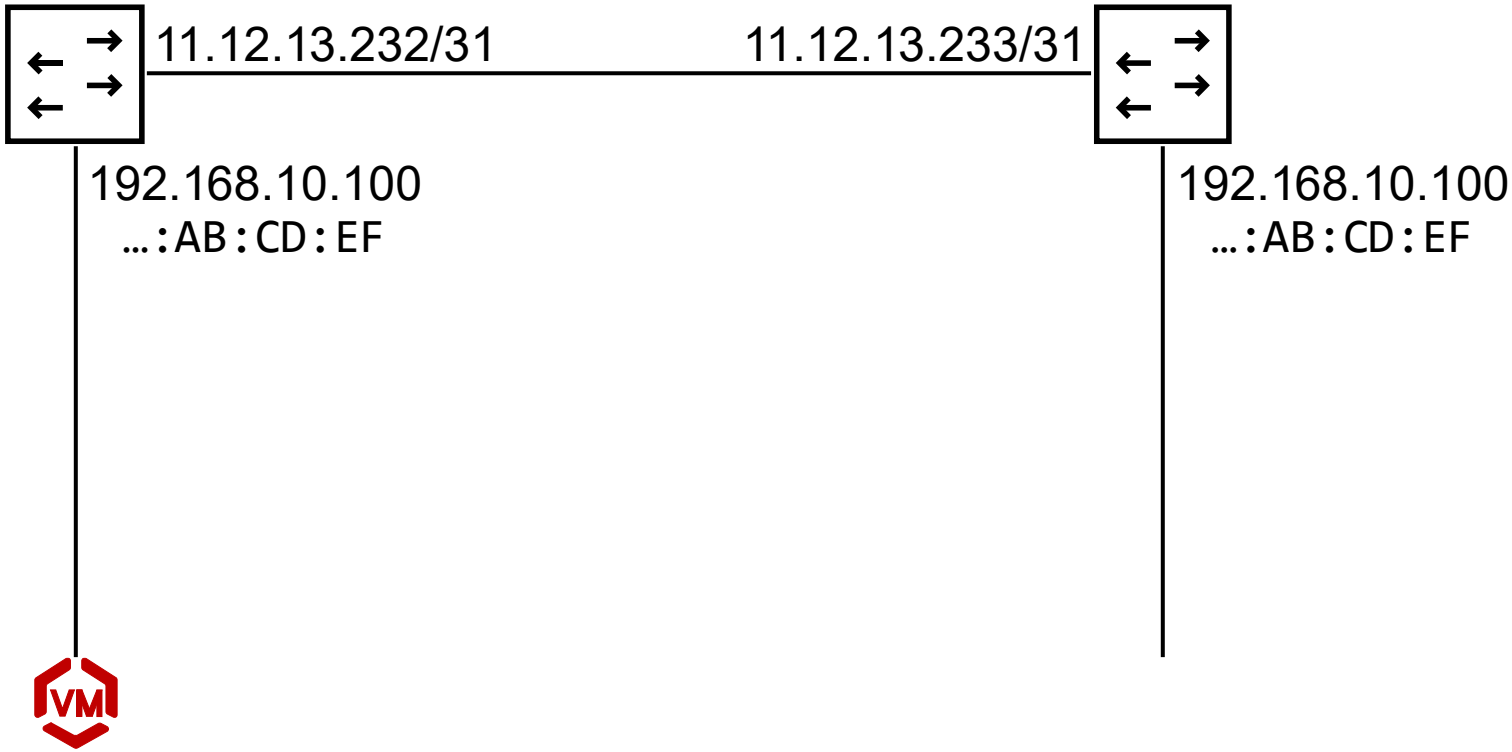
When VMs t… gateway w…

- In this exar… change its … 192.168.10…
- Once a VM … via DHCP, …

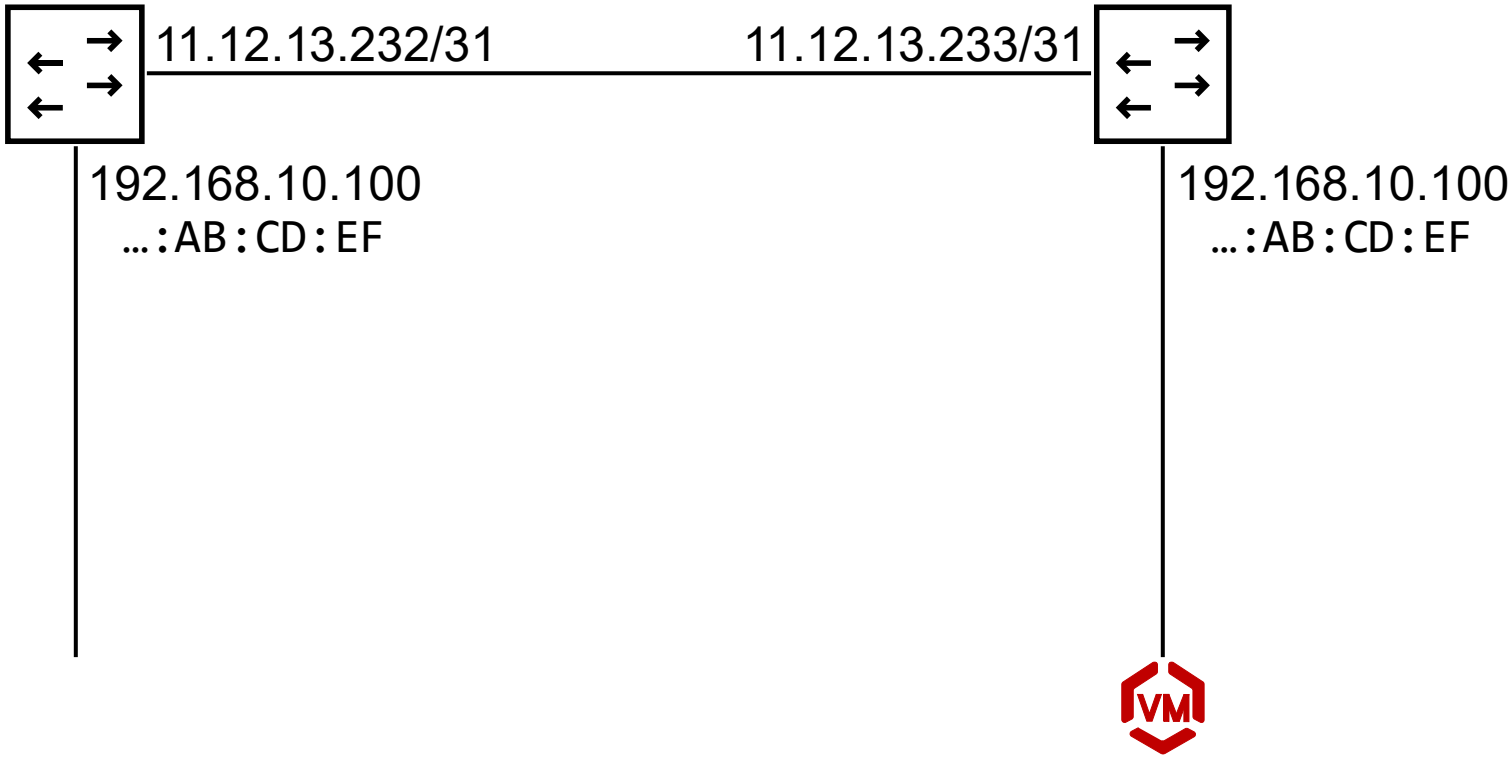
192.168.10.101



EVPN Data Plane – Anycast Gateways



EVPN Data Plane – Anycast Gateways



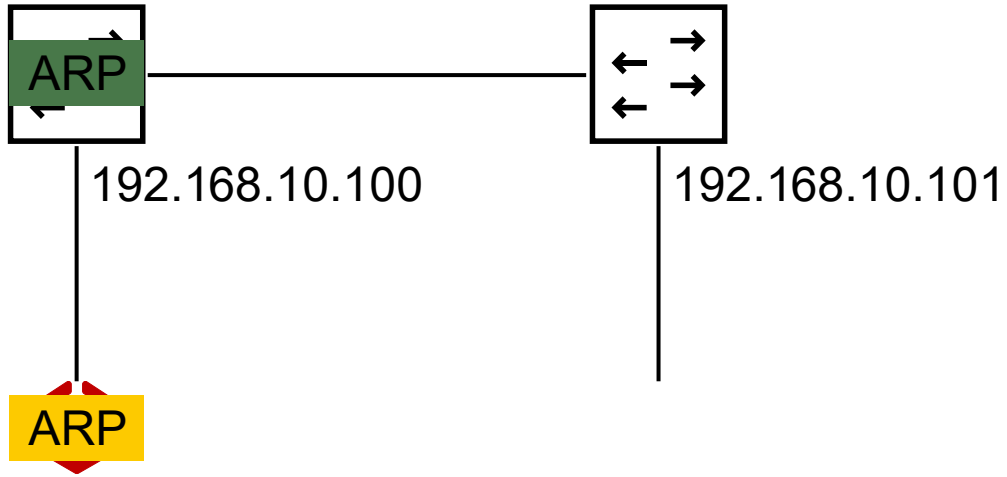
Why does this even work?

EVPN Data Plane – Anycast Gateways

TECHNISCHE UNIVERSITÄT DARMSTADT

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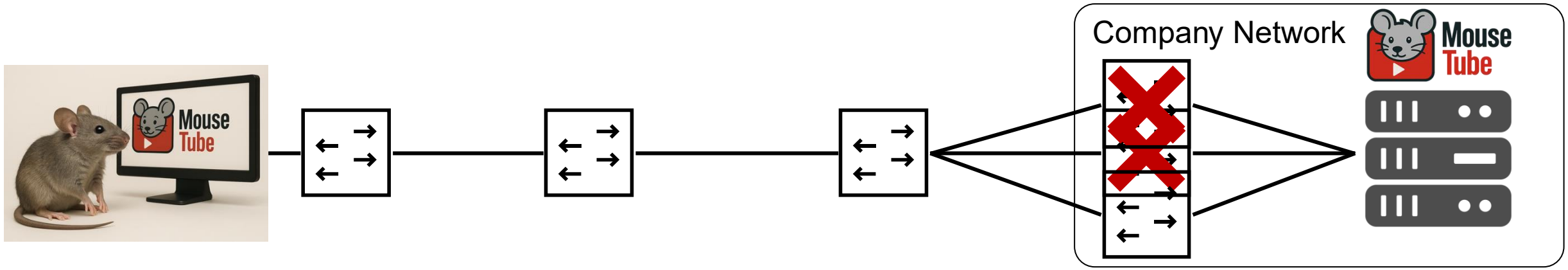


IP- and MAC-addresses of the gateway are not learned, but provided statically

- Gateway is always the first hop
- No ARP table instability

What if we want to achieve something similar in “traditional” networks?

Router Redundancy



If the edge router fails, the two networks become isolated

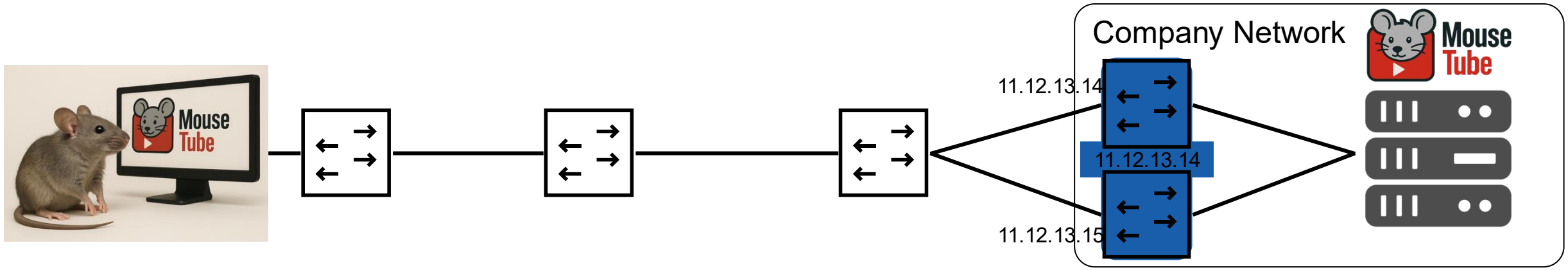
- No communication between client and server

To increase resilience, MouseTube has two routers to provide their service

- These routers need to be addressed when entering the network
- However, if any of the routers goes down, the IP and MAC address change, leading to service interruption

How can a failover between routers be performed?

Router Redundancy



General idea: Combine multiple physical routers to one logical router

IP-addresses exposed to other networks must be unique

Multiple well-known implementations

- Virtual Router Redundancy Protocol (VRRP)
- Common Address Redundancy Protocol (CARP)
- Hot Standby Router Protocol (HSRP)

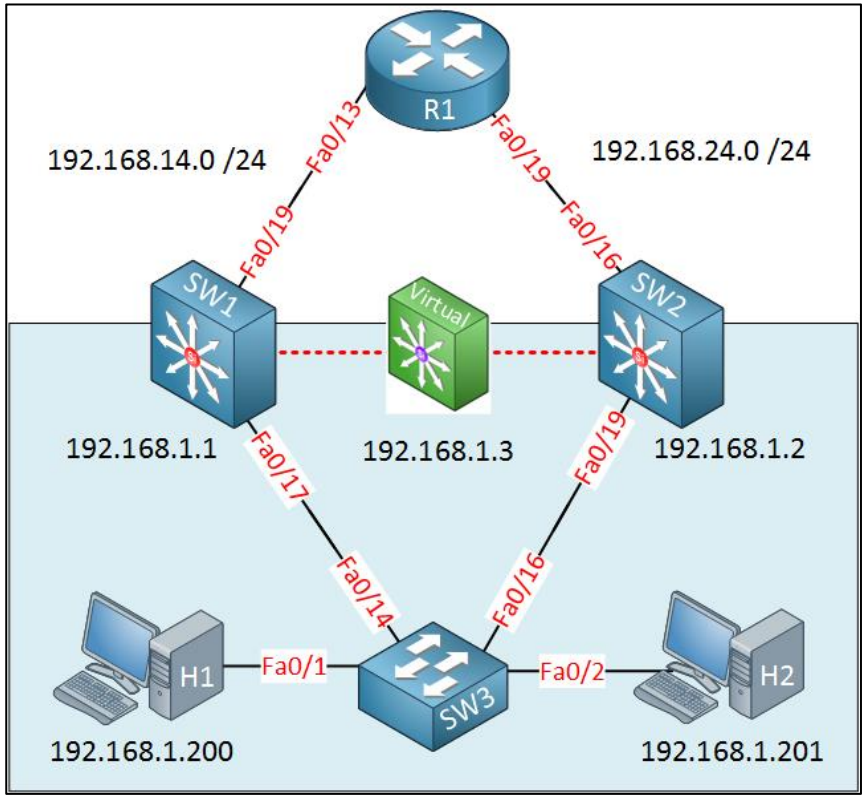
Virtual Router Redundancy Protocol (VRRP)

Open standard for resilient operation of important routers

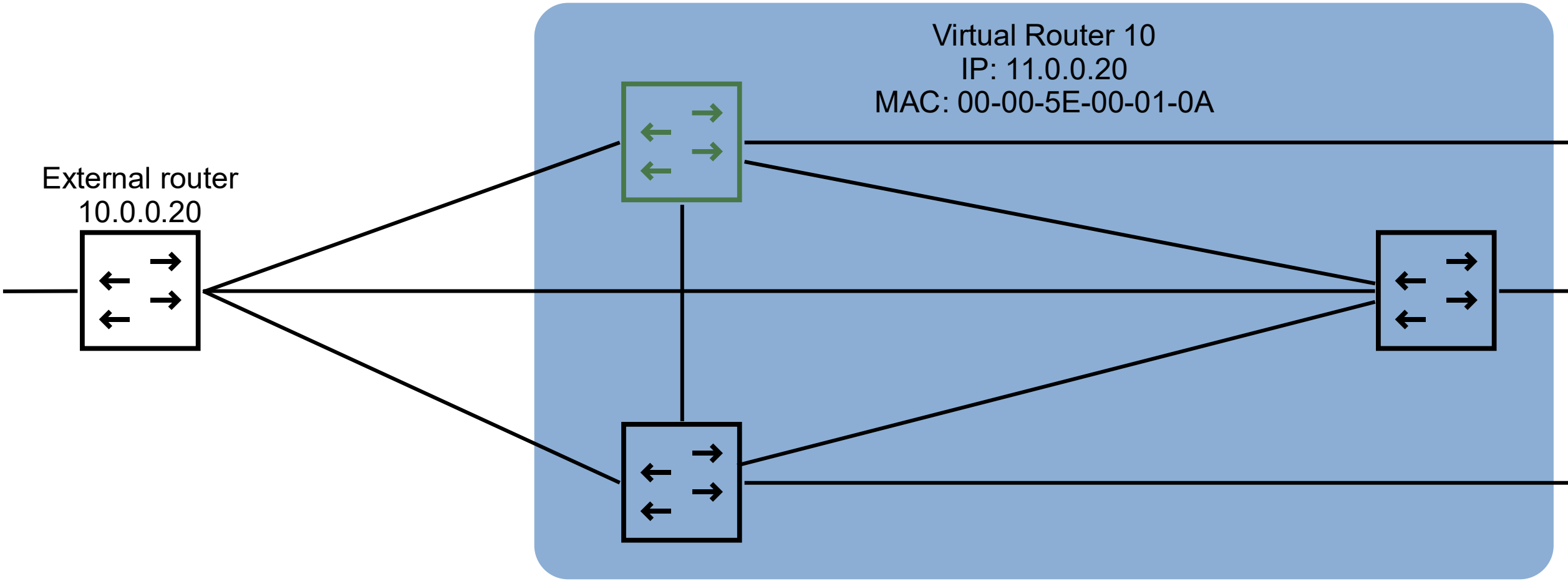
- Standardized in RFC 3769 and later refined in RFC 5798
- Implemented on hardware of vendors like Cisco, Juniper and HPE (Aruba)

Creates a virtual router that combines multiple physical routers

- Only one router is active at the same time
- The currently active router shares its current state with other routers forming the virtual router
- Once the currently active router fails, a new active router is selected and announces itself as active router



VRRP Explained (IPv4 case), only one master



Active Router depicted in green, failed routers in red

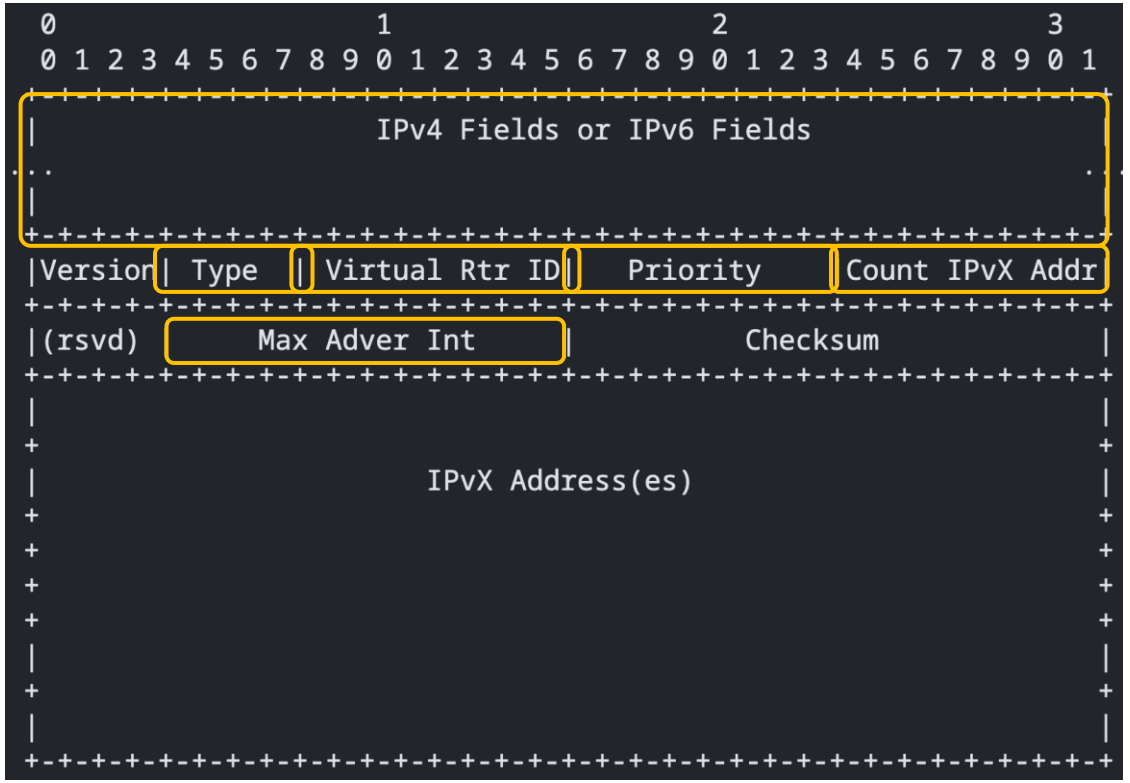
VRRP Packet Format (Version 3)

Not relevant for the exam

VRRP packets are used solely to coordinate between routers forming a virtual router

Important IPv4 fields

- Destination is set to 224.0.0.18
- TTL must be set to 255 (packets below 255 are dropped)



VRRP header fields

- Type must be 1 ADVERTISEMENT
- Virtual Router ID identifies the virtual router
- Priority specifies the priority of a physical router
  - The active router has priority 255
  - If a priority value of 0 is used, the master has stopped working
- Count is the number of IP addresses contained
- Time interval between two advertisements in centiseconds

State Transition Diagram of a VRRP Router

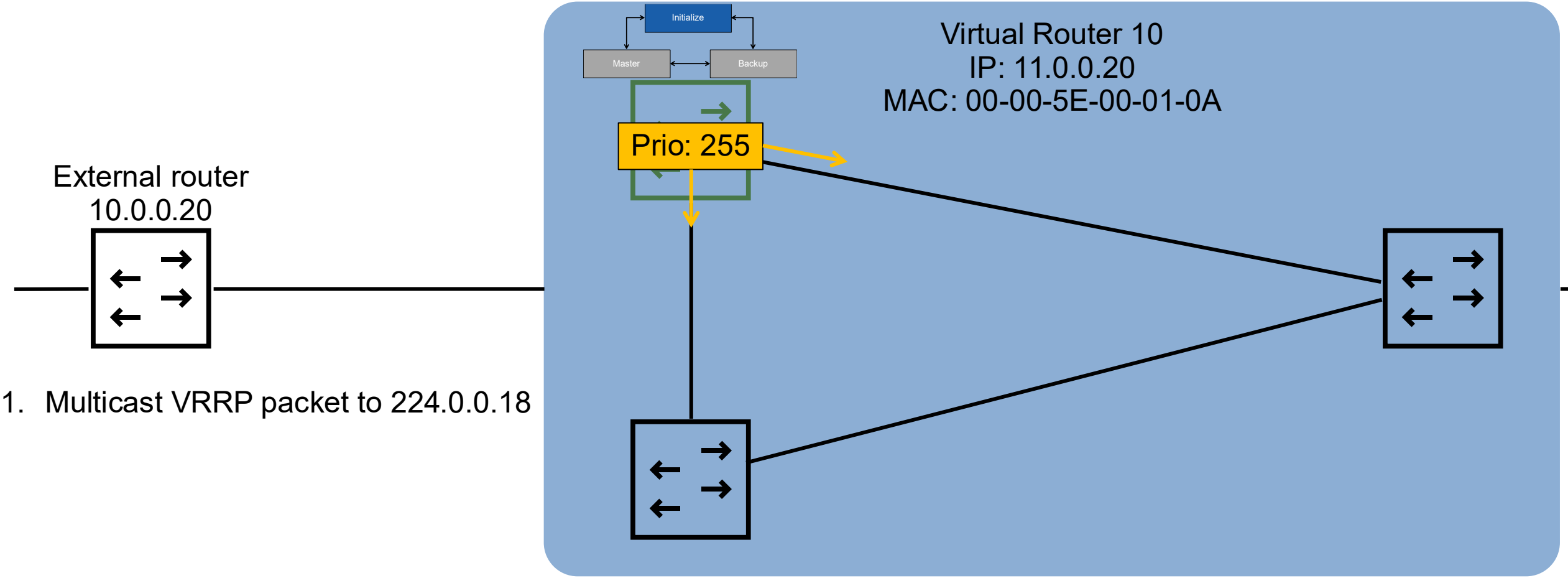
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Once a VRRP router is started, it start in initialize

- Transitions to either master or backup directly after
- States can be altered through failure of the master
  - Detected by an explicit failure notification
  - Detected by a timeout

VRRP Explained (IPv4 case), only one master

Initialize Master

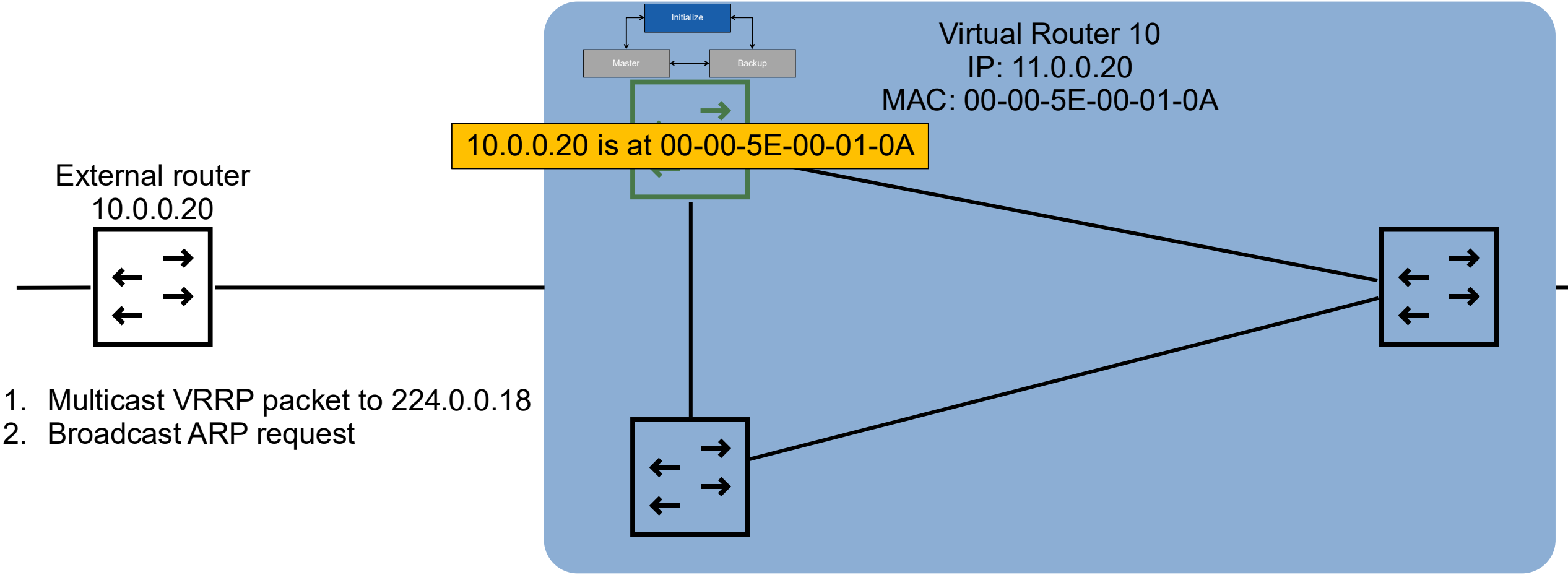


1. Multicast VRRP packet to 224.0.0.18

Active Router depicted in green, failed routers in red

VRRP Explained (IPv4 case), only one master

Initialize Master

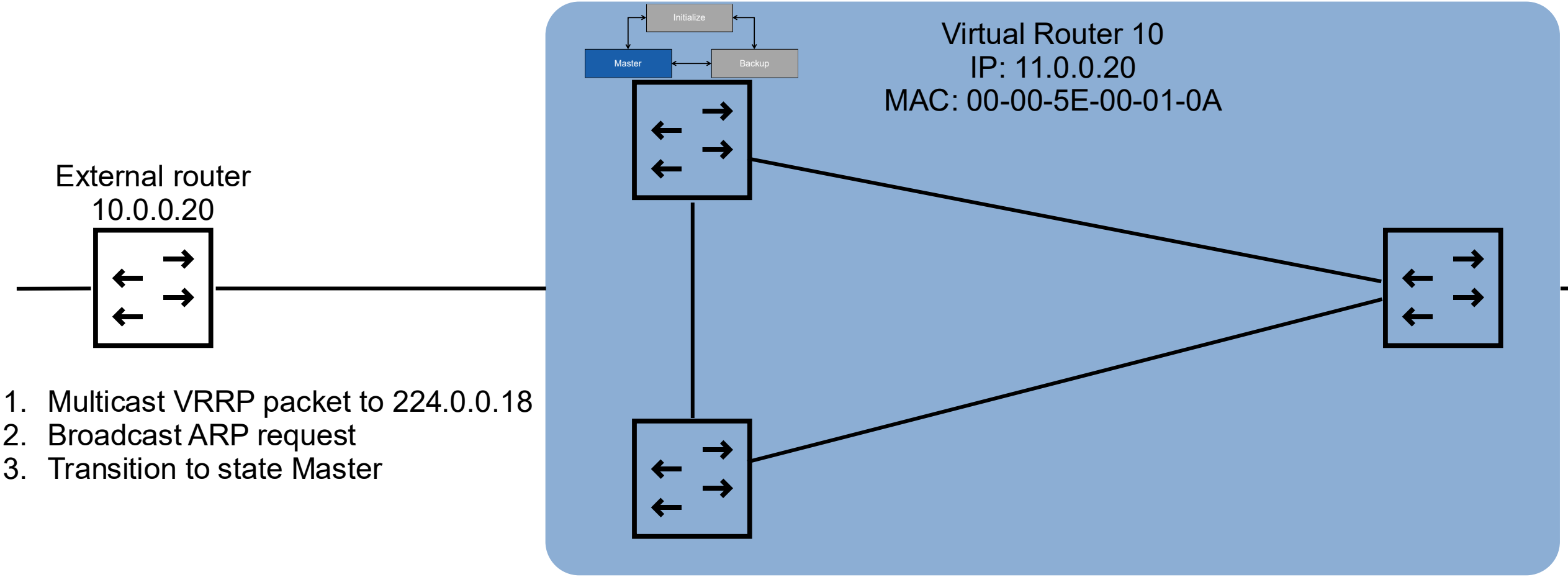


1. Multicast VRRP packet to 224.0.0.18
2. Broadcast ARP request

Active Router depicted in green, failed routers in red

VRRP Explained (IPv4 case), only one master

Initialize Master

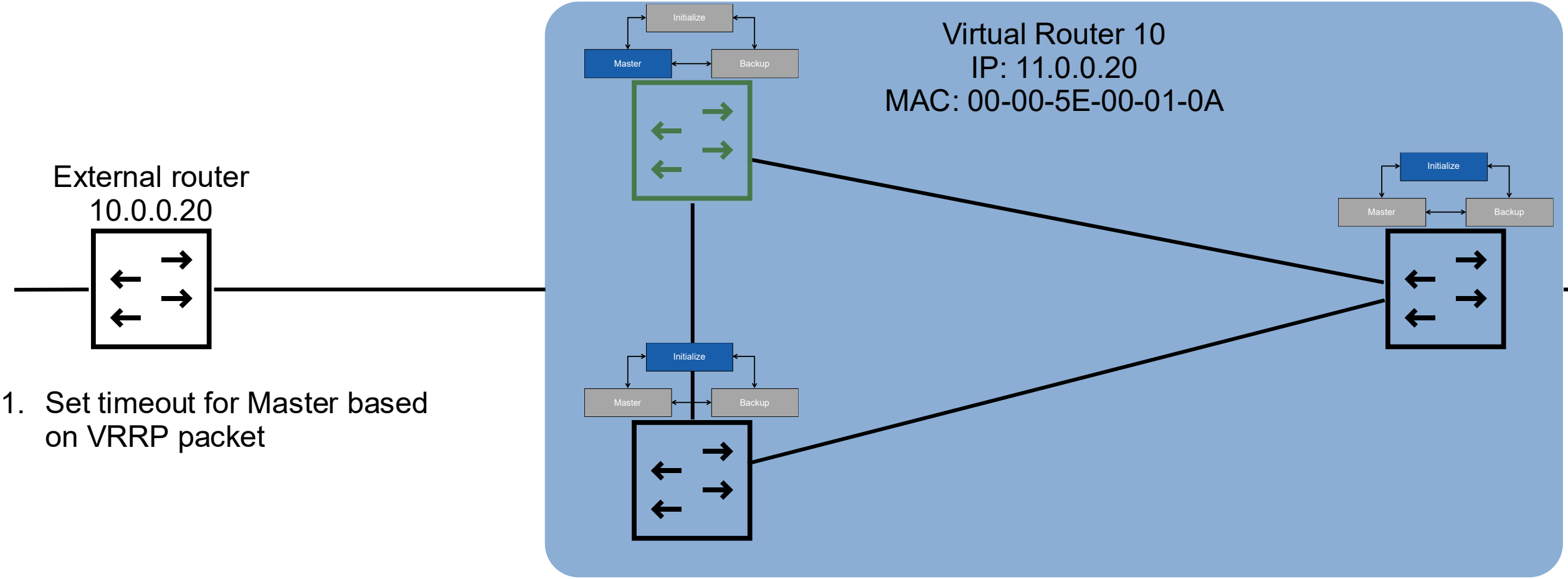


1. Multicast VRRP packet to 224.0.0.18
2. Broadcast ARP request
3. Transition to state Master

Active Router depicted in green, failed routers in red

VRRP Explained (IPv4 case), only one master

Initialize Backup

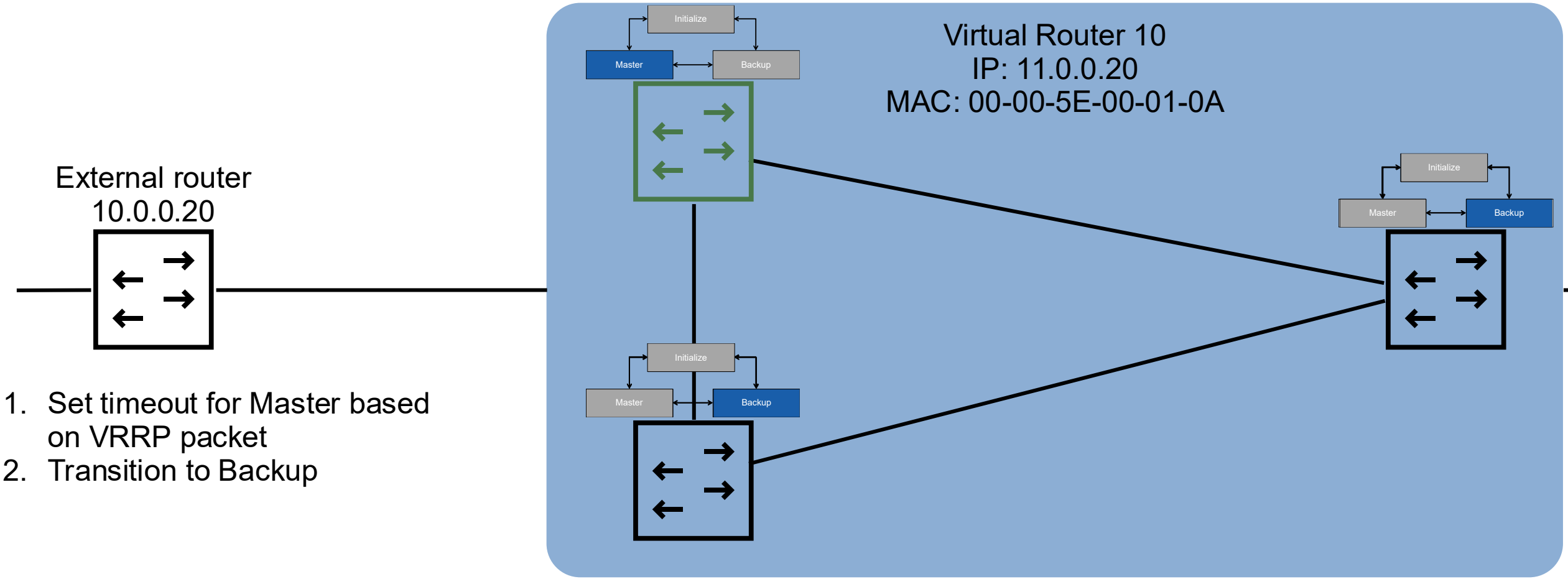


1. Set timeout for Master based on VRRP packet

Active Router depicted in green, failed routers in red

VRRP Explained (IPv4 case), only one master

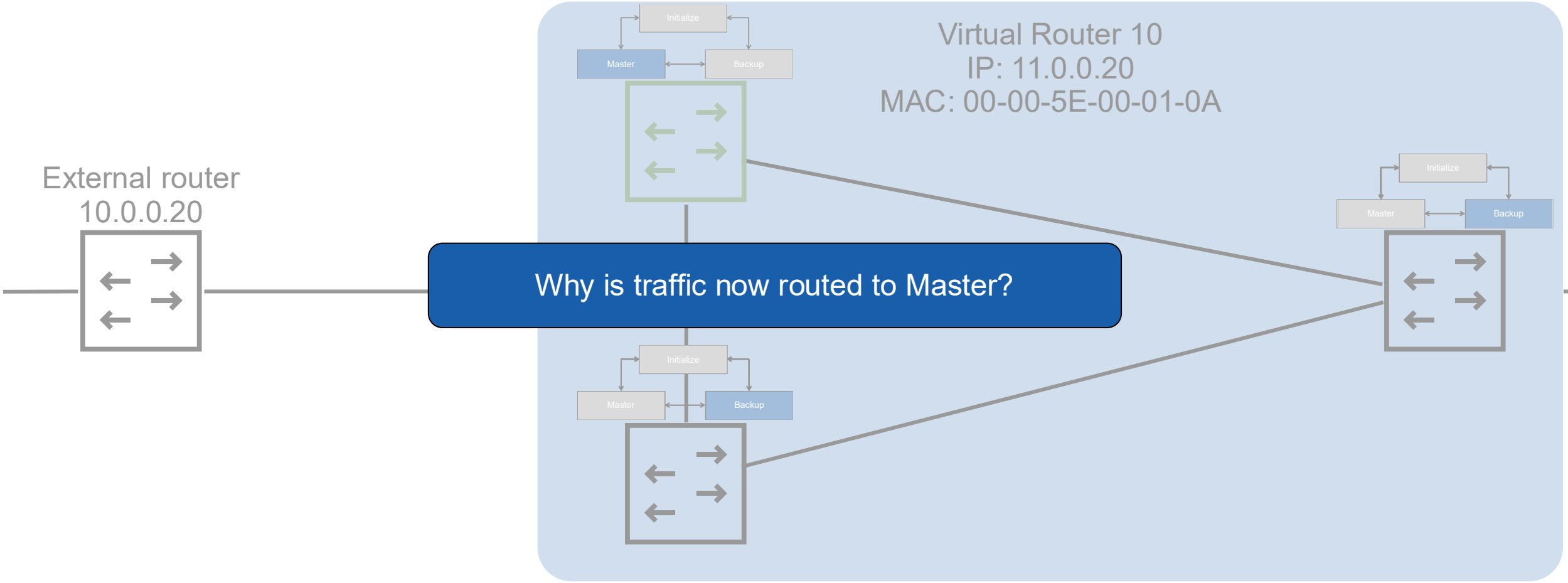
Initialize Backup



1. Set timeout for Master based on VRRP packet
2. Transition to Backup

Active Router depicted in green, failed routers in red

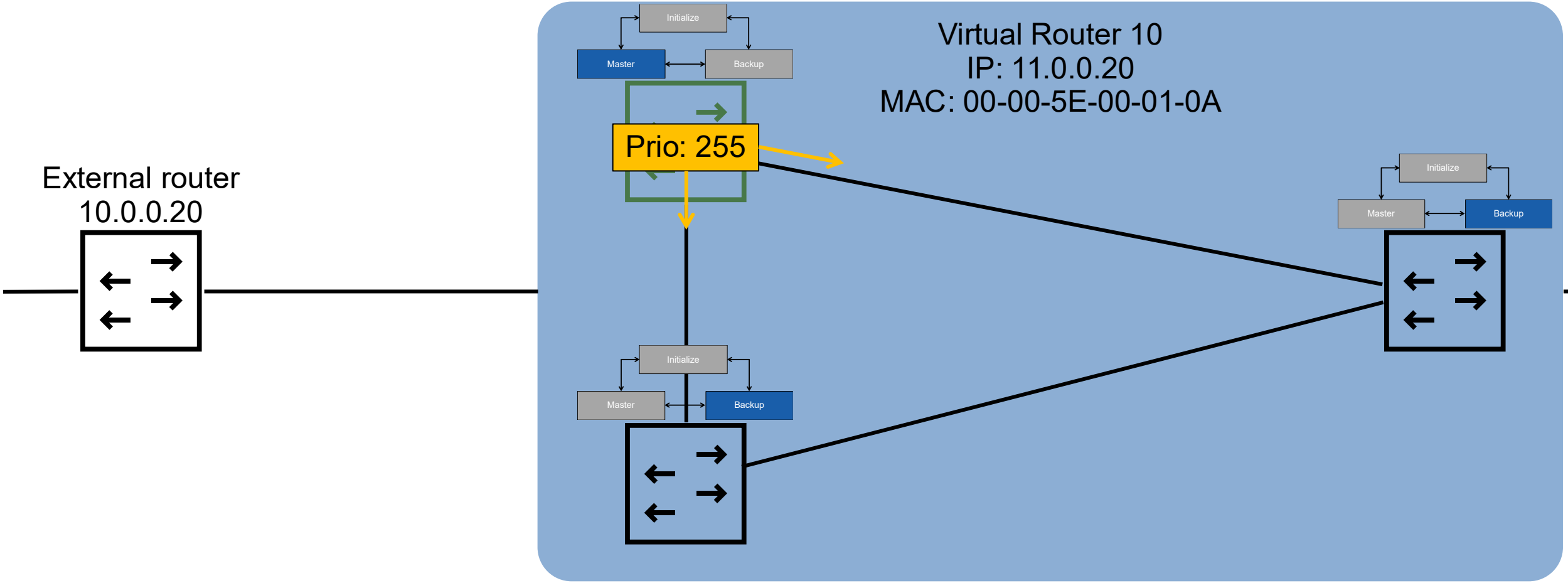
VRRP Explained (IPv4 case), only one master
Functionality



Active Router depicted in green, failed routers in red

VRRP Explained (IPv4 case), only one master

Normal operation



Active Router depicted in green, failed routers in red

VRRP Explained (IPv4 case), only one master

Master failed

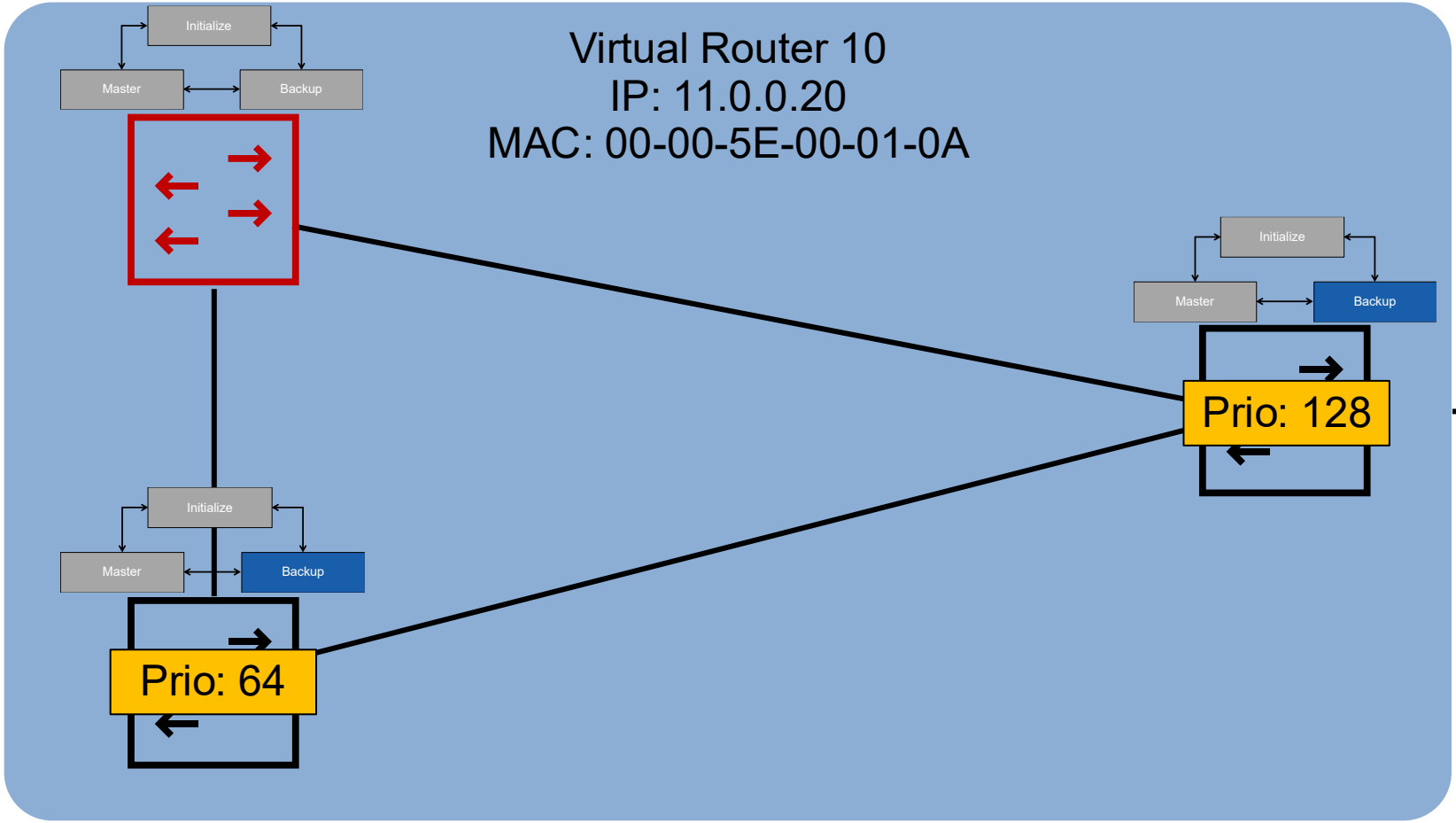
For now: All VRRP routers notice the failure simultaneously

How do we ensure that the higher-priority router becomes Master?

Solution: the waiting time depends on the priority p and the packet interval t_{master}

$$t_{up} = 3 * t_{master} + t_{skew}$$

$$t_{skew} = \frac{(256 - prio) * t_{master}}{256}$$



Active Router depicted in green, failed routers in red

VRRP Explained (IPv4 case), only one master

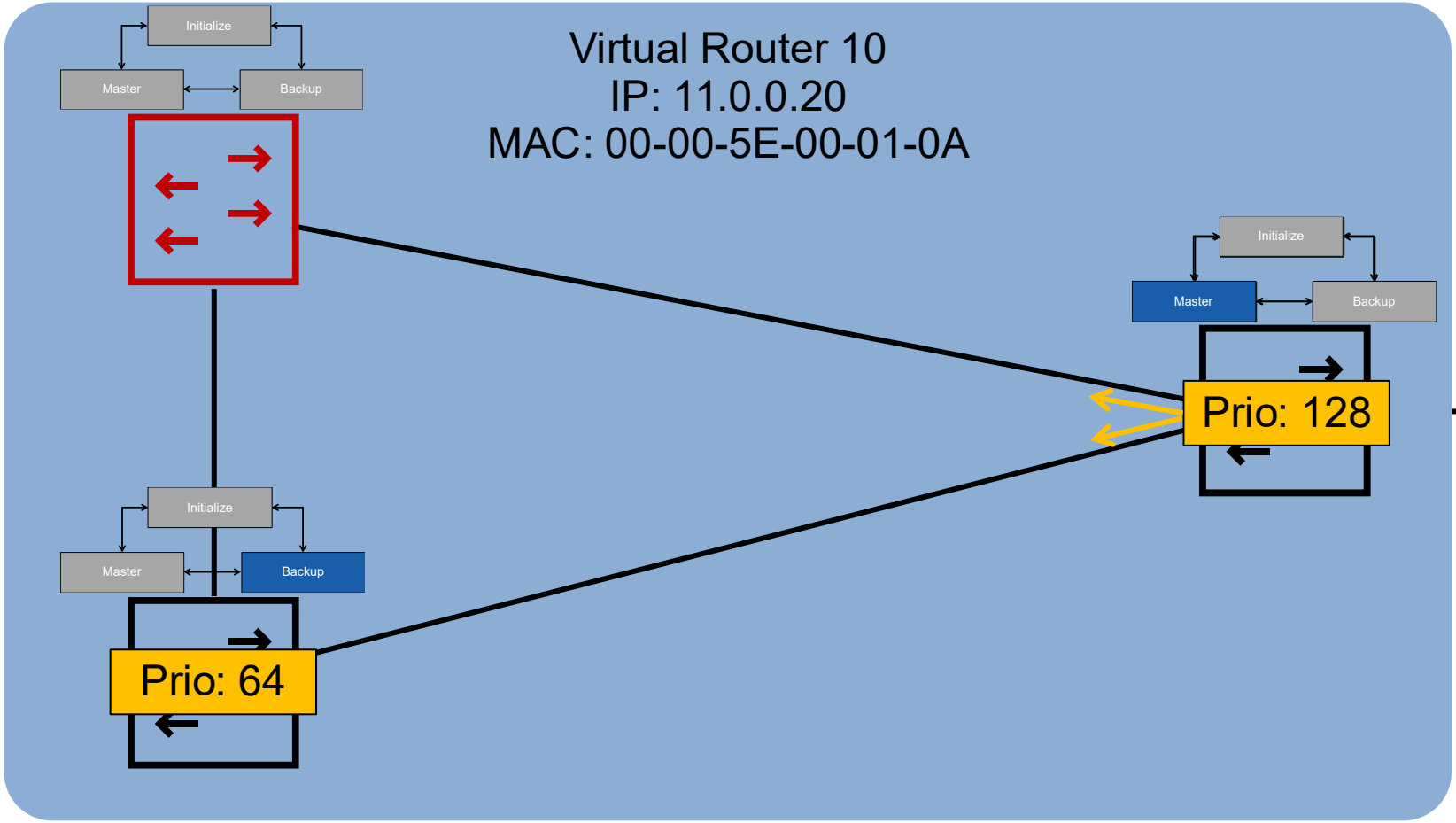
Master failed

$$t_{up}(p) = 3 * t_{master} + t_{skew}$$

$$t_{skew}(p) = \frac{(256 - p) * t_{master}}{256}$$

For example, with $t_{master} = 1s$

$$t_{up}(128) = 3{,}5s$$
$$t_{up}(64) = 3{,}75s$$



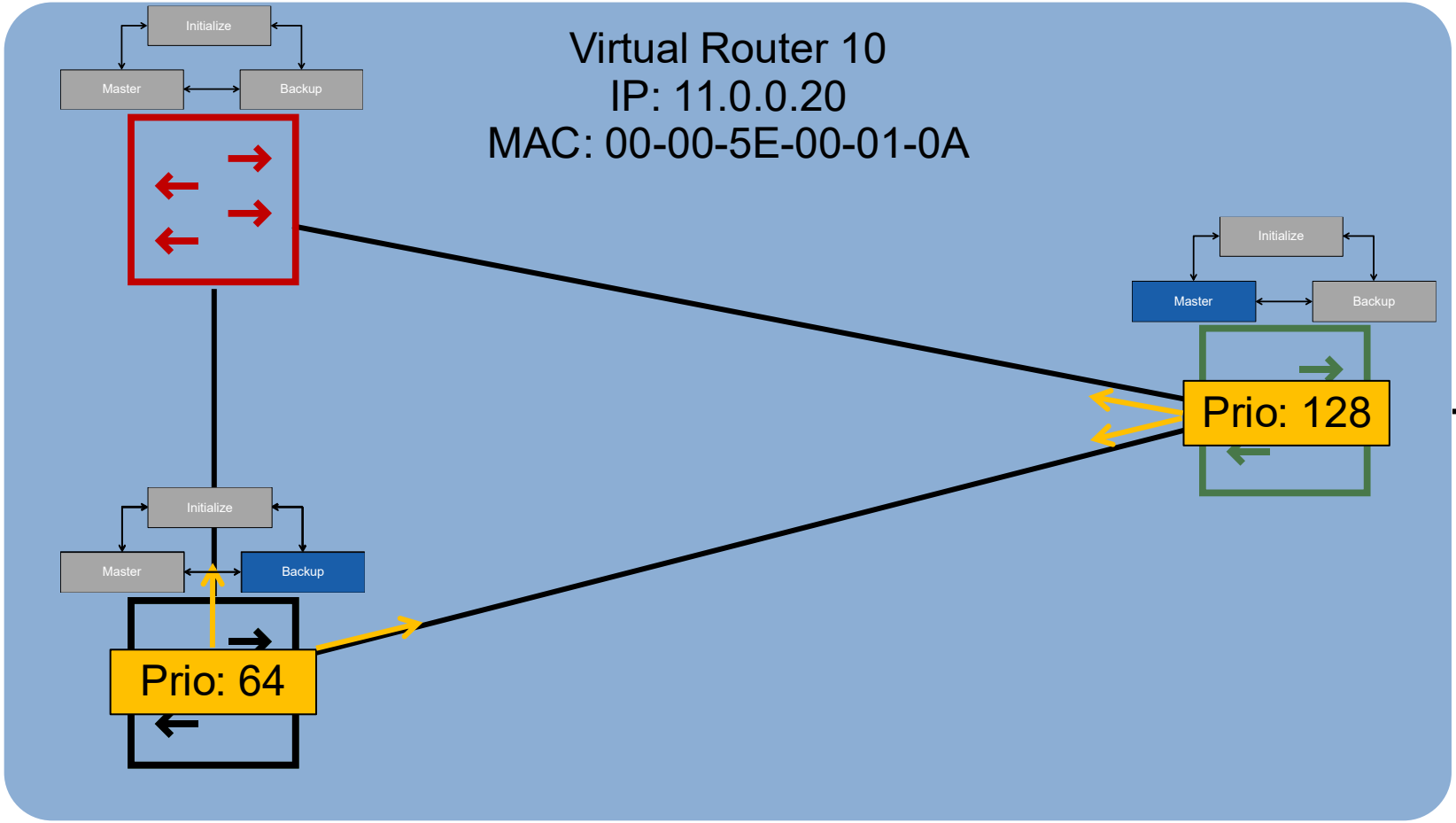
Active Router depicted in green, failed routers in red

VRRP Explained (IPv4 case), only one master
Two masters

What happens if two routers become new Master?

Once a VRRP router receives a VRRP packet, it checks the priority

- If the local priority is lower, then it switches to backup
- If the local priority is higher, the packet is ignored



Active Router depicted in green, failed routers in red